EDGE LAWS, A HADAMARD FORMULA AND KINK SINGULARITIES FOR EIGENFUNCTIONS OF THE LOCALIZED WEIL QUADRATIC FORM

JOHANN ALEXANDER BOTHA

ABSTRACT. Let Q_L be Weil's quadratic form for the Riemann zeta function on real test functions supported in $[-L, L]$ (pole terms, archimedean term, and all prime powers n with $\log n < 2L$); it is the form of a self-adjoint operator of order zero. We study its eigenfunctions ψ, with the pole terms carrying any real weight. For every eigenfunction the edge traces $A_\pm = \lim_{r\downarrow 0} \psi(\pm(L-r))(\log \frac{1}{r})^{1/2}$ exist and $|A_+| = |A_-|$, and a simple eigenvalue λ satisfies the Hadamard formula $d\lambda/dL = -(A_+^2 + A_-^2)$, also where a new prime power enters. These facts, with a limit formula for $A_\pm$, are not new: they were first proved, in unrefereed form, by the AI agent bramble_trace on the platform rStation (riemann_hypothesis paper 67), and the traces had been derived in substance earlier in the mathia research notes (details in Section 1.4). We give proofs with quantitative remainders that are uniform on compact parameter ranges, for the Hadamard formula by a different route (an exact two-window cross identity) and for eigenvalues that are only simple in their parity sector, and we prove the following. *Second-order edge law:* $\psi(\pm(L-r)) = A_\pm(\log \frac{1}{r} - D)^{-1/2} + O(\log^{-5/2} \frac{1}{r})$ with $D = \lambda + \log 2\pi + \gamma$. *Translation identity:* for an even ψ_e and an odd ψ_o with $\lambda_e \neq \lambda_o$, the overlap $\langle \psi_o, \psi_e(\cdot + h)\rangle$ has derivative $2A_+(\psi_e)A_+(\psi_o)/(\lambda_e - \lambda_o)$ at $h = 0$. *Kink law:* at $L - \log n$ and at every point reached from an edge by a chain of such shifts, ψ has a logarithmic singularity with an explicit two-sided profile; for $L > \frac{1}{2}\log 6$ these points are dense, and ψ is Hölder of no positive order on any interval when $A_\pm \neq 0$. The second-order and kink laws hold at fixed L, with constants that are not uniform near the windows $2L = \log n$. *Tail law:* $\sum_{|t_\rho|>T} \hat\psi(\gamma_\rho)\hat\psi(-\gamma_\rho) = (A_+^2 + A_-^2)/(\pi T) + O(1/(T\log T))$ over the nontrivial zeros $\rho = \frac{1}{2} + i\gamma_\rho$ of ζ, $t_\rho = \operatorname{Re}\gamma_\rho$, with no hypothesis on their location. The tools are the Wiener–Hopf factorisation of the archimedean symbol, a Krein-type cross identity and Landau's formula. None of these results proves or disproves the Riemann hypothesis, and the laws are asymptotic and say nothing where $A_\pm = 0$. Uncertified computations illustrate the results; two pre-registered tests failed in part as registered, and all failures are reported.

1. INTRODUCTION

1.1. **The localized Weil form.** For real $f \in L^2(\mathbb{R})$ write $F(w) = \hat f(w) = \int_{\mathbb{R}} f(x)e^{ixw}\,\mathrm{d}x$. Let Λ be the von Mangoldt function, $c_n := \Lambda(n)/\sqrt{n}$, $\psi_\Gamma = \Gamma'/\Gamma$, and

$$a(t) := \operatorname{Re}\psi_\Gamma\big(\tfrac{1}{4} + \tfrac{it}{2}\big) - \log\pi.$$

For $L > 0$ and real $f \in L^2(\mathbb{R})$ vanishing outside $[-L, L]$, Weil's explicit formula [79], evaluated at $f * \tilde f$ with $\tilde f(x) = f(-x)$, gives the quadratic form

$$Q_L(f) = \frac{1}{\pi}\int_0^\infty \Psi_L(t)\,|F(t)|^2\,\mathrm{d}t + 2F(\tfrac{i}{2})F(-\tfrac{i}{2}), \qquad \Psi_L(t) = a(t) - 2\sum_{\log n<2L} c_n\cos(t\log n). \tag{1.1}$$

For $f \in C_c^\infty(-L, L)$ one has $Q_L(f) = \sum_\rho F(\gamma_\rho)F(-\gamma_\rho)$, where $\rho = \frac{1}{2} + i\gamma_\rho$ runs over the nontrivial zeros of ζ with multiplicity (Lemma 8.1). Weil's criterion states that the Riemann hypothesis (RH) holds if and only if $Q_L(f) \geq 0$ for all such f and all $L > 0$ [79, 6]. Since $f * \tilde f$ is supported in $[-2L, 2L]$, only prime powers with $\log n < 2L$ enter, and n enters at $L = \frac{1}{2}\log n$. We use the

Date: 31 October 2026.

2020 *Mathematics Subject Classification.* 11M06 (primary); 11M26, 47A75, 47G20, 35P15, 60G51 (secondary).

Key words and phrases. Weil quadratic form, explicit formula, Riemann zeta function, nonlocal operator of order zero, boundary trace, Hadamard formula, Wiener–Hopf factorisation, complete Bernstein function, Landau's formula.

norm $\|f\|^2 = \int f^2 = \frac{1}{\pi}\int_0^\infty |F|^2$, as in [83] and [10, §2]. The form Q_L is the quadratic form of a self-adjoint operator on $L^2(-L,L)$, bounded below, with compact resolvent, which commutes with $x \mapsto -x$; its form domain is the form closure of $C_c^\infty(-L,L)$ (Section 2).

Connes and Consani studied the semilocal form on functions supported multiplicatively in $[e^{-L}, e^{L}]$ (their QW_λ with window parameter $\lambda = e^L$, not our eigenvalue): its very small eigenvalues, the approximation of its eigenvectors by prolate functions, and the loss of positivity when the contribution of a prime power is omitted as the window grows past it [17, §§2–3]. Connes, Consani and Moscovici [18] proposed a strategy based on this form; they show that the smallest eigenvalue is monotone in the window [18, Cor. 3.7, (3.27)]. Connes and van Suijlekom proved that if the lowest eigenvalue is simple with an even eigenfunction ξ, then $\hat{\xi}$ has only real zeros [19, Thm. 1.2]; for the Weil form this "simple-even" property is one of the missing steps listed in [16, §6.6]. Suzuki [72] realised the operator as a Friedrichs extension, proved that its lowest eigenvalue is continuous in L [72, Thm. 1.3] and simple with an even eigenfunction for small L [72, Thm. 1.4], and wrote the action of the operator, with its $\pm\log n$ shift structure, on smooth functions [72, eq. (2.11)]; see also [71]. Suzuki's window parameter a is our L. Earlier work on the form includes [82, 6, 7], and a numerical study of Suzuki's operator is [52]. Several unrefereed 2026 deposits give certified computations: per-sector lower bounds for the lowest eigenvalue up to $L = 1.495$ by Elston [27], positivity certificates of Liu [59] and Mori [61], simple-even at $L = 0.8$ by Zhu [83], a claim of simple-even on $(0, 0.8]$ by Kramarenko-Byrd [53], and the author's computer-assisted proof of simple-even for every $L \in (0, 1.19]$ [10]. On rStation, a platform on which AI research agents post manuscripts that are reviewed by automated reviewer agents, certificates of positivity of both parity sectors have been posted for $L \in (0, \frac{1}{2}\log 11]$ [58] and $L \in (0, 2]$ [15].

These works concern the spectrum. This paper is about the eigenfunctions: how they behave at the window edges and in the interior, and how this is reflected in the dependence of the eigenvalues on L and in the zero side of the explicit formula. The closest work on the eigenfunctions is the rStation manuscript [12], which gives proofs (unrefereed) of the existence of the edge traces and of the Hadamard formula of Theorem 5.6; the chronology and a detailed comparison are in Section 1.4.

1.2. **Main results.** By Stirling's formula $a(t) = \log\frac{|t|}{2\pi} + O(t^{-2})$, so Q_L has order zero: its principal part is a logarithm of the Laplacian. On a bounded interval such operators produce eigenfunctions that vanish at the boundary only logarithmically (for the logarithmic Laplacian, $|u| \le C\log(1/\operatorname{dist}(x,\partial\Omega))^{-1/2}$ is sharp [47, Thms. 1.1, 1.2]). We write the form as $\mathcal{E}+w_s$, where $\mathcal{E}$ is the Dirichlet form of a symmetric Lévy process whose Lévy density $J(u) = e^{-|u|/2}/(1-e^{-2|u|}) \sim \frac{1}{2|u|}$ is explicit (this representation, with the same J, is in [72, §2.4], [20], [63, §2.2] and [12, (3.1)–(3.3)]), and w_s collects the constant, the pole terms with weight s and the prime shifts; w_s is bounded on functions supported in a fixed window. We allow any pole weight $s \in \mathbb{R}$:

$$Q_L^{(s)}(f) = \frac{1}{\pi}\int_0^\infty \Psi_L(t)|F(t)|^2\,\mathrm{d}t + 2s\,F(\tfrac{i}{2})F(-\tfrac{i}{2}), \tag{1.2}$$

so $s = 1$ is Weil's form (1.1) and $s = 0$ removes the pole. Let $H_L^{(s)}$ be the associated operator, ψ a real L^2-normalised eigenfunction with eigenvalue λ, and $\ell(r) = \log(1/r)$. Eigenfunctions can be chosen even or odd; we speak of the even and odd *sectors*. All statements below hold for every $L > 0$, every $s \in \mathbb{R}$ and every eigenfunction, unless a hypothesis is stated.

Edge law (Theorems 4.1 and 7.1). ψ is bounded and continuous, the edge traces $A_\pm := \lim_{r\downarrow 0}\psi(\pm(L-r))\ell(r)^{1/2}$ exist, and A_+ is given by an explicit integral of the Wiener–Hopf kernel against ψ; these facts, with a limit formula of the same kind, are due to [12] (Section 1.4). Moreover $\psi(\pm(L-r)) = A_\pm\ell(r)^{-1/2} + O(\ell(r)^{-3/2})$, with constants uniform for L, s and λ in compact sets, and, at fixed L,

$$\psi\big(\pm(L-r)\big) = A_\pm\big(\ell(r) - D\big)^{-1/2} + O\big(\ell(r)^{-5/2}\big), \qquad D = \lambda + \log 2\pi + \gamma;$$

the constants of this second-order law depend on L and degenerate as $2L$ decreases to some $\log n$.

Hadamard formula and translation identity (Theorems 5.6, 5.4). $|A_+| = |A_-| =: A$ for every eigenfunction, and on each eigenspace the trace map $\psi \mapsto (A_+, A_-)$ has rank at most one. If $\lambda = \lambda_k^\sigma(L)$ is simple in its sector σ, then λ is C^1 in L and

$$\frac{d\lambda}{dL} = -\big(A_+^2 + A_-^2\big) = -2A^2,$$

also at the windows $2L = \log n$ where a new prime power enters. For simple eigenvalues these statements are [12, Thms. 2.1–2.2]; our proof takes a different route and gives the remainder $O(\delta\,\ell(\delta)^{-1/2})$ for a change δ of the window. For an even ψ_e and an odd ψ_o with $\lambda_e \neq \lambda_o$, $(\lambda_e - \lambda_o)\frac{d}{dh}\langle\psi_o, \psi_e(\cdot + h)\rangle|_{h=0} = 2A_+(\psi_e)A_+(\psi_o)$. Sector-simplicity is known for the ground state when $s \le 0$ (every L), for the odd ground state when $s \ge 0$ and $2L \le \log 2$, and, by the computer-assisted result [10], for the ground state of the true form ($s = 1$) for $L \in (0, 1.19]$. In these cases we also show that $A \neq 0$ (for $s = 1$: off a closed nowhere dense set of $L \in (0, 1.19)$; Proposition 5.8).

Kink law (Theorems 7.3, 7.5 and Corollaries 7.4, 7.6). The prime shifts transport the edge singularity into the interior. At $y_n = L - \log n$, n a prime power with $\log n < 2L$, and for generic L (no point is reached by chains of shifts from both edges; this holds for every nonzero algebraic L), with constants that depend on L and degenerate as $2L$ decreases to some $\log n$,

$$\psi(y_n - d) - \psi(y_n) = \tfrac{2}{3}c_n A_+ \ell(d)^{-3/2} + O(\ell^{-5/2}), \qquad \psi(y_n + d) - \psi(y_n) = -\tfrac{1}{3}c_n A_+ \ell(d)^{-3/2} + O(\ell^{-5/2}).$$

At a point reached from an edge by a chain of k shifts the profile has order $\ell^{-k-1/2}$ and split $\frac{k+1}{2k+1} : \frac{-k}{2k+1}$; the jump (odd part) has unit amplitude in every generation and, at second order, the same offset D as the edge. For $L > \frac{1}{2}\log 6$ the singular points are dense, so an eigenfunction with $A \neq 0$ is Hölder continuous of no positive order on any open interval.

Tail law (Theorems 8.3, 8.8, 9.1–9.3). Put $h(w) = F(w)F(-w)$ (so $h(t) = |F(t)|^2$ for real t) and $Z(T) = \sum_{|t_\rho| > T} h(\gamma_\rho)$, where $t_\rho = \operatorname{Re}\gamma_\rho = \operatorname{Im}\rho$. With no hypothesis on the location of the zeros,

$$\pi T\, Z(T) = A_+^2 + A_-^2 + O\Big(\frac{1}{\log T}\Big),$$

which equals $|\lambda'(L)| + O(1/\log T)$ when λ is sector-simple. An exact identity (Theorem 8.3) expresses $\sum_\rho h(\gamma_\rho)(1 - \cos z\gamma_\rho)$ through the translation overlap $\langle\psi, \psi(\cdot - z)\rangle$ and a contact integral at the edges. At second order, smoothed sums carry a per-sector term $(\lambda + \varepsilon c_X)/\log T$, where $X = e^{2L}$, $c_X = \Lambda(X)/\sqrt{X}$ (zero unless X is a prime power) and $\varepsilon = \pm 1$ is the parity. The error is $O(\log^{-3/2} T)$, so this term is resolved only by a factor $(\log T)^{1/2}$ (Theorem 9.2(d),(e)); the sharp-cutoff version is open (Conjecture 11.1).

The tools are the Wiener–Hopf factorisation of the archimedean symbol on a half-line, following Kwaśnicki [54]; an exact cross identity between eigenfunctions on two windows, an order-zero continuum analogue of Krein's resolvent identity [22, Lemma 2.3] and of the finite-dimensional commutator identities [18, Lemma 5.2(ii)], [19, Lemma 5.1(ii)]; second-order renewal asymptotics; and, on the zero side, the Riemann–von Mangoldt formula and Landau's formula [56].

1.3. **Strategy (heuristic).** The proofs rest on four mechanisms. We describe them informally; the precise statements are in Sections 3–9.

1. The edge germ. Reflect at the right edge, $r = L - x$. Near $r = 0$ the eigenvalue equation becomes a half-line problem $(q + \Psi_a)u = f$ on $(0, \infty)$, $u = 0$ on $(-\infty, 0)$, whose right side f is bounded: the primes and the pole enter only through f. The symbol factorises, $q + \Psi_a(t) = K(it)K(-it)$, with K a complete Bernstein function and $K(\xi)^2 \sim \log\xi$. If $1/K$ is the Laplace transform of v, the half-line solution is $u = v * (P_+(\tilde v * f))$, so $u(r) \approx (\int_0^\infty vf)\,V(r)$ with $V(r) = \int_0^r v \approx 1/K(1/r) \approx \ell(r)^{-1/2}$. This gives the edge law with $A_+ = \int vf$ (Theorem 4.1); expanding V and the function $R(x) = \int_0^x (v * \tilde v)$ to second order gives the offset D (Theorem 7.1).

2. The Hadamard formula. Compare normalised eigenfunctions ψ_1, ψ_2 on nested windows $(-L_1, L_1) \subset (-L_2, L_2)$, $\delta = L_2 - L_1$. A cross identity (Lemma 5.1) reduces $(\lambda_2 - \lambda_1)\langle\psi_1, \psi_2\rangle$ to integrals over the two layers $L_1 < |y| < L_2$ of the exterior field $J\psi_1$ against ψ_2. In the right layer, $(J\psi_1)(L_1 + \eta) \approx A_+ \int_0^{\rho_0} \frac{V(r)}{2(\eta+r)}\,\mathrm{d}r \approx A_+\ell(\eta)^{1/2}$, while $\psi_2(L_1 + \eta) \approx A_+\ell(\delta - \eta)^{-1/2}$, and

$$\int_0^\delta \ell(\eta)^{1/2}\ell(\delta - \eta)^{-1/2}\,\mathrm{d}\eta = \delta\big(1 + O(\ell(\delta)^{-1})\big)\,:$$

each edge contributes $A^2\delta$ (in the proof, the Wiener–Hopf form of the exterior field makes the corresponding integral exactly δ), so $\lambda' = -(A_+^2 + A_-^2)$. Translating the window instead of enlarging it gives $A_+^2 = A_-^2$.

3. Kinks. For x near $y_n = L - \log n$ the shift term $c_n\psi(x + \log n)$ of the eigenvalue equation equals $c_n\psi(L - (y_n - x)) \approx c_nA_+\ell(y_n - x)^{-1/2}$ for $x < y_n$ and vanishes for $x > y_n$: the source has a one-sided logarithmic singularity at y_n. Inverting the order-zero operator (convolution with $r_q = v * \tilde{v}$, whose primitive satisfies $2R(d) \approx \ell(d)^{-1}$) gains exactly one power of ℓ and spreads the singularity to both sides, in the ratio $2 : -1$. Iterating along chains of shifts gives the hierarchy, and the rotation $x \mapsto x + \log 2$ modulo $\log 6$ makes the singular points dense.

4. The tail. Put $f_z = \psi - \tau_z\psi$ $(z > 0)$ and apply the explicit formula to it. Three quantities must be kept apart. The form energy $Q^{(s)}(f_z) := Q^{(s)}_{L+z}(f_z)$ is $\lambda\|f_z\|^2$ plus the reduced energy $Q^{(s)}(f_z) - \lambda\|f_z\|^2 = 2\mathcal{L}(z)$, a contact integral at the edges, which by the same layer computation as in step 2 is $\approx (A_+^2 + A_-^2)z = 2A^2z$; since $\lambda\|f_z\|^2 = O(z/\ell(z))$, also $Q^{(s)}(f_z) \approx 2A^2z$. The zero sum $Z_{\cos}(z) = \sum_\rho h(\gamma_\rho)(1 - \cos z\gamma_\rho)$ is half the form energy, up to a pole term $O(z^2)$ that vanishes for $s = 1$, so $Z_{\cos}(z) \approx A^2z$ (Theorems 8.3 and 9.2(c)). On the Fourier side the edge germ gives, in the mean over t, and with the zero density from the Riemann–von Mangoldt formula,

$$|F(t)|^2 \approx \frac{2A^2}{t^2\log t}, \qquad \mathrm{d}N(t) \approx \frac{\log t}{2\pi}\,\mathrm{d}t, \qquad Z(T) \approx 2\int_T^\infty \frac{2A^2}{t^2\log t}\,\frac{\log t}{2\pi}\,\mathrm{d}t = \frac{2A^2}{\pi T},$$

and the same density gives $Z_{\cos}(z) \approx \frac{2A^2}{\pi}\int_0^\infty \frac{1-\cos zt}{t^2}\,\mathrm{d}t = A^2z$, in agreement with the position side. The logarithms cancel, which is why the tail law has the exact constant $(A_+^2 + A_-^2)/\pi$. The second-order terms come from $\log t$ being really $a(t) - \lambda$, and from Landau's formula, which adds εc_X to the zero density seen by the oscillating edge-times-edge term.

1.4. **Relation to previous work, credits and novelty.** *rStation.* Much of the closest work is recent and unrefereed. In particular, rStation (`https://rstation.ai`) is a platform on which AI research agents post manuscripts, which are reviewed and accepted by automated reviewer agents, and discuss them in public forums. The manuscript [12], posted there on 27 October 2026 by the agent bramble_trace, treats the operator of this paper in the same normalisation: its half-width a is our L, its traces $\tau_\pm$ are our $A_\pm$, its form (2.1) is $Q_L^{(1)}$, and its class (2.8) contains $H_L^{(s)}$ for every s. We cite rStation manuscripts and forum comments for the priority of public statements. They are unrefereed public claims written by AI agents; we have read the statements we cite and the parts of the proofs described below, but we have not checked those proofs fully, and no proof in this paper depends on them. The results of this paper were obtained without knowledge of rStation: we learned of [12] from an external review on 31 October 2026, after our proofs were written, and added the comparisons with rStation items then (a later revision, made after we learned of [12], shortened the proofs of Lemmas 5.2 and 5.3 and removed a factor $1 + \log\ell$ from their remainders, using only Appendix C). We knew parts of the mathia notes while working.

Edge law. The scale $\ell^{-1/2}$ is known: Kwaśnicki proved an exact boundary law for the generalised eigenfunctions of $\phi(-\Delta)$ on a half-line, ϕ a complete Bernstein function, by Wiener–Hopf and Tauberian methods [54, Lemma 4.27, Ex. 6.4] (first order, position side); our archimedean symbol is of this type. Two-sided bounds for log-type operators are in [45, 51, 5, 47]. For the Weil form,

the existence of $A_\pm$ for every eigenfunction, every real pole weight and every L, together with an L^∞ bound and a limit formula of the same kind as Theorem 4.1(iii), is proved in [12, (2.2), (3.8), §3.3, (3.19)]; it was first posted, as a lemma, in a forum comment of 18 October 2026 [14]. That manuscript also gives a uniform bound for the traces and their continuity in the window [12, (7.3), §7]; the only remainder it states is a relative $o(1)$ [12, §4]. Its route to existence (a variance-gamma splitting, the half-line renewal function of [45] and a bounded perturbation) is close to ours, which factorises the archimedean symbol directly (Section 3). Earlier, the mathia research notes, maintained by S. Garcia Murillo and written largely by AI agents, derived the trace in substance. PL-319 to PL-323 [34, 35] obtain the limit with a qualitative remainder $o(\ell^{-1})$ for each fixed eigenstate, assuming a pointwise realisation of a singular integral that is not written out (PL323-3, PL323-4). PL-329 [37] states, for endpoint profiles g that obey the exact endpoint identity of PL-319 with an envelope $|g| \le AQ^{-1/2}$ and a source bounded by S, the bound $|g(Q) - C\,Q^{-1/2}| \le K(1+S)/Q + Ke^{-c(Q-Q_1)}$, with K, c, Q_1 independent of the family parameter and $Q = \ell + \log 2$; its quantitative maximum-principle step is asserted, not written out (PL329-4, PL329-20). PL-336 [39] applies this to the odd eigenbranch at the first prime activation, conditionally on a uniform L^∞ bound, and obtains $u_0(1-2e^{-Q}) = C_0 Q^{-1/2} + O(Q^{-1})$ (PL336-6); PL-337 [40] proves such a bound for all normalised eigenpairs with windows in a compact set and bounded eigenvalue (PL337-3); and PL-338 [41] states the expansion, with an L^2 remainder on the newly active layer, for odd eigenstates near every prime-power activation (PL338-2, PL338-3). So a remainder $O(\ell^{-1})$, uniform on compact ranges, is stated in those notes for the true form, with a sketched step. Theorem 4.1(ii) gives a complete proof of such a remainder for the whole family $H_L^{(s)}$ and every eigenfunction, by explicit Wiener–Hopf inversion, with constants explicit in terms of the L^∞ bound of Lemma 3.6 (see (3.3)). New here, as far as we know, are the remainder $O(\ell^{-3/2})$, uniform on compact ranges as in Theorem 4.1 (Theorem 7.1(a)), and the second-order law with offset D and remainder $O(\ell^{-5/2})$ (Theorem 7.1(c), Proposition 6.2). The offset D appeared earlier, without proof, in the denominator of a heuristic fit for an eigenvalue ratio at an activation, posted on rStation by the agent span_of_dilates [69]; a second-order expansion of the Wiener–Hopf factor of the variance-gamma symbol was posted there by halasz_resonator [46]. Kim et al. list the boundary exponent of the eigenfunctions as an open problem requiring "a Wiener–Hopf analysis" [52, §§5.2, 5.4]; as we read their §5.2, a power fit gives an exponent of about 0.18–0.29, and a logarithmic correction to the boundary exponent is listed there as refuted. Theorem 4.1 shows that the profile is exactly logarithmic, $A_+\ell(r)^{-1/2}$, with no power of r. Its local power exponent $\frac{\mathrm{d}\log\psi}{\mathrm{d}\log r} = \frac{1}{2\ell(r)}$ is about 0.2 at $\ell = 2.5$, so a power fit on a finite range returns an apparent exponent of that size.

Hadamard formula. For the Weil form we found monotonicity [18, Cor. 3.7] and continuity [72, Thm. 1.3] (continuity is also asserted in [6, Thm. 5], [7, Thm. 4.4]; see [72, §1]). The question whether λ' is given by a zero-order Hadamard formula in squared, renormalised edge stresses was posed in the mathia notes, in a research clue of 13 September 2026 [28] and in PL-297 and PL-305 to PL-307 [30, 31, 32, 33], and reduced there to conditional statements: PL-306 derives the consequence $\lim(|\tau_+|^2 + |\tau_-|^2)/(2s) = -a\lambda_0'(a)$ of an assumed regularised Hadamard law, PL-307 recovers the same scalar stress from a fractional Pohozaev identity under regularity hypotheses on the branch, and PL-305 and PL-338 show that a prime activation adds no first-derivative singularity (for the lower-order part of the form at a fixed state in $C_0 \cap BV$, and for odd states with bounded traces). The formula itself was proved in [12, Thms. 2.1–2.2], and stated earlier the same day in a forum comment [13]. Under an enlargement δ of the window, an isolated eigenvalue cluster moves as $\lambda - \delta\,\mathrm{eig}(H) + o(\delta)$, where $H = B_- + B_+$ is the Gram matrix of the traces [12, (2.3), (2.4)]; translation invariance gives $B_- = B_+$, $H = 2B_+$ and $\operatorname{rank} H \le 1$ [12, (2.5)]; ordered eigenvalues are nonincreasing and locally Lipschitz, and a simple branch is C^1 with $\lambda' = -|\tau_-|^2 - |\tau_+|^2$, including at prime activations [12, Thm. 2.2, (2.6), (7.5)]. The proof there decomposes the enlarged window into the old interval and two collars, passes to a strong resolvent limit on the collars and eliminates

the spectral complement exactly (a Feshbach–Schur reduction). Theorem 2.4 of [12] also proves an activation formula: when a prime-power atom enters at $2a = \log n$, a simple eigenvalue moves by $-2wp|\tau_+|^2\delta/\log(1/\delta) + o(\delta/\log(1/\delta))$ relative to the branch without that atom, with p the parity and $w = \Lambda(n)/\sqrt{n}$. Accordingly, Theorem 5.6 for simple eigenvalues and Theorem 5.4(a),(b) are not new: they are [12, (2.5), (2.6)], and Theorem 5.4(d) is [12, (6.3)], spelled out for an even–odd pair in [70]. Our proof takes a different route: the exact cross identity of Lemma 5.1 between eigenfunctions on two windows, and a layer integral whose main term is exact because of the Wiener–Hopf form of the exterior field (Lemmas 5.2 and 5.3). The shared ingredient is the response $\approx A\,\ell^{1/2}$ of the eigenfunction just outside the edge (Lemma 5.2; [12, (4.7)]). Our route gives the remainder $C\delta\,\ell(\delta)^{-1/2}$, uniform on compact ranges of L, where [12] has $o(\delta)$, and C^1 for eigenvalues that are only sector-simple, through even–odd crossings, which is a minor extension. The manuscript [12] is stronger in other respects: it treats clusters and one-sided extensions and gives a Lipschitz bound. The translation identity with a rate (Theorem 5.4), its part (c) for $\lambda_e \neq \lambda_o$ (the mixed identity) and Corollary 5.7 are, to our knowledge, new. Hadamard formulas for the fractional Laplacian are in [23, 24], and for integrable kernels in [2]. The closest structural analogue is the prolate identity $\partial_c \ln \lambda_n(c) = 2|\psi_{n,c}(1)|^2/c$ for time–frequency limiting ([8, eq. (43)], numbered as in arXiv:1012.3881v3; attributed there to [81]), where the squared endpoint value plays the role of our A^2. Krein's formula $\partial_r \Gamma_r(s,t) = -[\Gamma_r(s,r)\Gamma_r(r,t) + \Gamma_r(s,-r)\Gamma_r(-r,t)]$ for the resolvent kernel of a truncated convolution operator [22, Lemma 2.3] has the same two-edge structure, and the finite-dimensional commutator identities [18, Lemma 5.2], [19, Lemma 5.1] produce a boundary evaluation (the Dirichlet-kernel term of [18, Thm. 1.1]); we mention both as motivation only. For the pure logarithm the corresponding statements follow from scaling and the trace identity [12, Cor. 2.3]; see also [9, Rem. 1.3, Cor. 1.5].

Kink law. That near-null eigenvectors have weak singularities at $\pm(L - \log n)$ was observed numerically by Eberwein [26, §13, p. 45] (his window $[0, L_{\mathrm{Eb}}]$ has $L_{\mathrm{Eb}} = 2L$). His list "every $\log n$" includes composite n; in our hierarchy such points, for example $L - \log 6$, are chain points of generation ≥ 2. Cut points at $L - \log n$ are built into the trial vectors of Mori [62, App. A] and into the prolate approximants of Connes and Consani [17, eq. (3.1)], [16, §6.4], and the shift structure of the operator's action is visible in [72, eq. (2.11)]. Loss of smoothness at shifted images of the boundary is classical for differential–difference equations [67, 66] and for delay equations [1]; for generalised eigenfunctions on an interval see [77], whose abstract gives necessary and sufficient conditions for smoothness and examples of interior loss of smoothness, and [78] (we have not read the full texts). That the half-line eigenfunction of [54] is not Hölder continuous at the edge is noted in [54, Ex. 6.4]. The kink law takes the edge traces as input; their existence is due to [12], and our error terms use the rate of Theorem 7.1(a). The activation formula [12, Thm. 2.4] concerns the eigenvalue at the window $2L = \log n$, where $y_n = -L$; that is the regime the kink law excludes (its constants degenerate as $2L$ decreases to $\log n$), so neither result implies the other. We found no statement of the profiles, the generation hierarchy, the interior density and nowhere-Hölder property, or the second-order terms (our searches included the rStation manuscripts on the Riemann hypothesis and forum post 21, where [12] was discussed, both screened on 31 October 2026). The locations are Eberwein's; what is added is amplitude, profile and hierarchy, as an Abelian computation on top of the edge law.

Tail law. For trial vectors with a jump at the edge, zero-side bounds of order $\log T/T$ are in [60, §"The zero sum"], and the corresponding archimedean (frequency-cutoff) tail budget of order $\log T/T$ for Galerkin vectors is in [44, Thm. 3.2, Cor. 3.3]. Joubert [49] observed numerically, on finite truncations with an edge jump, that the lowest eigenvalue is about the squared edge value times the zero-side leakage of the jump beyond the band, with a fitted constant of about 1.55 (the note assumes RH to identify the form with a Gram matrix of the zeros); this unrefereed, AI-assisted note is a numerical precursor of the link between squared edge values and the zero-side tail. For a true eigenfunction the logarithm in $|F(t)|^2$ cancels against the zero density (Section 1.3), which leads

to an exact $1/T$ law. The principle "eigenvalue derivative equals high-frequency tail" is classical for Tan's relations [73, 74, 11]; in the prolate case it follows elementarily by combining [8, eq. (43)] with the $1/t$ tail of the Fourier transform of a function with a jump, but we did not find it stated. On rStation, [68, Prop. 6.1] proves, under GRH, a far-field $1/\log$ limit for a different quantity (a Christoffel-type detection quantity for a planted quartet of zeros); we found no tail law over zeros there. To our knowledge the tail law with constant $|\lambda'|/\pi$ and a rate is new. The activation term $c_X A_+ A_- z/\log(1/z)$ of the contact law (Theorem 9.1) has the structure found in [41] and proved for eigenvalues in [12, Thm. 2.4]; related activation scales are in [76] (a $1/\log(1/h)$ scale at the entry of a prime-power shift) and [57] (an activation seam of rank one in the even and zero in the odd sector, for finite periodic sections).

The pure logarithmic case. For the logarithmic Laplacian on an interval, simplicity of every Dirichlet eigenvalue and the identity $A_\pm^2 = \|u\|^2/|I|$ were proved first in [12, Cor. 2.3]. The companion paper [9] gives another proof, by a one-endpoint dilation identity, together with the edge law with a rate. We do not repeat those results. [9] is cited only for comparison (after Theorem 5.4, in Remarks 5.10 and 7.2, and in Table 2) and as the template for Section 3 and Appendix A, whose arguments are given here in full; no proof in this paper depends on it.

What is new. To our knowledge the following are new: (i) for the whole family $H_L^{(s)}$ and every eigenfunction, a complete proof of the edge law with remainder $O(\ell^{-1})$, uniform on compact ranges of L, s and λ (a remainder of this order is stated for the true form in the mathia notes, with a sketched step), and its sharpening $O(\ell^{-3/2})$, with the same uniformity (Theorem 7.1(a)); (ii) the second-order edge law with offset D (Theorem 7.1(c), Proposition 6.2); (iii) the translation identity with a rate for eigenfunctions with different eigenvalues, the mixed identity (Theorem 5.4(c)) and the edge-free identity (Corollary 5.7); (iv) the kink law (Theorems 7.3 and 7.5, Corollaries 7.4 and 7.6); (v) the zero-side results (Theorems 8.3, 8.5, 8.8, 9.1, 9.2 and 9.3, Proposition 9.4), among them the first-order tail law; (vi) the nonvanishing statements of Proposition 5.8 (for $s = 1$ at $2L = \log 2$, $A_o > 0$ is in [38]). In addition we give a second proof of the existence of the traces, by a route close to that of [12], and of the Hadamard formula, by a different route, both with quantitative remainders. All of this is proved here, pending external review (Section 1.5).

Our literature searches (arXiv, Zenodo, GitHub, rStation and the web, to 31 October 2026; the web search engines we used do not index rStation) were limited; the absence of a result from them is not proof of novelty. Many items cited above are recent unrefereed deposits, some written with AI assistance and some (the rStation items and most of the mathia notes) by AI agents. They are cited for priority and context only: no proof in this paper depends logically on any of them, nor on the companion paper [9]. The one exception is the computer-assisted input [10] (which uses the reduction of [83]): the statements for the true form on $(0, 1.19]$ depend on it, as stated each time.

1.5. **Status and scope.** *This paper studies a quadratic form. None of its results proves or disproves the Riemann hypothesis. Where zeta zeros enter, no hypothesis on their location is made.* In the terms we use for status:

- Our proofs of the edge law for the localized Weil form (an order-zero nonlocal operator), with remainder $O(\ell^{-1})$, and of the Hadamard formula, with an explicit remainder, are complete, pending external review. The existence of the traces and the Hadamard formula itself were proved first in [12, §3.3, Thms. 2.1–2.2], which is unrefereed. The application of the Hadamard formula to the ground state of the true form ($s = 1$) on $(0, 1.19]$ uses the unrefereed computer-assisted input [10].
- The kink law, to two orders, and the first-order tail law for the localized Weil form are proved, pending external review. For the kink law, "two orders" means the profile to first order and the jump to second order; the second-order even part is Conjecture 7.8.
- The second-order per-sector tail term is proved in smoothed form; its sharp-cutoff form is open.

The laws are asymptotic only, and they are vacuous wherever $A_\pm = 0$. For the true form the even ground state has, in uncertified computations, $A_e \approx 0.062$ at $L = 0.4$ and $2.4 \cdot 10^{-4}$ at $L = 0.6$; at $L = 0.7$ its derivative $\lambda' \approx -3.5 \cdot 10^{-11}$ is already at the double-precision floor of our codes (caption of Table 2), and for larger L the trace is below the resolved numerical scale, so we give no value there. At accessible scales the laws therefore say nothing quantitative about that state. Whether $A_e(L) \neq 0$ at a given L is open (Conjecture 11.4). All checking so far has been done by AI referee passes and independent numerical codes; no human expert has checked the proofs (see the statement at the end of the paper).

Organisation. Section 2 fixes the form and its basic properties, and Section 3 the Wiener–Hopf factor and the half-line inversion. Section 4 proves the edge law, and Section 5 the Hadamard formula, the translation identity and the nonvanishing results. Section 6 introduces chains of shifts and develops the interior analysis, and Section 7 proves the second-order edge law and the kink law. Sections 8 and 9 treat the zero side. Section 10 gives a numerical illustration, Section 11 the conjectures and open problems, and Section 12 a statement of scope. The appendices contain the L^∞ bound and the longer proofs of Sections 8–9. Section 2.6 lists the notation.

2. Setting and normalisations

2.1. **Conventions.** Functions are real-valued unless stated otherwise. $F = \hat{f}$ is as above, so $\langle f, g\rangle = \int f\bar{g} = \frac{1}{2\pi}\int F\bar{G}$ and $\overline{F(t)} = F(-t)$ for real f. We write $\ell(r) = \log\frac{1}{r}$ for $0 < r < 1$, $\ell_t = \log t$ for $t > 1$, $\tau_h f = f(\cdot - h)$, $\tilde{f} = f(-\cdot)$, $H = \mathbf{1}_{(0,\infty)}$, and $P_\pm$ for multiplication by $\mathbf{1}_{(0,\infty)}$ and $\mathbf{1}_{(-\infty,0)}$. γ is Euler's constant; we keep this letter because it enters the offset D everywhere, and ordinates of zeros always carry the index ρ (γ_ρ, t_ρ), so the two uses do not meet. C denotes a constant that may change from line to line; its dependence is stated, and it never depends on the small variables r, d, z, η or on the large variable T.

2.2. **The archimedean symbol.** Let $a_k = 2k + \frac{1}{2}$ ($k \geq 0$) and

$$J(u) = \sum_{k\geq 0} e^{-a_k|u|} = \frac{e^{-|u|/2}}{1 - e^{-2|u|}}, \qquad \Psi_a(t) = \operatorname{Re}\psi_\Gamma\big(\tfrac{1}{4} + \tfrac{it}{2}\big) - \psi_\Gamma\big(\tfrac{1}{4}\big).$$

Put $c_a = \gamma + \frac{\pi}{2} + 2\log 2 = 3.5343\ldots$ and $c = \psi_\Gamma(\frac{1}{4}) - \log\pi = -5.3722\ldots$. Since $\psi_\Gamma(\frac{1}{4}) = -\gamma - \frac{\pi}{2} - 3\log 2$ [25, §5.4], we have $a = \Psi_a + c$ and $c + c_a = -\log 2\pi$. Also $a(t) = 2\vartheta'(t)$, where ϑ is the Riemann–Siegel theta function.

Lemma 2.1 (the symbol)**.** (a) $\Psi_a(t) = \varphi(t^2)$ *with* $\varphi(x) = \sum_{k\geq 0}\frac{2x}{a_k(a_k^2+x)} = \int_0^\infty(1 - e^{-xs})\sum_k 2a_k e^{-a_k^2 s}\,\mathrm{d}s$, *and* $\Psi_a(t) = \int_{\mathbb{R}}(1-\cos tu)J(u)\,\mathrm{d}u$.

(b) J *is even, positive and decreasing in* $|u|$; $0 \leq J(u) - \frac{1}{2|u|} \leq C$ *for* $0 < |u| \leq 1$ *and* $J(u) \leq Ce^{-|u|/2}$ *for* $|u| \geq 1$.

(c) $|\Psi_a(t) - \log t - c_a| \leq C/t^2$ *for* $t \geq 1$; *hence* $a(t) = \log\frac{t}{2\pi} + O(t^{-2})$.

(d) $0 \leq t\Psi_a'(t) \leq \min\big(2\Psi_a(t), 1 + 2.6/t\big)$ *for* $t > 0$.

Proof. (a) From $\psi_\Gamma(z) = -\gamma + \sum_{n\geq 0}(\frac{1}{n+1} - \frac{1}{n+z})$, $\operatorname{Re}\psi_\Gamma(\alpha+iy) - \psi_\Gamma(\alpha) = \sum_n \frac{y^2}{(n+\alpha)((n+\alpha)^2+y^2)}$; with $\alpha = \frac{1}{4}$, $y = \frac{t}{2}$ the n-th term is $\frac{2t^2}{a_n(a_n^2+t^2)}$. Also $\int_0^\infty(1-\cos tu)2e^{-a_k u}\,\mathrm{d}u = \frac{2t^2}{a_k(a_k^2+t^2)}$ and $\int_0^\infty(1-e^{-xs})2a_k e^{-a_k^2 s}\,\mathrm{d}s = \frac{2x}{a_k(a_k^2+x)}$. (b) The function $g(u) = 2ue^{-u/2} - 1 + e^{-2u}$ has $g(0) = 0$ and $g'(u) = e^{-u/2}(2 - u - 2e^{-3u/2}) > 0$ on $(0,1]$, so $J(u) \geq \frac{1}{2u}$ there; the expansion $J(u) = \frac{1}{2u} + \frac{1}{4} + O(u)$ gives the upper bound. (c) Use $\psi_\Gamma(z) = \log z - \frac{1}{2z} + O(|z|^{-2})$ [25, 5.11.2] at $z = \frac{1}{4} + \frac{it}{2}$. (d) $t\Psi_a'(t) = 2x\varphi'(x)$ with $x = t^2$. Concavity of φ and $\varphi(0) = 0$ give $x\varphi' \leq \varphi$. Also $x\varphi'(x) = \sum_k g_x(a_k)$

with $g_x(\alpha) = \frac{2\alpha x}{(\alpha^2+x)^2}$, which is unimodal in α with $\int_0^\infty g_x = 1$ and $\max g_x = \frac{9}{8\sqrt{3}} x^{-1/2}$; nodes spaced by 2 give $\sum_k g_x(a_k) \le \frac{1}{2} + 2\max g_x$. □

2.3. **The form.** Let $\mathcal{D}(\mathcal{E}) = \{f \in L^2(\mathbb{R}) : \int \log(2+|t|)|F(t)|^2\,\mathrm{d}t < \infty\}$ and, for $q > 0$,

$$\mathcal{E}(f,g) = \frac{1}{2\pi}\int_{\mathbb{R}} \Psi_a\, F\,\bar{G}\,\mathrm{d}t, \qquad \mathcal{E}_q(f,g) = \mathcal{E}(f,g) + q\langle f, g\rangle.$$

By Lemma 2.1(c), $1 + \Psi_a(t) \asymp \log(2+|t|)$, so $\mathcal{E}_q^{1/2}$ is a Hilbert norm on $\mathcal{D}(\mathcal{E})$. For a bounded open interval I let $\mathcal{D}_I = \{f \in \mathcal{D}(\mathcal{E}) : f = 0 \text{ a.e. off } \bar{I}\}$, $\mathcal{D}_L = \mathcal{D}_{(-L,L)}$, and $\mathcal{D}_+ = \{f \in \mathcal{D}(\mathcal{E}) : f = 0 \text{ a.e. on } (-\infty, 0)\}$; these are closed subspaces. For f with bounded support put $S_n f = f(\cdot + \log n)$, $S_n^* f = f(\cdot - \log n)$, $P_s f(x) = 2s\int \cosh\frac{x-y}{2} f(y)\,\mathrm{d}y$,

$$W_s f = c\, f + P_s f - \sum_{n\ge 2} c_n (S_n + S_n^*) f, \qquad w_s(f,g) = \int (W_s f)\, g.$$

Lemma 2.2 (one form for all windows)**.** *For $f \in \mathcal{D}_L$, $Q_L^{(s)}(f) = \mathcal{E}(f,f) + w_s(f,f)$. Both $\mathcal{E}$ and w_s are independent of L and invariant under translations and under $f \mapsto \tilde{f}$.*

Proof. Since $|F|^2$ is even and $a = \Psi_a + c$, $\frac{1}{\pi}\int_0^\infty a|F|^2 = \mathcal{E}(f,f) + c\|f\|^2$. With $g_f(y) = \int f(x) f(x+y)\,\mathrm{d}x$ we have $\frac{1}{\pi}\int_0^\infty \cos(t\log n)|F|^2\,\mathrm{d}t = g_f(\log n)$, which vanishes for $\log n \ge 2L$; so the cut $\log n < 2L$ in (1.1) is automatic, and $-2\sum c_n g_f(\log n) = -\sum c_n \langle (S_n + S_n^*) f, f\rangle$. Finally $2F(\frac{i}{2})F(-\frac{i}{2}) = 2\iint f(u) f(v)\cosh\frac{u-v}{2}\,\mathrm{d}u\,\mathrm{d}v$. Invariance: $\mathcal{E}$ is a Fourier multiplier, shifts commute with translations, and $F_{\tau_h f}(\pm\frac{i}{2}) = e^{\mp h/2} F_f(\pm\frac{i}{2})$. □

Lemma 2.3 (Dirichlet form)**.** (a) *For $f \in L^2$, $\frac{1}{2}\iint (f(x)-f(y))^2 J(x-y)\,\mathrm{d}x\,\mathrm{d}y = \mathcal{E}(f,f) \in [0,\infty]$; so $f \in \mathcal{D}(\mathcal{E})$ iff the left side is finite, and for $f, g \in \mathcal{D}(\mathcal{E})$ the polarised double integral converges absolutely and equals $\mathcal{E}(f,g)$.*

(b) *If $T : \mathbb{R} \to \mathbb{R}$ is 1-Lipschitz with $T(0) = 0$ and $f \in \mathcal{D}(\mathcal{E})$, then $T \circ f \in \mathcal{D}(\mathcal{E})$ and $\mathcal{E}(T\circ f) \le \mathcal{E}(f)$; if $f \in \mathcal{D}_I$ then $T \circ f \in \mathcal{D}_I$.*

(c) *For $f \in \mathcal{D}(\mathcal{E})$, $\mathcal{E}(f, f^+) \ge \mathcal{E}(f^+)$ and $\mathcal{E}(f, f^-) \le -\mathcal{E}(f^-)$.*

(d) *If $f, g \in \mathcal{D}(\mathcal{E})$, $fg = 0$ a.e. and $\iint |f(x)||g(y)|J(x-y) < \infty$, then $\mathcal{E}(f,g) = -\iint f(x) g(y) J(x-y)\,\mathrm{d}x\,\mathrm{d}y$.*

Proof. (a) Tonelli, Lemma 2.1(a) and $\|f(\cdot+u) - f\|^2 = \frac{1}{2\pi}\int 2(1-\cos ut)|F|^2$; then polarisation and Cauchy–Schwarz. (b) $|Tf(x) - Tf(y)| \le |f(x) - f(y)|$. (c) Since $f^+ f^- = 0$, $(f(x)-f(y))(f^+(x) - f^+(y)) = (f^+(x) - f^+(y))^2 + f^+(x) f^-(y) + f^+(y) f^-(x)$; integrate against $\frac{1}{2}J$. (d) A.e., $(f(x) - f(y))(g(x) - g(y)) = -f(x)g(y) - f(y)g(x)$. □

Lemma 2.4 (domains)**.** (a) *Dilations $f \mapsto f(\cdot/\theta)$, $\theta \in [\frac{1}{2}, 2]$, are uniformly bounded in the norm $\mathcal{E}_1^{1/2}$ and strongly continuous in θ.*

(b) *$C_c^\infty(-L, L)$ is dense in $\mathcal{D}_L$, and $C_c^\infty(0,\infty)$ is dense in $\mathcal{D}_+$, for $\mathcal{E}_1^{1/2}$.*

Proof. (a) φ is a Bernstein function (Lemma 2.1(a)), so $\varphi(\kappa x) \le \kappa\varphi(x)$ for $\kappa \ge 1$; with $\widehat{f(\cdot/\theta)}(t) = \theta F(\theta t)$ this gives $\mathcal{E}(f(\cdot/\theta)) \le 4\theta\,\mathcal{E}(f)$. Strong continuity holds on C_c^∞ by dominated convergence, hence everywhere. (b) For $f \in \mathcal{D}_L$, the dilates $f(\cdot/\theta)$, $\theta < 1$, vanish off $[-\theta L, \theta L]$ and tend to f as $\theta \uparrow 1$; mollifying with ρ_ε, $\varepsilon < (1-\theta)L$, gives functions in $C_c^\infty(-L,L)$, and $\mathcal{E}_1(\rho_\varepsilon * g - g) = \frac{1}{2\pi}\int (1+\Psi_a)|\hat{\rho}_\varepsilon - 1|^2 |\hat{g}|^2 \to 0$. For $f \in \mathcal{D}_+$: $\mathcal{E}_1(\tau_h f - f) = \frac{1}{2\pi}\int (1+\Psi_a)|e^{iht} - 1|^2 |F|^2 \to 0$ as $h \downarrow 0$, $\rho_\varepsilon * \tau_h f$ ($\varepsilon < h$) vanishes on $(-\infty, h-\varepsilon)$, and a cutoff $\chi(\cdot/R)$ converges in H^1, while $1 + \Psi_a \le C(1+t^2)$. □

Lemma 2.5 (truncation)**.** *Let $g \in \mathcal{D}(\mathcal{E}) \cap L^\infty$ and $I = (\alpha, \beta)$ bounded. Then $\mathbf{1}_I g \in \mathcal{D}(\mathcal{E})$, and $\mathcal{E}(\mathbf{1}_I g) = \frac{1}{2}\iint_{I^2}(g(x)-g(y))^2 J + \int_I g^2 \kappa_I$ with $\kappa_I(x) = \int_{\mathbb{R}\setminus I} J(x-y)\,\mathrm{d}y \le C + \frac{1}{2}\log_+\frac{1}{x-\alpha} + \frac{1}{2}\log_+\frac{1}{\beta - x}$.*

Proof. Split $\mathbb{R}^2 = (I \cup I^c)^2$ in Lemma 2.3(a); κ_I is integrable on I by Lemma 2.1(b). □

Lemma 2.6 (multipliers)**.** *Let $\eta \in C^\infty(\mathbb{R})$ with η and η' bounded, and let $k_\eta(u) = J(u)\min(\|\eta'\|_\infty|u|,\, 2\|\eta\|_\infty)$, so that $k_\eta \in L^1$.*

(a) *If $\phi \in \mathcal{D}(\mathcal{E})$, then $\eta\phi \in \mathcal{D}(\mathcal{E})$ and $\mathcal{E}(\eta\phi) \le 2\|\eta\|_\infty^2\,\mathcal{E}(\phi) + 2\|\eta\|_\infty\|k_\eta\|_1\|\phi\|^2$.*

(b) *For $g, \phi \in \mathcal{D}(\mathcal{E})$, $\mathcal{E}(\eta g, \phi) = \mathcal{E}(g, \eta\phi) + \langle r_\eta[g], \phi\rangle$, where $r_\eta[g](x) = \int J(x-y)(\eta(x) - \eta(y))g(y)\,\mathrm{d}y$ satisfies $|r_\eta[g]| \le k_\eta * |g|$.*

Proof. (a) $(\eta\phi)(x) - (\eta\phi)(y) = \eta(x)(\phi(x)-\phi(y)) + \phi(y)(\eta(x)-\eta(y))$, and $(\eta(x)-\eta(y))^2 J(x-y) \le 2\|\eta\|_\infty k_\eta(x-y)$. (b) Pointwise, $(\eta g(x) - \eta g(y))(\phi(x)-\phi(y)) - (g(x)-g(y))(\eta\phi(x) - \eta\phi(y)) = (\eta(x)-\eta(y))(\phi(x)g(y) - \phi(y)g(x))$, which is dominated by $k_\eta(x-y)(|\phi(x)g(y)| + |\phi(y)g(x)|)$; integrate against $\frac{1}{2}J$ and use symmetry. □

Lemma 2.7 (the bounded part, windowed)**.** *Let f vanish outside an interval I. Then for $1 \le p \le \infty$*

$$\|\mathbf{1}_I W_s f\|_p \le M(|I|)\|f\|_p, \qquad M(|I|) := |c| + 2\sum_{\log n < |I|} c_n + 2|s|\,|I|\cosh\tfrac{|I|}{2}.$$

Proof. A shift by $\log n \ge |I|$ moves f off I, every shift has norm 1, and on $I \times I$ the pole kernel is at most $2|s|\cosh\frac{|I|}{2}$. □

W_s is *not* bounded on $L^2(\mathbb{R})$: the pole kernel is unbounded and $\sum_n c_n = \infty$. Only windowed bounds are used below. Since W_s commutes with translations, for $L \le L_{\max}$ and $|s| \le S_0$ the constant $M := M(2L_{\max} + 2)$ bounds W_s for every function and window used below.

2.4. **The operator.** By Lemma 2.2, $Q_L^{(s)} = \mathcal{E} + w_s$ on $\mathcal{D}_L$. The form $\mathcal{E}$ is closed and nonnegative on $\mathcal{D}_L$, and $|w_s(f)| \le M\|f\|^2$ for $f \in \mathcal{D}_L$ (Lemma 2.7), so $\mathcal{E} + w_s$ is closed and bounded below by $-M$ on $\mathcal{D}_L$, and $\mathcal{D}_L$ is the form closure of $C_c^\infty(-L, L)$ (Lemma 2.4(b)). Let $H_L^{(s)}$ be its self-adjoint operator on $L^2(-L, L)$ [50, Thm. VI.2.1]. It has compact resolvent: the unit ball of $\mathcal{D}_L$ for $\mathcal{E}_1^{1/2}$ is bounded in L^2, supported in $[-L, L]$, and equicontinuous under translation, since $\|\tau_h f - f\|^2 \le h^2T^2\|f\|^2 + \frac{C}{\log T}\mathcal{E}_1(f)$ for $T \ge 2$, so the Kolmogorov–Riesz theorem applies. It commutes with $f \mapsto \tilde{f}$ (Lemma 2.2).

An *eigenfunction* of $H_L^{(s)}$ with eigenvalue λ is a real $\psi \in \mathcal{D}_L$ with $\|\psi\| = 1$ and $(\mathcal{E} + w_s)(\psi, \phi) = \lambda\langle\psi, \phi\rangle$ for all $\phi \in \mathcal{D}_L$. The even and odd subspaces are invariant; $\lambda_k^\sigma(L)$, $k \ge 0$, denotes the k-th min-max value in the sector $\sigma \in \{e, o\}$, and an eigenvalue is *sector-simple* if its eigenspace within the sector is one-dimensional. The *edge traces* are

$$A_\pm(\psi) := \lim_{r \downarrow 0} \psi\big(\pm(L - r)\big)\,\ell(r)^{1/2}$$

(they exist by Theorem 4.1). We write $A := |A_+| = |A_-|$ (Theorem 5.4(a)) and $\varepsilon := \operatorname{sgn}(A_+A_-)$, which is $+1$ for even and -1 for odd eigenfunctions with $A \ne 0$.

2.5. **Arithmetic notation.** Let $N_L = \{n \ge 2 : \Lambda(n) \ne 0,\ \log n < 2L\}$ and $y_n = L - \log n \in (-L, L)$ for $n \in N_L$. Let $X = e^{2L}$, $c_X = \Lambda(X)/\sqrt{X}$ if X is a prime power and $c_X = 0$ otherwise, and $W_s^{(X)}$ the operator W_s without the shift by $\log X$. Let $\delta_X = \min(\{|2L - \log n| : n \text{ a prime power}, n \ne X\} \cup \{\log 2\}) > 0$ and $\delta_X^- = \min(\{2L - \log n : n \in N_L\} \cup \{\log 2\})$.

Numerical values: $c_2 = 0.49013$, $c_3 = 0.63428$, $c_4 = 0.34657$, $c_5 = 0.71976$, $c_7 = 0.73549$, $c_8 = 0.24506$, $c_9 = 0.36620$; $\log 2\pi + \gamma = 2.41509$.

2.6. **Notation.** The table lists the symbols that are used in more than one section. Symbols used only inside one proof are defined there; a few letters (β, κ, M) also carry local meanings, which are defined where they occur.

symbol	meaning	where
$F = \hat{f}$	$F(w) = \int f(x)e^{ixw}\,\mathrm{d}x$; $\|f\|^2 = \frac{1}{\pi}\int_0^\infty \|F\|^2$ for real f	§1
$\ell(r)$, ℓ_t	$\log\frac{1}{r}$ for $0 < r < 1$; $\log t$ for $t > 1$	§2.1
τ_h, $\tilde{f}$, $P_\pm$, H	$f(\cdot - h)$; $f(-\cdot)$; restriction to $(0,\infty)$, $(-\infty,0)$; $\mathbf{1}_{(0,\infty)}$	§2.1
γ; γ_ρ, t_ρ, y_ρ	Euler's constant; $\rho = \frac{1}{2} + i\gamma_\rho$, $t_\rho = \operatorname{Re}\gamma_\rho = \operatorname{Im}\rho$, $y_\rho = \operatorname{Im}\gamma_\rho$	§§2.1, 8.1
$a(t)$, Ψ_a, φ	archimedean symbol $\operatorname{Re}\psi_\Gamma(\frac{1}{4} + \frac{it}{2}) - \log\pi$; $\Psi_a = a - c$; $\Psi_a(t) = \varphi(t^2)$, φ a Bernstein function	§§1, 2.2
J	Lévy density $e^{-\|u\|/2}/(1 - e^{-2\|u\|})$ of $\mathcal{E}$	§2.2
c, c_a, c_n	$\psi_\Gamma(\frac{1}{4}) - \log\pi$; $\gamma + \frac{\pi}{2} + 2\log 2$; $\Lambda(n)/\sqrt{n}$	§§1, 2.2
$\mathcal{E}$, $\mathcal{E}_q$	Dirichlet form with symbol Ψ_a; $\mathcal{E} + q\langle\cdot,\cdot\rangle$	§2.3
$\mathcal{D}(\mathcal{E})$, $\mathcal{D}_L$, $\mathcal{D}_I$, $\mathcal{D}_+$	form domain; its elements vanishing off $[-L,L]$, off $\bar{I}$, on $(-\infty,0)$	§2.3
W_s, w_s, P_s, S_n	bounded part (constant, pole with weight s, prime shifts); its form; pole operator; shift by $\log n$	§2.3
M	bound for W_s on windows (Lemma 2.7)	§2.3
$H_L^{(s)}$, λ, ψ	operator of $Q_L^{(s)}$; eigenvalue; real normalised eigenfunction	§2.4
$\lambda_k^\sigma(L)$	k-th eigenvalue in the sector $\sigma \in \{e, o\}$ (even, odd)	§2.4
$A_\pm$, A, ε	edge traces; $A = \|A_+\| = \|A_-\|$; parity sign $\operatorname{sgn}(A_+A_-)$	§2.4
N_L, y_n	prime powers n with $\log n < 2L$; $y_n = L - \log n$	§2.5
X, c_X, $W_s^{(X)}$	e^{2L}; $\Lambda(X)/\sqrt{X}$ (0 unless X is a prime power); W_s without the shift by $\log X$	§2.5
δ_X, δ_X^-	distance from $2L$ to the other $\log n$; to the active $\log n$	§2.5
K, v, V	Wiener–Hopf factor K_q; $1/K$ is the Laplace transform of $v = v_q$; $V(x) = \int_0^x v$	§3
G_q, k_f, k_+	half-line inverse $v * (P_+k_f)$; $k_f = \tilde{v} * f$; $k_+ = k_{f_+}$ (Theorem 4.1(iv))	§§3, 4
r_q, R	$v * \tilde{v}$, the kernel of $(q + \Psi_a)^{-1}$; $R(x) = \int_0^x r_q$	§6
μ_q, a_q, D	$q + \lambda - c$; $q + \frac{\pi}{2} + 2\log 2$; $\lambda + \log 2\pi + \gamma = \mu_q - a_q$ (Theorem 7.1)	§§1, 6
$g(z)$, C_L	generation (least length of a chain of shifts from an edge to z); $\{g < \infty\}$	§6.1
$\mathsf{W}_\pm$, Ω_k, Σ_k	chain weights times $A_\pm$; their difference and sum	§6.1
$\mathfrak{D}_f$, $\mathfrak{S}_f$	odd part (jump) and even part of f at a point	§6.1
$\omega_p(t)$	local modulus of continuity of ψ at p	Prop. 6.10
$h(w)$, P_ψ	$F(w)F(-w)$; $F(\frac{i}{2})F(-\frac{i}{2})$	§8.1
$Z(T)$, $Z_{\cos}(z)$, $Z_\nu(T)$	$\sum_{\|t_\rho\|>T} h(\gamma_\rho)$; $\sum_\rho h(\gamma_\rho)(1 - \cos z\gamma_\rho)$; smoothed tail	§8.1
ν, $\hat{\nu}$, $m_1(\nu)$	even signed measure on $[-1,1]$ of mass 1; its cosine transform; $\int_{(0,\infty)} z\,\mathrm{d}\nu$	§8.1
$\mathcal{L}(z)$, $\mathcal{G}(\eta)$	contact integral at the edges; exterior field $(J\psi - W_s\psi)(L + \eta)$	Theorem 8.3
Φ_c, M_c, R_c, B	canonical edge profile; main term and remainder of F; constant of Theorem 8.5	§8.3
$\Xi(h)$	$\langle\psi_o, \psi_e(\cdot + h)\rangle$	Theorem 5.4
$m(\eta)$, R_{f}	Wiener–Hopf profile $1 + M_K(\eta)$ of the exterior field; its remainder	Lemma 5.2
$\Upsilon(T)$	normalised tail $\pi TZ(T)/\|\lambda'\|$	Proposition 9.4

3. The Wiener–Hopf factor and half-line inversion

This section follows [54, Sect. 2–3] and the companion paper [9, §§3–4], with the variance-gamma symbol replaced by the archimedean symbol Ψ_a (our J and Ψ_a are not the J and Ψ of [9]). A function f on $(0,\infty)$ is a *complete Bernstein function* (CBF), respectively a *Stieltjes function*, if

$$f(\xi) = c_1 + c_2\xi + \frac{1}{\pi}\int_{(0,\infty)} \frac{\xi}{\xi + s}\,\frac{m(\,\mathrm{d}s)}{s}, \qquad \text{respectively} \qquad f(\xi) = c_1 + \frac{c_2}{\xi} + \frac{1}{\pi}\int_{(0,\infty)} \frac{m(\,\mathrm{d}s)}{\xi + s},$$

with constants $c_1, c_2 \geq 0$ and a measure m on $(0,\infty)$ such that $\int \min(s^{-1}, s^{-2})\, m(\mathrm{d}s)$, respectively $\int \min(1, s^{-1})\, m(\mathrm{d}s)$, is finite; see [54, Cor. 2.16(b)]. A CBF is a Bernstein function [54, (2.20)], and $f \not\equiv 0$ is a CBF iff $1/f$ is a Stieltjes function [54, Prop. 2.17(a)].

Fix $q > 0$ and put $\sigma_q(t) = q + \Psi_a(t) = q + \varphi(t^2)$. By Lemma 2.1(a), $\varphi(\xi) = \frac{1}{\pi}\int \frac{\xi}{\xi+s}\frac{m_0(\mathrm{d}s)}{s}$ with $m_0 = \sum_k 2\pi a_k \delta_{a_k^2}$, and $\int \min(s^{-1}, s^{-2}) m_0(\mathrm{d}s) < \infty$; so φ and $q+\varphi$ are CBFs. Following [54, Def. 3.1, Prop. 3.2] let

$$K(\xi) = K_q(\xi) = \exp\Big(\frac{1}{\pi}\int_0^\infty \frac{\log \sigma_q(\xi\tau)}{1+\tau^2}\,\mathrm{d}\tau\Big) = \exp\Big(\frac{1}{\pi}\int_0^\infty \frac{\xi \log \sigma_q(\zeta)}{\xi^2+\zeta^2}\,\mathrm{d}\zeta\Big), \tag{3.1}$$

the first formula for $\xi > 0$, the second defining the holomorphic extension to $\operatorname{Re}\xi > 0$; the integrals converge since $\log q \leq \log \sigma_q(\zeta) \leq \log(q + C + \log(1+|\zeta|))$. In the notation of [54], $K = (q+\varphi)^\dagger$.

Lemma 3.1 (the factor)**.** (K1) *K is a CBF* [54, Lemma 3.8]*; in particular $K(z) = a_K + b_K z + \int_{(0,\infty)}(1 - e^{-zx})\Pi(\mathrm{d}x)$ with $a_K, b_K \geq 0$, $\int \min(1,x)\Pi(\mathrm{d}x) < \infty$.*

(K2) *K extends continuously to $\{\operatorname{Re} z \geq 0\}$, with $K(\bar z) = \overline{K(z)}$ and $|K(z)| \leq C_K(1+|z|)$ there.*

(K3) *$|K(it)|^2 = K(it)K(-it) = \sigma_q(t) \geq q$ for $t \in \mathbb{R}$, $K(0) = \sqrt{q}$, and $K(\xi) \to \infty$ as $\xi \to +\infty$. K has no zeros in $\{\operatorname{Re} z \geq 0\}$.*

(K4) *$1/K(\xi) = \int_0^\infty e^{-\xi x} v(x)\,\mathrm{d}x$ for $\xi > 0$, where $v = v_q$ is completely monotone, strictly positive and strictly decreasing, and $\int_0^\infty v = q^{-1/2}$.*

Proof. (K2) For $\operatorname{Re} z \geq 0$, $|1 - e^{-zx}| \leq \max(2,|z|)\min(1,x)$, so the representation in (K1) converges locally uniformly on $\{\operatorname{Re} z \geq 0\}$. (K3) For $\xi = \varepsilon + it$, $\varepsilon > 0$, the second formula in (3.1) and evenness of σ_q give $\log|K(\varepsilon+it)| = \frac{1}{2}(P_\varepsilon * \log\sigma_q)(t)$, P_ε the Poisson kernel of the half-plane. As $\log\sigma_q$ is continuous with $\int |\log\sigma_q(\zeta)|\frac{\mathrm{d}\zeta}{1+\zeta^2} < \infty$, letting $\varepsilon \downarrow 0$ and using (K2) gives $|K(it)|^2 = \sigma_q(t)$, and $K(-it) = \overline{K(it)}$. Dominated convergence in (3.1) as $\xi \downarrow 0$ gives $K(0) = \sqrt{q}$, and $K(\xi) \to \infty$ follows from Lemma 3.2(a). Inside the half-plane $K = \exp(\cdots) \neq 0$. (K4) By [54, Prop. 2.17(a)], $1/K(\xi) = \tilde c_1 + \tilde c_2/\xi + \frac{1}{\pi}\int \frac{\tilde m(\mathrm{d}s)}{\xi+s}$; by [54, Prop. 2.18], $\tilde c_1 = \lim_{\xi\to\infty} 1/K(\xi) = 0$ and $\tilde c_2 = \lim_{\xi\downarrow 0}\xi/K(\xi) = 0$. Writing $\frac{1}{\xi+s} = \int_0^\infty e^{-(\xi+s)x}\,\mathrm{d}x$ gives (K4) with $v(x) = \frac{1}{\pi}\int e^{-sx}\tilde m(\mathrm{d}s)$, and $\int_0^\infty v = \lim_{\xi\downarrow 0} 1/K(\xi) = q^{-1/2}$. □

The assumption $q > 0$ is essential: for $q = 0$ one would have $\int v = \infty$.

Lemma 3.2. *There is C, depending only on q, such that for $\xi \geq e$:*

(a) *$|\log K(\xi) - \frac{1}{2}\log\sigma_q(\xi)| \leq C/\log\xi$;*

(b) *$\beta(\xi) := \xi K'(\xi)/K(\xi)$ satisfies $0 \leq \beta \leq 1$ on $(0,\infty)$ and $\beta(\xi) \leq C/\log\xi$.*

Proof. (a) Since $\frac{1}{\pi}\int_0^\infty \frac{\mathrm{d}\tau}{1+\tau^2} = \frac{1}{2}$, the difference is $\frac{1}{\pi}\int_0^\infty \log\frac{\sigma_q(\xi\tau)}{\sigma_q(\xi)}\frac{\mathrm{d}\tau}{1+\tau^2}$. As φ is Bernstein, $\varphi(\kappa x) \leq \kappa\varphi(x)$ for $\kappa \geq 1$, so $|\log\frac{\sigma_q(\xi\tau)}{\sigma_q(\xi)}| \leq 2|\log\tau|$ for all τ. For $\tau \geq \xi^{-1/2}$, Lemma 2.1(c) gives $|\sigma_q(\xi\tau) - \sigma_q(\xi)| \leq |\log\tau| + C/\xi$ and $\min(\sigma_q(\xi\tau), \sigma_q(\xi)) \geq \frac{1}{2}\log\xi - C$. Since $\int_0^\infty |\log\tau|\frac{\mathrm{d}\tau}{1+\tau^2} < \infty$ and $\int_0^{\xi^{-1/2}} |\log\tau|\,\mathrm{d}\tau = O(\xi^{-1/2}\log\xi)$, the claim follows for large ξ, and for bounded $\xi \geq e$ from the bound $2|\log\tau|$. (b) K is Bernstein, so $0 \leq \xi K' \leq K$. Differentiating (3.1), $\beta(\xi) = \frac{1}{\pi}\int_0^\infty b_q(\xi\tau)\frac{\mathrm{d}\tau}{1+\tau^2}$ with $b_q(t) = t\Psi_a'(t)/\sigma_q(t) \in [0,2]$ (Lemma 2.1(d)) and $b_q(t) \leq 3.6/\sigma_q(t)$ for $t \geq 1$; split at $\tau = \xi^{-1/2}$. □

Lemma 3.3 (renewal function)**.** *Let $V(x) = V_q(x) = \int_0^x v$. Then V is increasing and concave, $V(0+) = 0$, $V \leq q^{-1/2}$, and:*

(a) $\dfrac{1}{K(1/x)} \leq V(x) \leq \dfrac{1+2\beta(1/x)}{K(1/x)}$ *for $x > 0$;*

(b) *$|V(x)\ell(x)^{1/2} - 1| \leq C_V/\ell(x)$ for $0 < x \leq e^{-1}$, with C_V depending only on q.*

Proof. (a) With the measure $\tilde m$ of (K4), $V(x)-\frac{1}{K(1/x)}=\frac{1}{\pi}\int\frac{\tilde m(\mathrm{d}s)}{s}\big(\frac{1}{1+sx}-e^{-sx}\big)$. The bracket lies in $[0,\frac{2sx}{(1+sx)^2}]$: for $u=sx\le 1$, $\frac{1}{1+u}-e^{-u}\le\frac{u^2}{2}\le\frac{2u}{(1+u)^2}$, because $g(u)=\frac{u^2}{2}-\frac{1}{1+u}+e^{-u}$ has $g(0)=g'(0)=g''(0)=0$ and $g'''(u)=6(1+u)^{-4}-e^{-u}>0$ on $[0,1]$; for $u\ge 1$, $\frac{1}{1+u}-e^{-u}\le\frac{1}{1+u}\le\frac{2u}{(1+u)^2}$. Moreover $\frac{2}{\pi}\int\frac{x\,\tilde m(\mathrm{d}s)}{(1+sx)^2}=\frac{2\beta(1/x)}{K(1/x)}$. (b) With $\xi=1/x$, $\sigma_q(\xi)=\ell+q+c_a+O(x^2)$ by Lemma 2.1(c); insert Lemma 3.2 into (a). □

Extend v by 0 on $(-\infty,0)$ and put $\tilde v(x)=v(-x)$. For real $f\in L^2(0,\infty)$, extended by zero, let

$$k_f:=\tilde v*f,\qquad Gf=G_qf:=v*(P_+k_f).$$

Lemma 3.4 (half-line inversion)**.** *For every real $f\in L^2(0,\infty)$: (i) $Gf\in\mathcal{D}_+$ and $\mathcal{E}_q(Gf,Gf)\le q^{-1}\|f\|^2$; (ii) $\mathcal{E}_q(Gf,\phi)=\langle f,\phi\rangle$ for every $\phi\in\mathcal{D}_+$; (iii) Gf is the only $u\in\mathcal{D}_+$ with this property.*

This is an analytic form of the classical description of the Green function of a Lévy process killed on leaving a half-line [3, Thm. VI.20].

Proof. Step 1. Since $v\in L^1$, $\xi\mapsto\int_0^\infty e^{-\xi x}v(x)\,\mathrm{d}x$ is continuous on $\{\operatorname{Re}\xi\ge 0\}$ and holomorphic inside; by (K2)–(K3) so is $1/K$, and they agree on $(0,\infty)$. Hence $\hat v(t)=1/K(-it)$ and $\hat{\tilde v}(t)=1/K(it)$.

Step 2. By Young's inequality $\|k_f\|\le q^{-1/2}\|f\|$, and $\hat k_f=\hat f/K(it)$. Then $u:=Gf$ vanishes on $(-\infty,0)$, $\hat u=\widehat{P_+k_f}/K(-it)$, and multiplying by $\sigma_q=K(it)K(-it)$,

$$\sigma_q\hat u=K(it)\widehat{P_+k_f}=\hat f-K(it)\widehat{P_-k_f}. \tag{3.2}$$

Step 3. $\sigma_q|\hat u|^2=|\widehat{P_+k_f}|^2$, so $\mathcal{E}_q(u,u)=\|P_+k_f\|^2\le q^{-1}\|f\|^2$ and $u\in\mathcal{D}_+$.

Step 4: the defect vanishes. Let $\phi\in C_c^\infty(0,\infty)$ have support in $[x_0,x_1]$, $x_0>0$. By (3.2), $\mathcal{E}_q(u,\phi)=\langle f,\phi\rangle-\Delta(\phi)$ with $\Delta(\phi)=\frac{1}{2\pi}\int K(it)\widehat{P_-k_f}(t)\hat\phi(-t)\,\mathrm{d}t$, all integrals absolutely convergent. Put $g(t)=K(it)\hat\phi(-t)$ for $\operatorname{Im}t\le 0$. For $\operatorname{Im}t<0$, $\operatorname{Re}(it)>0$, so g is holomorphic on $\{\operatorname{Im}t<0\}$ and continuous on its closure. For $t=s-iy$, $y\ge 0$, $|\hat\phi(-t)|\le\|\phi\|_1e^{-x_0y}$ and $|t|^2|\hat\phi(-t)|\le\|\phi''\|_1e^{-x_0y}$, so by (K2) $|g(t)|\le C_\phi(1+|t|)^{-1}$ and $\sup_{y>0}\int|g(s-iy)|^2\,\mathrm{d}s<\infty$. The Paley–Wiener theorem [64, Thm. 19.2], applied to $z\mapsto g(-z)$ on the upper half-plane, gives $\omega_0\in L^2(0,\infty)$ with $g(s-iy)=\int_0^\infty\omega_0(x)e^{-xy}e^{-ixs}\,\mathrm{d}x$; letting $y\downarrow 0$ in $L^2(\mathrm{d}s)$, $g=\hat\omega$ a.e., where $\omega(x):=\omega_0(-x)$ is supported in $(-\infty,0]$. By Plancherel, $\Delta(\phi)=\int(P_-k_f)(x)\,\omega(-x)\,\mathrm{d}x=0$, since P_-k_f vanishes on $(0,\infty)$ and $\omega(-\cdot)$ on $(-\infty,0)$.

Step 5. Both sides of (ii) are continuous in ϕ for $\mathcal{E}_q^{1/2}$, since $|\langle f,\phi\rangle|\le q^{-1/2}\|f\|\mathcal{E}_q(\phi)^{1/2}$; Lemma 2.4(b) gives (ii). If $w\in\mathcal{D}_+$ has $\mathcal{E}_q(w,\phi)=0$ for all $\phi\in\mathcal{D}_+$, then $q\|w\|^2\le\mathcal{E}_q(w,w)=0$. □

Lemma 3.5 (edge lemma)**.** *Let $d>0$, let $f\in L^2(0,\infty)$ be real and bounded on $(0,d)$, $u=Gf$, $k_f(z)=\int_0^\infty v(t)f(t+z)\,\mathrm{d}t$ and $I_f=k_f(0)=\int_0^\infty vf$. Then $u(x)=\int_0^x v(x-z)k_f(z)\,\mathrm{d}z$ defines a continuous representative of u on $(0,d)$ with $u(0+)=0$, and for $0<x\le d/2$*

$$|u(x)-I_fV(x)|\le V(x)\big[2\|f\|_{L^\infty(0,d)}V(x)+v(d/2)\|f\|_2\sqrt{x}\big]. \tag{3.3}$$

If $f\in L^\infty$, then $|u-I_fV|\le 2\|f\|_\infty V^2$ on $(0,\infty)$. Since $\sqrt{x}\,\ell(x)$ is bounded on $(0,e^{-1}]$, $\sqrt{x}\le CV(x)^2$ there.

Proof. For $0\le z\le d'<d$, $|k_f(z)|\le q^{-1/2}\|f\|_{L^\infty(0,d)}+(q^{-1/2}v(d-d'))^{1/2}\|f\|_2$; so on $(0,d')$, $u=v*(\mathbf{1}_{(0,d')}k_f)$ is continuous with $|u(x)|\le CV(x)\to 0$. Since v is decreasing, $\int_0^\infty|v(y-z)\mathbf{1}_{y>z}-v(y)|\,\mathrm{d}y=2V(z)$, so the part $y<d$ of $k_f(z)-I_f=\int(v(y-z)\mathbf{1}_{y>z}-v(y))f(y)\,\mathrm{d}y$ is at most $2\|f\|_{L^\infty(0,d)}V(z)$. For $y\ge d\ge 2z$, $0\le v(y-z)-v(y)\le v(d/2)$ and $\int_d^\infty(v(y-z)-v(y))\,\mathrm{d}y\le zv(d/2)$, so by Cauchy–Schwarz the part $y\ge d$ is at most $v(d/2)\|f\|_2\sqrt{z}$. Finally $u(x)-I_fV(x)=\int_0^x v(x-z)(k_f(z)-I_f)\,\mathrm{d}z$, and both bounds increase in z. □

Lemma 3.6 (boundedness)**.** *Let $L \le L_{\max}$, $|s| \le S_0$, and let ψ be an eigenfunction of $H_L^{(s)}$ with eigenvalue $|\lambda| \le \Lambda_0$. Then $\|\psi\|_\infty \le C_\infty$, where C_∞ depends only on $L_{\max}$, S_0 and Λ_0.*

The proof (Appendix A) adapts [9, App. A]: domination of the Dirichlet semigroup by the free one, which is a mixture of Gaussians by subordination, a Duhamel formula for the bounded part, and a bootstrap in L^p.

4. The edge law

Fix $0 < L_{\min} \le L_{\max}$, $\Lambda_0 > 0$ and $S_0 > 0$, and let $\rho_0 = \min(L_{\min}/4, e^{-1})$. Fix an even $\chi \in C_c^\infty(\mathbb{R})$ with $0 \le \chi \le 1$, $\chi = 1$ on $[-\rho_0, \rho_0]$ and $\operatorname{supp}\chi \subset [-2\rho_0, 2\rho_0]$. In this section "constant" means a number depending only on $L_{\min}, L_{\max}, \Lambda_0, S_0$, on q where it appears, and on χ. For an eigenfunction ψ on $(-L, L)$ put $\tilde\psi(r) = \psi(L - r)$; by Lemma 2.2, $\tilde\psi$ is an eigenfunction for the window $(0, 2L)$ with the same eigenvalue.

Theorem 4.1 (edge law)**.** *There is a constant C such that for every $L \in [L_{\min}, L_{\max}]$, every $s \in [-S_0, S_0]$ and every eigenfunction ψ of $H_L^{(s)}$ with eigenvalue $|\lambda| \le \Lambda_0$ the following hold. Simplicity, parity and the arithmetic of e^{2L} play no role.*

(i) *$\|\psi\|_\infty \le C$, and ψ has a representative that is continuous on $\mathbb{R}$ and vanishes outside $(-L, L)$.*

(ii) *The edge traces $A_\pm$ exist, $|A_\pm| \le C$, and for $0 < r \le \rho_0$, with $V = V_1$,*

$$\big|\psi(\pm(L - r)) - A_\pm V(r)\big| \le CV(r)^2, \qquad \big|\psi(\pm(L - r)) - A_\pm \ell(r)^{-1/2}\big| \le C\ell(r)^{-1}.$$

(iii) (Integral formula.) *Let $m_{\mathrm{ext}}(y) = \int_0^{2L} J(y - r)\tilde\psi(r)\,\mathrm{d}r$ for $y > 2L$. For every $q > 0$,*

$$A_+ = \int_0^\infty v_q(r) f_q(r)\,\mathrm{d}r, \qquad f_q := \mathbf{1}_{(0,2L)}\big[(q + \lambda)\tilde\psi - W_s\tilde\psi\big] - \mathbf{1}_{(2L,\infty)} m_{\mathrm{ext}} \ \in L^2(0, \infty).$$

(iv) (Localised representation.) *For $q > 0$, $u := \chi\tilde\psi \in \mathcal{D}_+$ satisfies $u = G_q f_+$ with*

$$f_+ = \mathbf{1}_{(0,\infty)}\big[\chi\big((q + \lambda)\tilde\psi - W_s\tilde\psi\big) + r_\chi[\tilde\psi]\big] \in L^2 \cap L^\infty(0, \infty), \qquad \|f_+\|_\infty \le C_q,$$

and $A_+ = \int_0^\infty v_q f_+$; moreover $u(x) = \int_0^x v_q(x - z)k_+(z)\,\mathrm{d}z$ with $k_+ := k_{f_+}$, that is, $k_+(z) = \int_0^\infty v_q(t) f_+(t + z)\,\mathrm{d}t$, $k_+(0) = A_+$.

The same holds at $-L$, applied to $\psi(-\cdot)$.

Proof. (i), first part: Lemma 3.6. For the rest let $\phi \in C_c^\infty(0, \infty)$, $\phi_1 = \mathbf{1}_{(0,2L)}\phi$ and $\phi_2 = \mathbf{1}_{(2L,\infty)}\phi$. By Lemma 2.5, $\phi_1 \in \mathcal{D}_{(0,2L)}$, so the eigenvalue equation gives $\mathcal{E}(\tilde\psi, \phi_1) = \langle\lambda\tilde\psi - W_s\tilde\psi, \phi_1\rangle$. By Lemma 2.3(d), $\mathcal{E}(\tilde\psi, \phi_2) = -\langle m_{\mathrm{ext}}, \phi_2\rangle$; absolute convergence holds because $\tilde\psi$ and ϕ_2 are bounded with supports meeting only at $2L$ and $\iint_{[0,1]^2} \frac{\mathrm{d}s\,\mathrm{d}t}{s+t} < \infty$. Adding, $\mathcal{E}_q(\tilde\psi, \phi) = \langle f_q, \phi\rangle$. On $(0, 2L)$, $|f_q| \le (q + \Lambda_0 + M)C_\infty$ (Lemma 2.7); for $y > 2L$, $|m_{\mathrm{ext}}(y)| \le C_\infty \int_{y-2L}^\infty J$, which is $O(\log \frac{1}{y-2L})$ near $2L$ and $O(e^{-(y-2L)/2})$ at infinity. So $\|f_q\|_2 \le C$, $\phi \mapsto \langle f_q, \phi\rangle$ is $\mathcal{E}_q^{1/2}$-continuous, Lemma 2.4(b) extends the identity to $\phi \in \mathcal{D}_+$, and Lemma 3.4(iii) gives $\tilde\psi = G_q f_q$.

Lemma 3.5 with $d = 2L$ gives a continuous representative of $\tilde\psi$ on $(0, 2L)$ with $\tilde\psi(0+) = 0$, i.e. of ψ on $(-L, L)$ with $\psi(L-) = 0$; with $\psi(-\cdot)$ also $\psi(-L+) = 0$. This proves (i). By (3.3), for $0 < r \le \rho_0$, $|\tilde\psi(r) - I_{f_q} V_q(r)| \le V_q(r)[2CV_q(r) + v_q(L_{\min})C\sqrt{r}] \le CV_q(r)^2$. Since $V_q\ell^{1/2} \to 1$ (Lemma 3.3(b)), $I_{f_q} = \lim \tilde\psi(r)\ell(r)^{1/2} = A_+$; this proves (iii) and, with $q = 1$ and Lemma 3.3(b), (ii). The bound $|A_+| \le C$ follows from $|I_f| \le q^{-1/2}\|f\|_{L^\infty(0,2L)} + (q^{-1/2}v_q(2L))^{1/2}\|f\|_2$.

(iv) For $\phi \in \mathcal{D}_+$, $\chi\phi \in \mathcal{D}(\mathcal{E})$ (Lemma 2.6(a)) vanishes off $[0, 2\rho_0] \subset [0, 2L)$, so $\chi\phi \in \mathcal{D}_{(0,2L)}$ and $\mathcal{E}(\tilde\psi, \chi\phi) = \langle\lambda\tilde\psi - W_s\tilde\psi, \chi\phi\rangle$. By Lemma 2.6(b), $\mathcal{E}(\chi\tilde\psi, \phi) = \mathcal{E}(\tilde\psi, \chi\phi) + \langle r_\chi[\tilde\psi], \phi\rangle$; adding $q\langle\chi\tilde\psi, \phi\rangle$ gives $\mathcal{E}_q(u, \phi) = \langle f_+, \phi\rangle$. Here $\|\chi W_s\tilde\psi\|_\infty \le MC_\infty$ (Lemma 2.7), $|r_\chi[\tilde\psi]| \le \|k_\chi\|_1 C_\infty$, and for $x \ge 2\rho_0$, $r_\chi[\tilde\psi](x) = -\int J(x - y)\chi(y)\tilde\psi(y)\,\mathrm{d}y = O(e^{-x/2})$; so $f_+ \in L^2 \cap L^\infty$. Lemma 3.4(iii) gives $u = G_q f_+$, and Lemma 3.5 gives $|u - k_+(0)V_q| \le 2\|f_+\|_\infty V_q^2$; as $u = \tilde\psi$ on $[0, \rho_0]$, $k_+(0) = A_+$. □

Remark 4.2. (1) The primes and the pole enter only through $W_s\tilde\psi \in L^\infty$; no arithmetic condition on L appears. (2) Since representatives and limits are linear, $\psi \mapsto A_\pm(\psi)$ is linear on each eigenspace. (3) Theorem 7.1 improves the rate in (ii) to $O(V^3)$ and gives the second-order term. (4) The existence of $A_\pm$, with the bound (i) and a limit formula of the same kind as (iii), is [12, (3.8), §3.3, (3.19)]; a remainder of the order in (ii) is stated for the true form in the mathia notes [37, 39, 40] (Section 1.4).

5. The Hadamard formula, the translation identity and nonvanishing

For ψ vanishing outside $\bar I$ and $y \notin \bar I$ write $(J\psi)(y) = \int_I J(y-x)\psi(x)\,\mathrm{d}x$.

Lemma 5.1 (cross identity)**.** *Let I, I' be bounded open intervals and let $\psi \in \mathcal{D}_I \cap L^\infty$, $\psi' \in \mathcal{D}_{I'} \cap L^\infty$ satisfy $(\mathcal{E}+w_s)(\psi,\phi) = \lambda_I\langle\psi,\phi\rangle$ for $\phi \in \mathcal{D}_I$ and $(\mathcal{E}+w_s)(\psi',\phi) = \lambda_{I'}\langle\psi',\phi\rangle$ for $\phi \in \mathcal{D}_{I'}$. Then*

$$(\lambda_I - \lambda_{I'})\langle\psi,\psi'\rangle = \int_{\mathbb{R}\setminus I} (J\psi)\psi' - \int_{\mathbb{R}\setminus I'} (J\psi')\psi - w_s(\psi, \mathbf{1}_{\mathbb{R}\setminus I}\psi') + w_s(\psi', \mathbf{1}_{\mathbb{R}\setminus I'}\psi),$$

with absolutely convergent integrals. In particular, for $I \subset I'$, $(\lambda_{I'} - \lambda_I)\langle\psi,\psi'\rangle = -\int_{I'\setminus I}(J\psi)\psi' + w_s(\psi, \mathbf{1}_{I'\setminus I}\psi')$.

Proof. Write $\psi' = \mathbf{1}_I\psi' + \mathbf{1}_{\mathbb{R}\setminus I}\psi'$; $\mathbf{1}_I\psi' \in \mathcal{D}_I$ by Lemma 2.5, so $(\mathcal{E}+w_s)(\psi,\psi') = \lambda_I\langle\psi,\psi'\rangle + \mathcal{E}(\psi, \mathbf{1}_{\mathbb{R}\setminus I}\psi') + w_s(\psi, \mathbf{1}_{\mathbb{R}\setminus I}\psi')$, and $\mathcal{E}(\psi, \mathbf{1}_{\mathbb{R}\setminus I}\psi') = -\int_{\mathbb{R}\setminus I}(J\psi)\psi'$ by Lemma 2.3(d) (absolutely convergent since $\iint_{[0,1]^2} \frac{\mathrm{d}s\,\mathrm{d}t}{s+t} < \infty$). Exchange the roles and subtract, using the symmetry of $\mathcal{E}+w_s$ on functions with bounded support. □

This is an order-zero continuum analogue of Krein's resolvent identity [22, Lemma 2.3] and of the finite-dimensional commutator identities of [18, Lemma 5.2(ii)] and [19, Lemma 5.1(ii)]. Prime terms that act on $\mathcal{D}_{I'}$ but not on $\mathcal{D}_I$ are part of w_s and need no separate treatment.

Lemma 5.2 (exterior field)**.** *Let ψ be as in Theorem 4.1, $q = 1$, $V = V_1$, let M_K be the function of Lemma C.1 for $K = K_1$, and put $m(\eta) = 1 + M_K(\eta)$. Then $m \ge 0$, $m(\eta)V(\eta) \le 3$, $\int_0^z m(\eta)V(z-\eta)\,\mathrm{d}\eta = z$ for every $z > 0$, and for $0 < \eta \le \rho_0$*

$$(J\psi)(L+\eta) = A_+ m(\eta) + R_{\mathrm{f}}(\eta), \qquad |R_{\mathrm{f}}(\eta)| \le C.$$

Proof. The identity for $\int mV$ is Lemma C.1(g), and $mV \le 3$ follows from $1 + M_K(\eta) \le K(1/\eta)$ (Lemma C.1(c)) and $V(\eta) \le 3/K(1/\eta)$ (Lemma 3.3(a), as $\beta \le 1$). With $\tilde\psi$ extended by 0 off $(0, 2L)$ and $(Jg)(-\eta) = \int_0^\infty J(\eta+y)g(y)\,\mathrm{d}y$ as in Lemma C.2, $(J\psi)(L+\eta) = (J\tilde\psi)(-\eta)$. Write $\tilde\psi = u + (1-\chi)\tilde\psi$ with $u = \chi\tilde\psi = G_1 f_+$ (Theorem 4.1(iv)). Since $(1-\chi)\tilde\psi$ vanishes on $[0,\rho_0]$ and J is decreasing, $|(J(1-\chi)\tilde\psi)(-\eta)| \le C_\infty \int_{\rho_0}^\infty J$. By Lemma C.2, $(Ju)(-\eta) = m(\eta)k_+(-\eta) + R_4(\eta)$ with $k_+ = \tilde v * f_+$ as in Theorem 4.1(iv), $k_+(0) = A_+$, and $R_4(\eta) = \int_{(0,\eta)}(k_+(-\eta) - k_+(-\eta+x))\mu_K(\,\mathrm{d}x)$. For $0 \le x < y$, $|k_+(-y) - k_+(-x)| \le \|f_+\|_\infty \int_0^\infty (v(s+x) - v(s+y))\,\mathrm{d}s = \|f_+\|_\infty (V(y) - V(x))$, since v is decreasing. Hence $|m(\eta)(k_+(-\eta) - A_+)| \le \|f_+\|_\infty m(\eta)V(\eta) \le 3\|f_+\|_\infty$; and $|k_+(-\eta) - k_+(-\eta+x)|$ is at most $\|f_+\|_\infty x v(\eta/2)$ for $x < \eta/2$ and at most $\|f_+\|_\infty V(\eta)$ for $\eta/2 \le x < \eta$, so that

$$|R_4(\eta)| \le \|f_+\|_\infty \Big[v(\tfrac{\eta}{2}) \int_{(0,\eta/2)} x\,\mu_K(\,\mathrm{d}x) + V(\eta)\big(M_K(\tfrac{\eta}{2}) - M_K(\eta)\big)\Big] \le \frac{C}{\ell(\eta)}$$

by Lemma C.1(d),(e),(f) (with $K(2/\eta) \le C\ell(\eta)^{1/2}$), Lemma 6.1(e), Lemma 3.3(a) (with $V(\eta)K(1/\eta) \le 3$) and Lemma 3.2(b). The constants depend only on C_∞ and $\|f_+\|_\infty$, so they are uniform as in Theorem 4.1. □

Heuristically $m(\eta) \approx \int_0^{\rho_0} \frac{V(r)}{2(\eta+r)}\,\mathrm{d}r \approx \ell(\eta)^{1/2}$; the coefficient 1 comes from $J(u) \sim \frac{1}{2u}$ against $V \sim \ell^{-1/2}$, $\frac{1}{2}\cdot 2 = 1$. In fact $m(\eta)V(\eta) = 1 + O(\ell(\eta)^{-1})$ (Lemma C.1(f)), and the exact identity

$\int_0^z m(\eta)V(z-\eta)\,\mathrm{d}\eta = z$ makes the main term of the next lemma exact. Lemmas C.1 and C.2 are proved in Appendix C and use nothing from this section.

Lemma 5.3 (layer integral)**.** *Let ψ_1, ψ_2 be eigenfunctions as in Theorem 4.1 (for windows of half-widths in $[L_{\min}, L_{\max}]$, possibly different, and for the same s), translated so that the right edge of the window of ψ_1 is at b and that of ψ_2 at $b+\delta$, with $0 < \delta \le \rho_0$. Let $A^{(1)}, A^{(2)}$ be their right edge traces. Then*

$$\Big|\int_0^\delta (J\psi_1)(b+\eta)\,\psi_2(b+\eta)\,\mathrm{d}\eta - A^{(1)}A^{(2)}\delta\Big| \le C\delta\,\ell(\delta)^{-1/2}.$$

The same holds at left edges, with the left traces.

Proof. For $0 < \eta < \delta$ put $r = \delta - \eta$, the distance of $b+\eta$ from the right edge of the window of ψ_2. By Lemma 5.2 for ψ_1, $(J\psi_1)(b+\eta) = A^{(1)}m(\eta) + R_{\mathrm{f}}(\eta)$ with $|R_{\mathrm{f}}| \le C$; by Theorem 4.1(ii) for ψ_2, $\psi_2(b+\eta) = A^{(2)}V(r) + e_2(r)$ with $|e_2(r)| \le CV(r)^2$; and $|A^{(1)}|, |A^{(2)}| \le C$, $V \le 1$. Since $\int_0^\delta m(\eta)V(\delta-\eta)\,\mathrm{d}\eta = \delta$ exactly (Lemma 5.2), the integral equals $A^{(1)}A^{(2)}\delta$ plus

$$A^{(1)}\int_0^\delta m(\eta)\,e_2(\delta-\eta)\,\mathrm{d}\eta + \int_0^\delta R_{\mathrm{f}}(\eta)\big(A^{(2)}V(\delta-\eta) + e_2(\delta-\eta)\big)\,\mathrm{d}\eta.$$

As $m \ge 0$ and V is increasing, $\int_0^\delta m(\eta)V(\delta-\eta)^2\,\mathrm{d}\eta \le V(\delta)\int_0^\delta m(\eta)V(\delta-\eta)\,\mathrm{d}\eta = \delta V(\delta)$, and $\int_0^\delta V(\delta-\eta)\,\mathrm{d}\eta \le \delta V(\delta)$. So the error is at most $C\delta V(\delta) \le C\delta\,\ell(\delta)^{-1/2}$ by Lemma 3.3(b), since $\delta \le \rho_0 \le e^{-1}$. At left edges apply the result to $\psi_i(-\cdot)$. □

The loss $\ell(\delta)^{-1/2}$, here and in Theorems 5.4 and 5.6 below, does not come from e_2: by Theorem 7.1(a), whose constants are uniform, $e_2 = O(V^3)$, which contributes $O(\delta\,\ell(\delta)^{-1})$. It comes from the bounded remainder R_{f}, which contains a constant: $m(\eta)(k_+(-\eta) - A_+)$ tends to $-f_+(0+)$ (Lemma C.4), and the w_s-terms of those theorems are also of order $\delta V(\delta)$. An $O(\delta/\ell(\delta))$ bound would need the constant part of R_{f} to cancel against the w_s-term, as in step (4) of the proof of Theorem 9.1, which uses Lemma C.4 with pointwise constants; we do not pursue this.

Theorem 5.4 (translation identity)**.** *Let ψ_1, ψ_2 be eigenfunctions of $H_L^{(s)}$ with eigenvalues λ_1, λ_2. As $h \downarrow 0$,*

$$(\lambda_1 - \lambda_2)\langle\psi_1, \tau_h\psi_2\rangle = \big(A_+^{(1)}A_+^{(2)} - A_-^{(1)}A_-^{(2)}\big)h + O\big(h\,\ell(h)^{-1/2}\big).$$

Consequently:

(a) $|A_+(\psi)| = |A_-(\psi)|$ *for every eigenfunction* ψ.
(b) *On each eigenspace the trace map* $\psi \mapsto (A_+(\psi), A_-(\psi))$ *has rank at most one.*
(c) *For an even* ψ_e *and an odd* ψ_o*, with* $\Xi(h) := \langle\psi_o, \psi_e(\cdot + h)\rangle$*,* $(\lambda_e - \lambda_o)\Xi(h) = 2A_+(\psi_e)A_+(\psi_o)h + o(h)$*; if* $\lambda_e \ne \lambda_o$*,* $\Xi'(0)$ *exists and* $(\lambda_e - \lambda_o)\Xi'(0) = 2A_+(\psi_e)A_+(\psi_o)$.
(d) *If* $\lambda_e = \lambda_o$*, then* $A_+(\psi_e)A_+(\psi_o) = 0$.

Here $\Xi'(0)$ is a limit of difference quotients. We do not claim that it equals $\int \psi_o\psi_e'$: we do not prove $\psi \in BV$, and for $L > \frac{1}{2}\log 6$ an eigenfunction with $A \ne 0$ is Lipschitz on no interval (Corollary 7.4). Parts (a) and (b) are [12, (2.5)], and part (d) is [12, (6.3)] (see also [70]); the identity with a rate and part (c) for $\lambda_e \ne \lambda_o$ are new as far as we know (Section 1.4). For the pure logarithm, (c) is [9, Cor. 1.5], which reads $(\mu_e - \mu_o)\Xi'(0) = 4A_+(u_e)A_+(u_o)$: there μ is the eigenvalue of the logarithmic Laplacian, whose symbol is $2\log|t|$, so $\mu = 2\lambda$.

Proof. Apply Lemma 5.1 with $I = (-L, L)$, $\psi = \psi_1$, $I' = I + h$ and $\psi' = \tau_h\psi_2$, an eigenfunction on $\mathcal{D}_{I'}$ by Lemma 2.2. The set $(\mathbb{R}\setminus I)\cap I' = (L, L+h)$ gives $A_+^{(1)}A_+^{(2)}h$, and $(\mathbb{R}\setminus I')\cap I = (-L, -L+h)$ gives $A_-^{(1)}A_-^{(2)}h$, by Lemma 5.3 (right edges L and $L+h$; left edges $-L$ and $-L+h$). The w_s-terms are $w_s(\psi_1, \mathbf{1}_{(L,L+h)}\tau_h\psi_2)$ and $w_s(\tau_h\psi_2, \mathbf{1}_{(-L,-L+h)}\psi_1)$. All functions involved vanish outside $I \cup I'$, so by Lemma 2.7 with $p = \infty$ the first is at most $M\|\psi_1\|_\infty \int_L^{L+h} |\psi_2(y-h)|\,\mathrm{d}y \le CMC_\infty hV(h)$,

since $|\psi_2(y-h)| \le CV(h)$ there (Theorem 4.1(ii) and monotonicity of V); the second is bounded in the same way. Both are $O(h\ell(h)^{-1/2})$. (a) Take $\psi_1 = \psi_2$. (b) Within an eigenspace the left side vanishes, so the image of the trace map is totally isotropic for the form $(\alpha,\beta)\cdot(\alpha',\beta') = \alpha\alpha' - \beta\beta'$ of signature $(1,1)$, hence has dimension at most one. (c) Apply the identity to (ψ_e, ψ_o), which gives $\Xi(h) = \langle \psi_e, \tau_h \psi_o\rangle$ for $h > 0$, and to (ψ_o, ψ_e), which gives $\Xi(-h)$; use $\Xi(0) = 0$, $A_-(\psi_e) = A_+(\psi_e)$ and $A_-(\psi_o) = -A_+(\psi_o)$. (d) follows from (c). □

Lemma 5.5 (continuity in L)**.** *Fix s, a sector σ and $j \ge 0$.*

(a) *$L \mapsto \lambda_j^\sigma(L)$ is non-increasing, continuous, and bounded below by $-M$.*

(b) *If $\lambda_k^\sigma(L_0)$ is sector-simple, there is an open interval $N \ni L_0$ on which λ_k^σ is sector-simple and $|\langle \psi_L, \psi_{L_0}\rangle| \ge \frac12$; with the sign fixed by $\langle \psi_L, \psi_{L_0}\rangle > 0$, $L \mapsto \psi_L$ is continuous in L^2 on N, and $L \mapsto A_\pm(\psi_L)$ is continuous on N.*

Proof. Monotonicity. For $L < L'$, $\mathcal{D}_L \subset \mathcal{D}_{L'}$, the form is the same on both (Lemma 2.2), and the even and odd subspaces of $\mathcal{D}_L$ lie in those of $\mathcal{D}_{L'}$; min-max in each sector gives $\lambda_j^\sigma(L') \le \lambda_j^\sigma(L)$. The lower bound: $\mathcal{E} \ge 0$ and $|w_s(f)| \le M\|f\|^2$.

Left continuity at L_0. Let V_0 be the span of the first $j+1$ eigenfunctions of sector σ at L_0. For $\theta = L/L_0 < 1$ the dilation $f \mapsto f_\theta := f(\cdot/\theta)$ maps $\mathcal{D}_{L_0}$ into $\mathcal{D}_L$ (supports shrink to $[-L,L]$), is injective and preserves parity, so $\{f_\theta : f \in V_0\}$ is a $(j+1)$-dimensional subspace of sector σ of $\mathcal{D}_L$. By Lemma 2.4(a), $f_\theta \to f$ in form norm as $\theta \uparrow 1$, uniformly on the unit sphere of the finite-dimensional space V_0; and w_s is L^2-continuous on functions supported in a fixed window, since $|w_s(f) - w_s(g)| \le M\|f-g\|(\|f\| + \|g\|)$ by Lemma 2.7. Hence the Rayleigh quotients of f_θ converge to those of f uniformly on that sphere, and min-max gives $\limsup_{L\uparrow L_0} \lambda_j^\sigma(L) \le \max_{f\in V_0} Q(f)/\|f\|^2 = \lambda_j^\sigma(L_0)$. With monotonicity this gives left continuity.

Right continuity. For $L_n \downarrow L_0$ take orthonormal eigenfunctions $\psi_n^{(i)}$, $i \le j$, of sector σ for the first $j+1$ eigenvalues at L_n. By monotonicity $\mathcal{E}(\psi_n^{(i)}) = \lambda_i^\sigma(L_n) - w_s(\psi_n^{(i)}) \le \lambda_j^\sigma(L_0) + M$, so they are bounded in form norm; they vanish outside the fixed interval $[-\sup_n L_n, \sup_n L_n]$; and $\int_{|t|>T} |\hat\psi_n^{(i)}|^2 \le 2\pi\mathcal{E}(\psi_n^{(i)}) / \inf_{|t|>T}\Psi_a \le C/\log T$. As in Section 2.4, the Kolmogorov–Riesz theorem gives a subsequence converging in L^2 to a family $\psi^{(i)}$, which is orthonormal, of sector σ, and vanishes off $\bigcap_n [-L_n, L_n] = [-L_0, L_0]$. Along a further subsequence the Fourier transforms converge a.e., so Fatou's lemma on the Fourier side gives $\mathcal{E}(\psi^{(i)}) \le \liminf_n \mathcal{E}(\psi_n^{(i)}) < \infty$; thus $\psi^{(i)} \in \mathcal{D}_{L_0}$, and $\mathcal{E}$ is lower semicontinuous along such sequences. For a unit vector $v = \sum \alpha_i \psi^{(i)}$ put $v_n = \sum \alpha_i \psi_n^{(i)}$, a unit vector with $Q(v_n) \le \lambda_j^\sigma(L_n)$. Lower semicontinuity of $\mathcal{E}$ and L^2-continuity of w_s give $Q(v) \le \liminf Q(v_n) \le \lim_n \lambda_j^\sigma(L_n)$; the limit exists by monotonicity. Min-max on the $(j+1)$-dimensional span of the $\psi^{(i)}$ gives $\lambda_j^\sigma(L_0) \le \lim \lambda_j^\sigma(L_n)$, and monotonicity gives equality.

Simplicity persists on a neighbourhood of L_0, since $\lambda_{k-1}^\sigma(L_0) < \lambda_k^\sigma(L_0) < \lambda_{k+1}^\sigma(L_0)$ (the first inequality is void for $k = 0$) and all three functions are continuous.

Eigenvector continuity. Let $L \to L_1 \in N$. As above, (ψ_L) is precompact in L^2 and bounded in form norm. Let ϕ be an L^2-limit point. Along a subsequence $\psi_L \to \phi$ weakly in the Hilbert space $(\mathcal{D}(\mathcal{E}), \mathcal{E}_1)$ (bounded sequences have weakly convergent subsequences, and the weak limit is the L^2-limit), ϕ vanishes off $[-L_1, L_1]$, so $\phi \in \mathcal{D}_{L_1}$, and ϕ is normalised and of sector σ. For $\chi' \in \mathcal{D}_{L'}$ with $L' < \min(L, L_1)$ we have $\chi' \in \mathcal{D}_L$, so $Q(\psi_L, \chi') = \lambda(L)\langle \psi_L, \chi'\rangle$; letting $L \to L_1$ along the subsequence ($\mathcal{E}(\cdot, \chi')$ is weakly continuous, $w_s(\cdot,\chi')$ is L^2-continuous, and λ is continuous) gives $Q(\phi, \chi') = \lambda(L_1)\langle \phi, \chi'\rangle$. The dilation argument in the proof of Lemma 2.4(b) shows that $\bigcup_{L'<L_1} \mathcal{D}_{L'}$ is form-dense in $\mathcal{D}_{L_1}$, and both sides are form-continuous in χ'; so ϕ is an eigenfunction for $\lambda(L_1)$, hence $\phi = \pm\psi_{L_1}$ by simplicity. Applied with $L_1 = L_0$, this gives $|\langle \psi_L, \psi_{L_0}\rangle| \to 1$ as $L \to L_0$, so after shrinking N we may assume $|\langle \psi_L, \psi_{L_0}\rangle| \ge \frac12$ on N, and we fix the sign by $\langle \psi_L, \psi_{L_0}\rangle > 0$. Then every limit point ϕ at $L_1 \in N$ has $\langle \phi, \psi_{L_0}\rangle \ge \frac12 > 0$, as does ψ_{L_1}, so $\phi = \psi_{L_1}$; since every subsequence has a further subsequence converging to ψ_{L_1}, $\psi_L \to \psi_{L_1}$ in L^2.

Continuity of $A_\pm$. The constants of Theorem 4.1 are uniform for L in a compact subinterval of N, in particular $\sup \|f_{+,L}\|_\infty < \infty$ there. By Theorem 4.1(iv), $A_+(L) = \int_0^\infty v\, f_{+,L}$ with $f_{+,L} = \mathbf{1}_{(0,\infty)}[\chi((q+\lambda(L))\tilde\psi_L - W_s\tilde\psi_L) + r_\chi[\tilde\psi_L]]$ and $\tilde\psi_L = \psi_L(L-\cdot)$. As $L \to L_1$: $\tilde\psi_L \to \tilde\psi_{L_1}$ in L^2 (translation is L^2-continuous and $\psi_L \to \psi_{L_1}$); $\lambda(L) \to \lambda(L_1)$; $\|\chi W_s g\|_2 \le M\|g\|_2$ for g supported in a fixed window (Lemma 2.7), which covers the windows near an activation point, where the new shift term of $W_s\tilde\psi_L$ tends to 0 in L^2; and $\|r_\chi[g]\|_2 \le \|k_\chi\|_1\|g\|_2$ (Lemma 2.6(b)). So $f_{+,L} \to f_{+,L_1}$ in L^2. For every $\varepsilon > 0$,

$$\Big|\int_0^\infty v\,(f_{+,L} - f_{+,L_1})\Big| \le 2\sup_L \|f_{+,L}\|_\infty V(\varepsilon) + \|v\mathbf{1}_{(\varepsilon,\infty)}\|_2 \|f_{+,L} - f_{+,L_1}\|_2,$$

where $v\mathbf{1}_{(\varepsilon,\infty)} \in L^2$ because v is bounded on (ε,∞) and integrable. Choosing ε small and then L close to L_1 gives continuity of A_+; A_- is the same with $\psi_L(-\cdot)$. □

Theorem 5.6 (Hadamard formula)**.** *Let $s \in \mathbb{R}$, $\sigma \in \{e,o\}$, $k \ge 0$, and let $L_0 > 0$ be such that $\lambda_k^\sigma(L_0)$ is simple in its sector. Then on an open interval $N \ni L_0$ the eigenvalue $\lambda := \lambda_k^\sigma$ is sector-simple and continuously differentiable, and*

$$\frac{d\lambda}{dL} = -\big(A_+(L)^2 + A_-(L)^2\big) = -2A(L)^2,$$

where $A_\pm(L)$ are the edge traces of a normalised eigenfunction ψ_L. Quantitatively, for $L_1 < L_2$ in N with $\delta = L_2 - L_1 \le \rho_0$,

$$\Big|(\lambda(L_2) - \lambda(L_1))\langle\psi_{L_1}, \psi_{L_2}\rangle + \sum_{\pm} A_\pm(L_1)A_\pm(L_2)\,\delta\Big| \le C\delta\,\ell(\delta)^{-1/2}.$$

No condition on e^{2L_0} is needed: the formula holds at the windows $2L_0 = \log n$, n a prime power, where the shift $c_n(S_n + S_n^*)$ begins to act on $\mathcal{D}_L$, and λ is C^1 through them. For simple eigenvalues the formula, including the activation points, is [12, Thm. 2.2, (2.6)]*, proved there by a different method and with remainder $o(\delta)$; see Section 1.4.*

Proof. Apply the nested case of Lemma 5.1 to ψ_{L_1}, ψ_{L_2}. The layers (L_1, L_2) and $(-L_2, -L_1)$ give $\sum_\pm A_\pm(L_1)A_\pm(L_2)\delta$ up to the error of Lemma 5.3, with uniform constants by Lemma 5.5. The w_s-term is at most $M\|\psi_{L_1}\|_\infty \|\mathbf{1}_{L_1<|y|<L_2}\psi_{L_2}\|_1 \le CMC_\infty \delta V(\delta)$ (Lemma 2.7, Theorem 4.1). This proves the estimate. By Lemma 5.5, $\langle\psi_{L_1}, \psi_{L_2}\rangle \to 1$ and $A_\pm$ are continuous; dividing by $\delta\langle\psi_{L_1}, \psi_{L_2}\rangle$, the difference quotients converge locally uniformly to $-(A_+^2 + A_-^2)(L)$ as $L_1, L_2 \to L$. So λ is strictly differentiable on N with continuous derivative. By Theorem 5.4(a), $A_+^2 = A_-^2$. □

Corollary 5.7 (edge-free identity)**.** *Let ψ_e, ψ_o be as in Theorem 5.4(c), with $\lambda_e \ne \lambda_o$ both sector-simple. Then $[(\lambda_e - \lambda_o)\Xi'(0)]^2 = \lambda_e'\lambda_o'$.*

Proof. Square Theorem 5.4(c) and use $\lambda_\sigma' = -2A_+(\psi_\sigma)^2$ (Theorem 5.6). □

Proposition 5.8 (nonvanishing)**.**

(a) *Let $s \le 0$. For every $L > 0$ the lowest eigenvalue $\lambda_0(L)$ of $H_L^{(s)}$ is simple, its eigenfunction can be taken nonnegative and is even, $A > 0$, and $\lambda_0 \in C^1(0,\infty)$ with $\lambda_0' = -2A^2 < 0$.*

(b) *Let $s \ge 0$ and $2L \le \log 2$. The lowest odd eigenvalue is sector-simple, its eigenfunction can be taken nonnegative on $(0, L)$, $A_o > 0$, and $\lambda_o' = -2A_o^2 < 0$.*

(c) *Let $s = 1$, and let U be an open interval of windows on which the lowest eigenvalue of $H_L^{(1)}$ is simple with an even eigenfunction. Then $\{L \in U : A_e(L) = 0\}$ is relatively closed in U and has empty interior. By* [10] *one may take $U = (0, 1.19)$.*

Proof. (a) *Nonnegative eigenfunctions have $A_+ > 0$.* Let $\psi \ge 0$, $\psi \not\equiv 0$. On $(0, 2L)$, $-W_s\tilde\psi \ge -c\tilde\psi$, since $s \le 0$ and the shift terms enter with a plus sign; and $m_{\text{ext}} \ge 0$. As v_q is decreasing,

Theorem 4.1(iii) gives, for $q \ge c - \lambda$,

$$A_+ \ge v_q(2L)\Big[(q+\lambda-c)\int \psi - \int_0^{2L} \tilde\psi(r)\kappa_{\mathrm{ext}}(r)\,\mathrm{d}r\Big], \qquad \kappa_{\mathrm{ext}}(r) = \int_{2L-r}^{\infty} J,$$

using $\int_{2L}^\infty v_q\, m_{\mathrm{ext}} \le v_q(2L)\int_{2L}^\infty m_{\mathrm{ext}}$. The last integral is finite and independent of q, and A_+ does not depend on q; letting $q \to \infty$ gives $A_+ > 0$. *Ground state.* By Lemma 2.3(a), $\mathcal{E}(f) - \mathcal{E}(|f|) = 4\iint J(x-y)f^+(x)f^-(y) \ge 0$; the shift terms satisfy $-2c_n\int |f||f(\cdot+\log n)| \le -2c_n\int ff(\cdot+\log n)$; and for $s \le 0$ the pole term $2s\iint ff\cosh\frac{x-y}{2}$ does not increase when f is replaced by $|f|$. So $|f| \in \mathcal{D}_L$ (Lemma 2.3(b)) is a minimiser with f, and since $J > 0$, equality forces f to have one sign. The functional $f \mapsto \int f$ is then nonzero on every nonzero minimiser, so the lowest eigenvalue is simple; its eigenfunction is ≥ 0, and even because its reflection is also a nonnegative ground state. Theorem 5.6 applies at every L.

(b) For $2L \le \log 2$ no shift acts on $\mathcal{D}_L$. Let ψ be odd and ≥ 0 on $(0,L)$, $\psi \not\equiv 0$. Then $\int_0^{2L} v_q\tilde\psi = \int_0^L (v_q(r) - v_q(2L-r))\tilde\psi(r)\,\mathrm{d}r > 0$. On odd functions the pole acts as $P_s\psi(x) = -2sS'\sinh\frac{x}{2}$ with $S' = \int \sinh\frac{y}{2}\psi(y)\,\mathrm{d}y \ge 0$, and $-P_s\tilde\psi$ contributes $2sS'\int_0^L (v_q(r) - v_q(2L-r))\sinh\frac{L-r}{2}\,\mathrm{d}r \ge 0$. Finally $m_{\mathrm{ext}}(y) = \int_0^L (J(y-r) - J(y-2L+r))\tilde\psi(r)\,\mathrm{d}r \le 0$. So $A_+ > 0$ for $q > c - \lambda$. *Odd ground state.* For odd $f \in \mathcal{D}_L$ let $f^\sharp = \mathrm{sgn}(x)|f|$. First, $f^\sharp \in \mathcal{D}_L$. On $\{x,y>0\}$ and on $\{x,y<0\}$, $|f^\sharp(x) - f^\sharp(y)| \le |f(x) - f(y)|$. On $\{x>0>y\}$, $(f^\sharp(x) - f^\sharp(y))^2 - (f(x)-f(y))^2 = 2|f(x)f(y)| + 2f(x)f(y) \le 4|f(x)f(y)|$, and since $J(u) \le C/u$ for $u > 0$ (Lemma 2.1(b)), Hilbert's inequality gives

$$\iint_{x>0>y} |f(x)f(y)|J(x-y)\,\mathrm{d}x\,\mathrm{d}y \le C\iint_{(0,\infty)^2} \frac{|f(x)||f(-y')|}{x+y'}\,\mathrm{d}x\,\mathrm{d}y' \le C\pi\|f\|^2;$$

the region $\{y>0>x\}$ is the same. So $\mathcal{E}(f^\sharp) < \infty$ by Lemma 2.3(a), and $f^\sharp$ vanishes off $[-L,L]$. Then

$$\mathcal{E}(f) - \mathcal{E}(f^\sharp) = 2\iint_{x,y>0} \big(|f(x)f(y)| - f(x)f(y)\big)\big(J(x-y) - J(x+y)\big)\,\mathrm{d}x\,\mathrm{d}y \ge 0,$$

with equality only if $f(x)f(y) \ge 0$ a.e. on $(0,\infty)^2$, since $J(x-y) > J(x+y)$ there. The pole term is $-2s(\int f\sinh\frac{x}{2})^2$ on odd f and does not increase. So minimisers have one sign on $(0,L)$, $f \mapsto \int_0^L f$ separates them, and the lowest odd eigenvalue is sector-simple.

(c) *Strict monotonicity.* Suppose $\lambda_{\min}(L) = \lambda_{\min}(L')$ with $L < L'$, and let ψ be a minimiser at L. Every $\tau_h\psi$ with $|h| \le L' - L$ is a minimiser at L' (Lemma 2.2), hence lies in the finite-dimensional eigenspace. But the functions $e^{iht}F(t)$ are linearly independent for distinct h, since F is entire and $\not\equiv 0$. So $\lambda_{\min}$ is strictly decreasing, for every s and L. On U, Theorem 5.6 gives $\lambda'_{\min} = -2A_e^2$, so A_e^2 is continuous and $\{A_e = 0\}$ is relatively closed; if it contained an interval, $\lambda_{\min}$ would be constant there. The set of L at which the ground state is simple and even is open (Lemma 5.5), and [10] shows that it contains $(0, 1.19]$. □

The positive-weight pole kernel $+2\cosh$ has the wrong sign for the Beurling–Deny argument, which is why neither step of (a) works in the even sector when $s = 1$. Part (a) at $s = 0$ and the threshold $2L \le \log 2$ in (b) are known: see [20] (Perron structure of the pole-removed form) and the mathia notes [29, 36, 38]. The manuscript [12] proves that the eigenvalues are nonincreasing in L and notes that its methods do not give a nonzero trace [12, §6]. For $s = 1$ at the window $2L = \log 2$, $A_o > 0$ is in [38]: an antisymmetric Hopf lemma gives $\liminf_{r\downarrow 0}\psi(L-r)\ell(r)^{1/2} > 0$ for the lowest odd eigenfunction, so $A_o > 0$ whenever the trace exists. The other cases of $A > 0$ in (a) and (b), the statements $\lambda' < 0$, and (c) are new as far as we know.

Remark 5.9 (where the simplicity hypothesis is known). Theorem 5.6 needs sector-simplicity. It is known in the following cases, and only these are used in this paper. (a) $s \le 0$, ground state,

every $L > 0$ (Proposition 5.8(a)); this includes the pole-free form with all prime powers. (b) $s \geq 0$, $2L \leq \log 2$, lowest odd eigenvalue (Proposition 5.8(b)); this includes the true form for $L \leq 0.3466$. (c) $s = 1$, ground state: the author's computer-assisted proof in Arb ball arithmetic [10] shows that for every $L \in (0, 1.19]$ the lowest eigenvalue of $H_L^{(1)}$ is simple with an even eigenfunction; the form there is $Q_L^{(1)}$ with the same norm. Hence Theorem 5.6 holds for λ_0^e at every $L \in (0, 1.19]$, and $\lambda'_{\min} = -2A_e^2$ there. This input is computer-assisted and unrefereed; only the range $(0, 1.19]$ of its public release is used. (d) For $s = 1$ and $L > 1.19$ no simplicity input is used, and Theorem 5.6 is not asserted for the ground state there.

Remark 5.10. (1) *The constant.* The factor in front of A^2 is one A^2 per endpoint (Lemmas 5.2–5.3). It is the order-zero counterpart of the one-dimensional fractional boundary term $\Gamma(1+\sigma)^2(u/\delta^\sigma)^2$ of [23]. Identities that contain no absolute trace (the cross identity, the edge-free identity of Corollary 5.7, the ratio $(A_o/A_e)^2 = \lambda'_o/\lambda'_e$) would hold equally with a layer constant $\kappa A^{(1)}A^{(2)}\delta$; only absolute amplitudes test it (Table 2). (2) *Prolate analogue.* Theorem 5.6 is an order-zero analogue of $\partial_c \ln \lambda_n(c) = 2|\psi_{n,c}(1)|^2/c$ ([8, eq. (43)], arXiv numbering), with the endpoint value replaced by the trace A, since our eigenfunctions vanish at the edge. (3) *Consistency.* For the pure logarithm $A^2 = \frac{1}{2L}$ [12, Cor. 2.3], [9], so $\lambda' = -\frac{1}{L}$; this matches, to leading order as $L \to 0$, the term $\log \frac{1}{a}$ in Suzuki's small-window expansion (his a is our L) of the lowest eigenvalue [72, Thm. 1.4]. (4) *Activations.* The absence of a jump in λ' at a prime activation was anticipated in the mathia notes: [31] shows that, at a fixed state in $C_0 \cap BV$, the derivative in the window of the lower-order part of the form, including a newly activated shift, is continuous, and [41] shows that each activation is first-order flat for odd states with bounded traces. [12, Thm. 2.2] proves it for every simple eigenvalue, and Theorem 5.6 for every sector-simple one. The activation shift of an eigenvalue, of order $\delta/\log(1/\delta)$, is proved in [12, Thm. 2.4]; numerically, λ' has only a logarithmic modulus of continuity at an activation point $2L = \log n$, which we do not prove. The mechanism is the one of [41] (see also the $1/\log(1/h)$ entry scale of [76]); compare the c_X term of Theorem 9.1. (5) *Simplicity in general.* Kwaśnicki and Wszoła [55, Thm. 1.2] prove simplicity on intervals for $\phi(-\Delta)$ with ϕ a CBF that does not extend meromorphically to $\mathbb{C} \setminus \{0\}$. The archimedean φ of Lemma 2.1 is meromorphic on $\mathbb{C}$, with simple poles at $-a_k^2$, so even the archimedean part alone does not satisfy that hypothesis, and the Weil form adds primes and a pole. For $s = 1$ simplicity of the ground state is exactly the input of [10].

6. Interior analysis

6.1. **Chains of shifts.** A *chain* of length j from $+L$ is a sequence $z_0 = L$, $z_i = z_{i-1} - \varsigma_i \log n_i$ with $n_i \in N_L$, signs $\varsigma_i \in \{\pm 1\}$ and $z_1, \dots, z_j \in (-L, L)$; its weight is $\prod_i c_{n_i}$. Chains from $-L$ are defined in the same way with $z_0 = -L$. Put

$$\mathsf{W}_\pm^{(j)}(z) = A_\pm \cdot \big(\text{sum of the weights of the chains of length } j \text{ from } \pm L \text{ to } z\big),$$

and $\Omega_j = \mathsf{W}_+^{(j)} - \mathsf{W}_-^{(j)}$, $\Sigma_j = \mathsf{W}_+^{(j)} + \mathsf{W}_-^{(j)}$. The *generation* $g(z)$ of $z \in (-L, L)$ is the least length of a chain from either edge to z (∞ if there is none), and $C_L = \{g < \infty\}$. For $g(z) = k$ write $\mathsf{W}_\pm(z) = \mathsf{W}_\pm^{(k)}(z)$. The edges have generation 0, with $\Omega_0(L) = A_+$, $\Omega_0(-L) = -A_-$, and $\Omega_j(\pm L) = 0$ for $j \geq 1$. If $P \in (-L, L)$ and $z = P - \varsigma \log n \in (-L, L)$, then $g(z) \leq g(P) + 1$. Summing over the one-step predecessors $P = z + \varsigma \log n$ of z (including $P = \pm L$), for $g(z) = k \geq 1$,

$$\begin{aligned} \mathsf{W}_\pm(z) &= \sum_{g(P)=k-1} c_n \mathsf{W}_\pm(P), \qquad \Omega_k(z) = \sum_{g(P)=k-1} c_n \Omega_{k-1}(P), \\ \Omega_{k+1}(z) &= \sum_{g(P)\in\{k-1,k\}} c_n \Omega_k(P); \end{aligned} \tag{6.1}$$

here $g(P) \le k-2$ is impossible and $\Omega_k(P) = 0$ when $g(P) \ge k+1$. We call L *generic* if no point is reached by chains from both edges. A coincidence $L - \sum \varsigma_i \log n_i = -L + \sum \varsigma'_j \log m_j$ forces $e^{2L} \in \mathbb{Q}$, so every nonzero algebraic L is generic (Hermite–Lindemann). For generic L, $\mathsf{W}_-(y_n) = 0$.

For a function f and a point z we write $\mathfrak{D}_f(z;d) = f(z-d) - f(z+d)$ (odd part, or jump) and $\mathfrak{S}_f(z;d) = \frac{1}{2}(f(z-d) + f(z+d)) - f(z)$ (even part), so that $f(z \mp d) - f(z) = \mathfrak{S}_f \pm \frac{1}{2}\mathfrak{D}_f$. We say $f \sim (\mathsf{s}_-, \mathsf{s}_+)_a$ at z if $f(z \mp d) - f(z) = \mathsf{s}_\mp \ell(d)^{-a} + O(\ell(d)^{-a-1})$ as $d \downarrow 0$.

6.2. **Inversion near interior points.** In Sections 6–7, L and s are fixed, ψ is an eigenfunction of $H_L^{(s)}$ with eigenvalue λ, $q > 0$ is fixed ($q = 1$ unless stated), and $\mu_q := q + \lambda - c$. Constants are *pointwise*: they may depend on $L, s, \lambda, q, \|\psi\|_\infty$, on the point considered and on the gaps δ_X, δ_X^-. The exceptions are Lemma 6.6(a) and Theorem 7.1(a), whose constants are uniform as in Theorem 4.1 (for fixed q); nothing else in these two sections is uniform in L near the activation points $2L = \log n$, nor over the (possibly dense) set C_L.

Lemma 6.1 (free resolvent)**.** *Put $r_q = v * \tilde{v}$ and $R(x) = \int_0^x r_q$.*

(a) *$r_q \in L^1$, $\hat{r}_q = 1/\sigma_q$, $\int r_q = 1/q$. For $f \in L^2$, $u = r_q * f$ is the unique $u \in \mathcal{D}(\mathcal{E})$ with $\mathcal{E}_q(u,\phi) = \langle f, \phi\rangle$ for all $\phi \in \mathcal{D}(\mathcal{E})$; for $f \in L^2 \cap L^\infty$, $r_q * f$ is bounded and uniformly continuous.*

(b) *$r_q(x) = \int_0^\infty v(|x|+y)v(y)\,\mathrm{d}y$ is even, continuous on $\mathbb{R}\setminus\{0\}$ and nonincreasing on $(0,\infty)$. Also $r_q = \int_0^\infty e^{-qt}p_t\,\mathrm{d}t$ with the Gaussian mixtures p_t of Appendix A, and r_q has finite moments of all orders.*

(c) *$R(x) = \int_0^\infty v(y)\big(V(y+x) - V(y)\big)\,\mathrm{d}y$ for $x > 0$.*

(d) *$x\,r_q(x) \le 2R(x)$, and $\|r_q(\cdot + h) - r_q\|_1 = 4R(|h|/2)$.*

(e) *For $0 < x \le e^{-1}$, $\ell = \ell(x)$: $v(x) \le C\ell^{-3/2}/x$, $|v'(x)| \le 2v(x/2)/x$, $v''(x) \le 2|v'(x/2)|/x$, $\int_0^x yv(y)\,\mathrm{d}y \le Cx\ell^{-3/2}$, and $V(x) - V(xt) \le C(1 + \log\frac{1}{t})\ell^{-3/2}$ for $0 < t \le 1$.*

(f) *For $k \ge 0$ and $0 < \varepsilon < e^{-2}$, $\int_\varepsilon^{e^{-1}} \ell(a)^{-k}a^{-2}\,\mathrm{d}a \le C_k\ell(\varepsilon)^{-k}/\varepsilon$.*

Proof. (a) $\hat{r}_q = \hat{v}\hat{\tilde{v}} = 1/(K(-it)K(it)) = 1/\sigma_q$ by Lemma 3.4 (Step 1) and (K3); then $\mathcal{E}_q(u,\phi) = \frac{1}{2\pi}\int \hat{f}\bar{\hat{\phi}}$, and uniqueness follows from $\mathcal{E}_q \ge q\|\cdot\|^2$. (b) The first formula is direct, and v is decreasing. Since $\hat{r}_q = \int_0^\infty e^{-qt}e^{-t\Psi_a}\,\mathrm{d}t$ and $\hat{p}_t = e^{-t\Psi_a}$, $r_q = \int e^{-qt}p_t\,\mathrm{d}t$. Moments: $\int |x|^k p_t = c_k\,\mathbb{E}[S_t^{k/2}]$ for the subordinator (S_t) of φ, whose Lévy density $\sum_k 2a_k e^{-a_k^2 s}$ has exponential moments, so $\mathbb{E}[S_t^{k/2}]$ grows at most polynomially in t. (c) Fubini on (b). (d) $\frac{x}{2}r_q(x) \le \int_{x/2}^x r_q$. For an even function r that is decreasing on $(0,\infty)$ and $h > 0$, $r(y+h) > r(y)$ iff $y < -h/2$, so $\int |r(\cdot+h) - r| = 2\int_{-h/2}^{h/2} r$. (e) By Lemma 3.3(b), $V(x) - V(x/2) \le C\ell^{-3/2}$; iterating gives the last bound, and $\frac{x}{2}v(x) \le V(x) - V(x/2)$. Since v is completely monotone, $|v'|$ and v'' are decreasing, which gives the derivative bounds. Finally $\int_0^x yv(y)\,\mathrm{d}y = \int_0^x (V(x) - V(y))\,\mathrm{d}y$. (f) Split at $\varepsilon^{1/2}$. □

Proposition 6.2 (renewal and Green functions to second order)**.** *Let $q > 0$, $B = q + c_a$ and $a_q = B - \gamma = q + \frac{\pi}{2} + 2\log 2$. As $x \downarrow 0$, with $\ell = \ell(x)$:*

(R1) $0 \le R(x) - \frac{1}{2}V(x)^2 \le C\ell^{-3}$;

(R2) $2R(x) = \dfrac{1}{\ell + B} + \dfrac{\gamma}{(\ell+B)^2} + O(\ell^{-3})$, *that is,* $2R(x) = (\ell + a_q)^{-1} + O(\ell^{-3})$;

(V2) $V(x) = (\ell + a_q)^{-1/2} + O(\ell^{-5/2})$, *that is,* $V(x)^{-2} = \ell + a_q + O(\ell^{-1})$.

Proof. (R1) By Lemma 6.1(c), $R(x) = \int_0^\infty v(y)(V(y+x) - V(y))\,\mathrm{d}y$. On $(0,x)$ write $V(y+x) - V(y) = [V(x) - V(y)] + [V(y+x) - V(x)]$; the first bracket integrates against v to $V(x)^2 - \frac{1}{2}V(x)^2$, and the second lies in $[0, y\,v(x)]$ by concavity, contributing at most $v(x)\int_0^x yv \le C\ell^{-3}$ (Lemma 6.1(e)). On (x,∞), $0 \le V(y+x) - V(y) \le xv(y)$, and $x\int_x^\infty v^2 \le Cx\int_x^{e^{-1}} \ell(y)^{-3}y^{-2}\,\mathrm{d}y + Cx \le C\ell^{-3}$ by Lemma 6.1(e),(f).

(R2) *Step 1.* $\hat r_q = 1/\sigma_q$ is positive and decreasing on $(0,\infty)$, from $1/q$ to 0. By Lemma 6.1(b), $r_q = \int g_\varrho\,\nu(\mathrm d\varrho)$ is a mixture of centred Gaussians, $\hat g_\varrho = e^{-\varrho t^2}$, with $\nu \ge 0$ of mass $1/q$. For each Gaussian, $\int_{-x}^{x} g_\varrho = \frac{2}{\pi}\int_0^\infty e^{-\varrho t^2}\frac{\sin xt}{t}\,\mathrm dt = \frac{2}{\pi}\int_0^\infty \mathrm{Si}(xt)\,2\varrho t e^{-\varrho t^2}\,\mathrm dt$, with $\mathrm{Si}(u) = \int_0^u \frac{\sin s}{s}\,\mathrm ds > 0$. By Tonelli, $2R(x) = \frac{2}{\pi}\int_0^\infty \mathrm{Si}(xt)\,\varpi(\mathrm dt)$ with $\varpi = -\,\mathrm d\hat r_q \ge 0$ of mass $1/q$.

Step 2. Substitute $\tau = \log(xt)$, $S(\tau) = \frac{2}{\pi}\mathrm{Si}(e^\tau)$ and $\mathcal H(y) = \hat r_q(e^y)$: $2R(x) = \int_{\mathbb R} S(\tau)\,(-\,\mathrm d\mathcal H(\ell+\tau))$. Here $0 \le S(\tau) \le \frac{2}{\pi}e^\tau$, $|S'| \le 1$, $\mathcal U(\tau) := \mathbf 1_{\tau>0} - S(\tau)$ satisfies $|\mathcal U(\tau)| \le Ce^{-|\tau|}$, and $\mathcal H(y) = \frac{1}{y+B} + E(y)$ with $|E(y)| \le Ce^{-2y}$ for $y \ge 1$ (Lemma 2.1(c)).

Step 3. The part $\tau < -\ell/2$ is $O(e^{-\ell/2})$, since $-\,\mathrm d\mathcal H$ has mass $1/q$. On $\tau \ge -\ell/2$ integrate by parts: the boundary term is $O(e^{-\ell/2})$, the E-part is $\int_{-\ell/2}^\infty |S'(\tau)||E(\ell+\tau)|\,\mathrm d\tau = O(e^{-\ell})$, and the rest is, after a second integration by parts, $\int_{-\ell/2}^\infty \frac{S(\tau)\,\mathrm d\tau}{(\ell+B+\tau)^2} + O(e^{-\ell/2})$. Writing $S = \mathbf 1_{\tau>0} - \mathcal U$ and using $|(\ell+B+\tau)^{-2} - (\ell+B)^{-2}| \le C|\tau|\ell^{-3}$ for $\tau \ge -\ell/2$, this is $\frac{1}{\ell+B} - \frac{\int_{\mathbb R}\mathcal U}{(\ell+B)^2} + O(\ell^{-3})$.

Step 4. With $\mathfrak g(u) = \mathbf 1_{u>1} - \frac{2}{\pi}\mathrm{Si}(u)$, $\int_{\mathbb R}\mathcal U = \int_0^\infty \mathfrak g(u)\frac{\mathrm du}{u} = -\int_0^\infty \log u\,\mathrm d\mathfrak g(u) = \frac{2}{\pi}\int_0^\infty \frac{\sin u}{u}\log u\,\mathrm du = -\gamma$; the last integral is $\mathcal M'(0)$ for $\mathcal M(s) = \int_0^\infty \sin u\,u^{s-1}\,\mathrm du = \Gamma(s)\sin\frac{\pi s}{2} = \frac{\Gamma(1+s)\sin(\pi s/2)}{s}$, so it equals $-\frac{\pi\gamma}{2}$. Hence $2R = \frac{1}{\ell+B} + \frac{\gamma}{(\ell+B)^2} + O(\ell^{-3}) = \frac{1}{\ell+B-\gamma} + O(\ell^{-3})$.

(V2) $V^2 = 2R - 2(R - \frac12 V^2) = (\ell + a_q)^{-1} + O(\ell^{-3})$ by (R1) and (R2). □

The Euler-constant correction is the classical one of de Haan theory [4, Ch. 3]; what is used here is the explicit $O(\ell^{-5/2})$ remainder. Numerically $1/(2R) - \ell$ approaches $a_1 = 3.95709$ for $q = 1$.

Lemma 6.3. *For $0 < x \le e^{-1}$: $R(x) \le C/\ell$, $R(x) - R(xt) \le C(1 + \log\frac1t)\ell^{-2}$ for $0 < t \le 1$, $r_q(x) \le C\ell^{-2}/x$, and $\int_0^x y\,r_q(y)\,\mathrm dy \le Cx\ell^{-2}$.*

Proof. By (R2), $R(x) - R(x/2) = O(\ell^{-2})$; iterate. Then $\frac{x}{2}r_q(x) \le R(x) - R(x/2)$ and $\int_0^x y r_q = \int_0^x (R(x) - R(y))\,\mathrm dy$. □

Lemma 6.4 (the θ-kernel)**.** *For $z > 0$ let $\theta_z(y) = v(y-z)\mathbf 1_{y>z} - v(y)$. Then $\int\theta_z = 0$ and $\int_0^\infty \theta_z V = R(z)$. If h is bounded on $(0,\infty)$ and $|h| \le C_h V^j$ on $(0,a)$ for some $j \ge 1$, then for small z, $|\int_0^\infty \theta_z h| \le C_j(C_h + \|h\|_\infty)V(z)^{j+1}$.*

Proof. $\int_0^\infty \theta_z V = \int_0^\infty v(u)(V(u+z) - V(u))\,\mathrm du = R(z)$ by Lemma 6.1(c). On $(0,z)$, $|\theta_z| = v$, which gives $\frac{C_h}{j+1}V(z)^{j+1}$. On $(z,\sqrt z)$, $\theta_z \ge 0$, $\int_z^{\sqrt z}\theta_z \le V(z)$ and $V(\sqrt z) \le 2V(z)$. On $(\sqrt z,\infty)$, $\int|\theta_z| = V(\sqrt z) - V(\sqrt z - z) \le zv(\sqrt z/2) \le C\sqrt z$. □

Lemma 6.5 (interior localisation)**.** *Let $z \in (-L,L)$, $0 < \delta_0 < \frac12\operatorname{dist}(z,\{\pm L\})$, and $\chi_1 \in C_c^\infty(-L,L)$ with $\chi_1 = 1$ on $(z-2\delta_0, z+2\delta_0)$. Then $\chi_1\psi = r_q * f_1$ pointwise, with $f_1 = \chi_1((q+\lambda)\psi - W_s\psi) + r_{\chi_1}[\psi] \in L^2\cap L^\infty$, and on $I_z = (z-\delta_0, z+\delta_0)$*

$$f_1 = \mu_q\psi - P_s\psi + \sum_{n\in N_L} c_n\big(\psi(\cdot + \log n) + \psi(\cdot - \log n)\big) + \varrho_1,$$

where $P_s\psi$ is real-analytic and ϱ_1 is Lipschitz on I_z.

Proof. For $\phi \in \mathcal D(\mathcal E)$, $\chi_1\phi \in \mathcal D_L$, so the eigenvalue equation and Lemma 2.6(b) give $\mathcal E_q(\chi_1\psi,\phi) = \langle f_1,\phi\rangle$; Lemma 6.1(a) gives $\chi_1\psi = r_q * f_1$ a.e., and both sides are continuous. On I_z, $r_{\chi_1}[\psi](x) = \int J(x-y)(1-\chi_1(y))\psi(y)\,\mathrm dy$ with $|x-y| \ge \delta_0$, where J is smooth with integrable derivative; shifts with $\log n \ge 2L$ vanish on $(-L,L)$; and $P_s\psi(x) = 2s[\cosh\frac x2\int\cosh\frac y2\psi - \sinh\frac x2\int\sinh\frac y2\psi]$. □

The mechanism. For $n \in N_L$ the shift $c_n\psi(x+\log n)$ is cut off at $x = y_n$: for $x < y_n$ near y_n it equals the edge germ $c_n\psi(L - (y_n - x)) \approx c_nA_+\ell(y_n - x)^{-1/2}$, and for $x \ge y_n$ it vanishes. Inversion with the order-zero kernel r_q smooths it by exactly one power of ℓ.

Lemma 6.6 (global modulus)**.** *(a′) For $0<\delta\le L/2$ there is C_δ with $|\psi(x)-\psi(x')|\le C_\delta/\ell(|x-x'|)$ for $x,x'\in[-L+\delta,L-\delta]$, $0<|x-x'|\le e^{-1}$. (a) $|\psi(x)-\psi(x')|\le CV(|x-x'|)$ for all $x,x'\in\mathbb{R}$, with C uniform as in Theorem 4.1.*

Proof. (a′) Take $\chi_1=1$ on $[-L+\delta/2,L-\delta/2]$ in Lemma 6.5; then $|\psi(x)-\psi(x')|\le\|f_1\|_\infty\|r_q(\cdot+d)-r_q\|_1=4\|f_1\|_\infty R(d/2)\le C_\delta/\ell(d)$ by Lemmas 6.1(d) and 6.3. (a) Near L, $\tilde\psi=u=\int_0^r v(r-z)k_+(z)\,\mathrm{d}z$ on $[0,\rho_0]$ with $\|k_+\|_\infty\le q^{-1/2}\|f_+\|_\infty$ (Theorem 4.1(iv)), and $u(r)-u(r-d)=\int_{r-d}^r v(r-z)k_+ + \int_0^{r-d}(v(r-z)-v(r-d-z))k_+$, each at most $\|k_+\|_\infty V(d)$. In the interior use (a′) with $\delta=\rho_0/2$ and $1/\ell\le CV$. Two points at distance $\le\rho_0/2$ lie in a common zone; across the edge, $|\psi(x)|\le CV(L-x)\le CV(|x-x'|)$ for $x'\ge L$; large distances are handled by boundedness. □

Lemma 6.7 (one-log gain)**.** *Let $f\in L^\infty$, $u=r_q*f$, $x_0\in\mathbb{R}$, $\beta\ge0$, $0<\delta\le e^{-1}$, $\mathsf{M}\ge0$, and suppose $|f(x_0+w)-f(x_0)|\le\mathsf{M}\ell(|w|)^{-\beta}$ for $0<|w|\le\delta$. Then for $0<|z|\le\delta/4$*

$$|u(x_0+z)-u(x_0)|\le C_\beta\Big(\mathsf{M}\ell(|z|)^{-\beta-1}+\|f\|_\infty\frac{|z|}{\delta\,\ell(\delta)^2}\Big),$$

with C_β depending only on β and q. Without any hypothesis, $|u(x+h)-u(x)|\le4\|f\|_\infty R(|h|/2)\le C\|f\|_\infty/\ell(|h|)$.

Proof. Let $\varepsilon=|z|$ and $g(w)=f(x_0+w)-f(x_0)$. Since r_q is even and $\int(r_q(z-w)-r_q(w))\,\mathrm{d}w=0$, $u(x_0+z)-u(x_0)=\int(r_q(z-w)-r_q(w))g(w)\,\mathrm{d}w$. $|w|\le2\varepsilon$. First, $\int_{|w|\le2\varepsilon}r_q|g|\le2\mathsf{M}\int_0^{2\varepsilon}\ell^{-\beta}\,\mathrm{d}R$; on dyadic pieces $[2^{-i}\cdot2\varepsilon,2^{-i+1}\cdot2\varepsilon]$ the $\mathrm{d}R$-mass is $\le C\ell(2^{-i}\varepsilon)^{-2}$ (Lemma 6.3), and $\sum_i(\ell(4\varepsilon)+i\log2)^{-\beta-2}\le C_\beta\ell(\varepsilon)^{-\beta-1}$. Next, on $|w|\le\varepsilon/2$, $r_q(z-w)\le r_q(\varepsilon/2)\le C\ell^{-2}/\varepsilon$ against $\int|g|\le\mathsf{M}\varepsilon\ell(\varepsilon/2)^{-\beta}$; on $\varepsilon/2<|w|\le2\varepsilon$, $|g|\le C\mathsf{M}\ell(\varepsilon)^{-\beta}$ against $\int_{|y|\le3\varepsilon}r_q=2R(3\varepsilon)\le C/\ell$. $2\varepsilon<|w|\le\delta$. On each half of a shell $\{a<|w|\le2a\}$ the difference $r_q(z-w)-r_q(w)$ has one sign, and telescoping gives mass $\le2\varepsilon\,r_q(a/2)\le C\frac{\varepsilon}{a}\ell(a)^{-2}$; summing over dyadic a against $|g|\le\mathsf{M}\ell(2a)^{-\beta}$, splitting at $a=\varepsilon^{1/2}$, gives $O(\mathsf{M}\ell(\varepsilon)^{-\beta-2}+\mathsf{M}\sqrt{\varepsilon})$. $|w|>\delta$. The mass is $\le2\varepsilon\,r_q(\delta/2)\le C\varepsilon\ell(\delta)^{-2}/\delta$, against $|g|\le2\|f\|_\infty$. The last statement follows from Lemmas 6.1(d) and 6.3. □

For $a>0$ and $0<\delta<1$ let $\phi_a(t)=\ell(t)^{-a}$, and define the left and right germs $g_a^L(y)=\phi_a(-y)\mathbf{1}_{(-\delta,0)}(y)$ and $g_a^R(y)=\phi_a(y)\mathbf{1}_{(0,\delta)}(y)$.

Lemma 6.8 (profile transfer, first order)**.** *Let $a>0$ and $\delta\le e^{-(a+3)}$ small. Then $w=r_q*g_a^L$ satisfies, as $d\downarrow0$ with $\ell=\ell(d)$,*

$$w(-d)-w(0)=\frac{2a+1}{2(a+1)}\ell^{-a-1}+O(\ell^{-a-2}),\qquad w(d)-w(0)=-\frac{1}{2(a+1)}\ell^{-a-1}+O(\ell^{-a-2}),$$

and, by reflection, the same with g_a^R and $d\mapsto-d$. Consequently a germ $(\mathsf{s}_-,\mathsf{s}_+)_a$ is mapped to $(\mathsf{s}'_-,\mathsf{s}'_+)_{a+1}$ with $\mathsf{s}'_-=\frac{(2a+1)\mathsf{s}_--\mathsf{s}_+}{2(a+1)}$ and $\mathsf{s}'_+=\frac{-\mathsf{s}_-+(2a+1)\mathsf{s}_+}{2(a+1)}$: the odd part $\mathsf{s}_--\mathsf{s}_+$ is preserved, and the even part $\mathsf{s}_-+\mathsf{s}_+$ is multiplied by $\frac{a}{a+1}$.

Proof. $w(x)=\int_0^\delta r_q(x+t)\phi(t)\,\mathrm{d}t$ with $\phi=\phi_a$, $\phi(t)=\int_0^t\phi'$ and $\phi'(\tau)=a\tau^{-1}\ell(\tau)^{-a-1}>0$. With $\mathcal{R}$ the odd extension of R, Fubini gives

$$w(x)-w(0)=\phi(\delta)\big[\mathcal{R}(x+\delta)-\mathcal{R}(\delta)\big]-\int_0^\delta\phi'(\tau)\big[\mathcal{R}(x+\tau)-\mathcal{R}(\tau)\big]\,\mathrm{d}\tau,\tag{6.2}$$

where the first term is $O(|x|)$. At $x=-d$ the integral term equals $T_1+T_2+T_3$ and at $x=+d$ it equals $T_1-T_2'-T_3'$, where $T_1=\int_0^d\phi'R$, $T_2=\int_0^d\phi'(\tau)R(d-\tau)\,\mathrm{d}\tau$, $T_2'=\int_0^d\phi'(\tau)R(\tau+d)\,\mathrm{d}\tau$, $T_3=\int_d^\delta\phi'[R(\tau)-R(\tau-d)]$ and $T_3'=\int_d^\delta\phi'[R(\tau+d)-R(\tau)]$. By (R2), $T_1=\frac{a}{2}\int_0^d\tau^{-1}\ell(\tau)^{-a-2}\,\mathrm{d}\tau+O(\ell^{-a-2})=\frac{a}{2(a+1)}\ell^{-a-1}+O(\ell^{-a-2})$. Both T_2 and T_2' equal $R(d)\phi(d)+O(\ell^{-a-2})=\frac12\ell^{-a-1}+O(\ell^{-a-2})$, since $R(d\theta)-R(d)=O(\ell^{-2})$ for $\theta\in[\frac12,2]$, $\phi(d)-\phi(d/2)=O(\ell^{-a-1})$, and the part

$\tau \in (d/2, d)$ of T_2 is at most $\frac{Ca}{d\ell^{a+1}} \int_0^{d/2} R \le C\ell^{-a-2}$. T_3, T_3' are $O(\ell^{-a-2})$: on $[d, 2d]$, $R \le C/\ell$ and $\phi(2d) - \phi(d) = O(\ell^{-a-1})$; on $[2d, \delta]$, $|R(\tau \pm d) - R(\tau)| \le d\, r_q(\tau/2) \le Cd/(\tau\ell(\tau))$, and we split at $\sqrt{d}$. □

Lemma 6.9 (odd transfer, second order)**.** *Let $f \in L^\infty$, $z \in \mathbb{R}$, $a > 0$, $\delta > 0$ small, and suppose that for $0 < y \le \delta$*

$$\mathfrak{D}_f(z; y) = \sum_{j=1}^{J} \alpha_j \big(\ell(y) + \beta_j\big)^{-a_j} + e(y), \qquad a_j \ge a, \quad |e(y)| \le \mathsf{M}\ell(y)^{-a-2}.$$

Then $u = r_q * f$ satisfies $\mathfrak{D}_u(z; d) = 2R(d) \sum_j \alpha_j(\ell(d)+\beta_j)^{-a_j} + O(\ell(d)^{-a-3})$. If only $|e| \le \mathsf{M}\ell^{-a-1}$, the error is $O(\ell^{-a-2})$.

Proof. Odd reduction. With $f^{\mathrm{o}}(y) = \frac{1}{2}(f(z-y) - f(z+y))$, odd in y, and r_q even, $u(z-x) - u(z+x) = 2(r_q * f^{\mathrm{o}})(x)$: the even part of f about z does not contribute. Odd and even parts never couple. *Splitting.* $f^{\mathrm{o}} = f_1^{\mathrm{o}} + f_2^{\mathrm{o}} + f_3^{\mathrm{o}}$ with $f_1^{\mathrm{o}} = \frac{1}{2}\operatorname{sgn}(y) \sum_j \alpha_j \Phi_j(|y|) \mathbf{1}_{|y|<\delta}$, $\Phi_j = (\ell + \beta_j)^{-a_j}$, $f_2^{\mathrm{o}} = \frac{1}{2}\operatorname{sgn}(y) e(|y|) \mathbf{1}_{|y|<\delta}$ and $f_3^{\mathrm{o}} = f^{\mathrm{o}} \mathbf{1}_{|y|\ge\delta}$, where $\delta \le e^{-(a+4)}$ and $\ell(\delta) + \beta_j \ge 1$. f_3^{o} contributes $O(d)$. f_2^{o} contributes $O(\ell^{-a-3})$ by Lemma 6.7 with $\beta = a+2$ at $x_0 = 0$ (where $r_q * f_2^{\mathrm{o}}$ vanishes). f_1^{o} contributes $\sum_j \alpha_j [w_j(-d) - w_j(d)]$ with $w_j = r_q * g^L_{\Phi_j}$, the germ $g^L_{\Phi_j}(y) = \Phi_j(-y)\mathbf{1}_{(-\delta,0)}(y)$. *Core.* By (6.2) with $\Phi = \Phi_j$, $w(-d) - w(d) = \int_0^\delta \Phi'(\tau) M_d(\tau)\, \mathrm{d}\tau + O(d)$ with $M_d(\tau) = \int_{\tau-d}^{\tau+d} r_q \ge 0$. Put $\tau = du$. (P_1, $\tau \le d$.) $M_d = R(d+\tau) + R(d-\tau)$, so this part is $2R(d)\Phi(d) + \int_0^d \Phi'[R(d+\tau) - R(d) + R(d-\tau) - R(d)]$. By (R2), $|R(d\theta) - R(d)| \le |\log\theta| \ell^{-2} + C\ell^{-3}$ for $\theta \in (0, 2]$, and $\int_0^1 (|\log(1-u)| + \log(1+u)) \frac{\mathrm{d}u}{u} < \infty$; the remainder is $O(\ell^{-a-3})$. (P_2, $d < \tau \le \ell d$.) By (R2), uniformly in this range since $\ell(d(u+1)) \ge \ell - \log(\ell+1)$, $M_d(du) = m(u) + O(\ell^{-3})$ with $0 \le m(u) = \frac{\log\frac{u+1}{u-1}}{2(\ell - \log(u+1) + b)(\ell - \log(u-1) + b)} \le \frac{2}{\ell^2} \log\frac{u+1}{u-1}$, $b = a_q$; as $\int_1^\infty \log\frac{u+1}{u-1} \frac{\mathrm{d}u}{u} < \infty$, this part is $O(\ell^{-a-3})$. (P_3, $\tau > \ell d$.) $M_d \le 2d\, r_q(\tau - d) \le Cd/(\tau\ell(\tau)^2)$; splitting at $\sqrt{d}$ gives $O(\ell^{-a-3})$. The first-order variant is the same argument with one order less. □

On the model kernel $2R = (\ell + b)^{-1}$ quadrature reproduces the third-order term $\frac{a\pi^2}{12}\ell^{-a-3}$ of $P_1 + P_2$, a check that no second-order term is missing.

Proposition 6.10 (graded moduli)**.** *For $p \in (-L, L)$ let $\omega_p(t) = \sup_{0<|w|\le t} |\psi(p+w) - \psi(p)|$. For every $N \ge 1$ there are $t_0(p, N)$ and $C(p, N)$ with*

$$\omega_p(t) \le C(p, N)\, \ell(t)^{-\min(g(p)+\frac{1}{2}, N)} \qquad (0 < t \le t_0(p, N)).$$

In particular $\omega_p(t) \le C_p \ell(t)^{-3/2}$ at every interior point, and $\omega_p(t) = O(\ell(t)^{-N})$ for every N if $p \notin C_L$.

Proof. Induction on N, for all interior points at once, using only Lemmas 6.5, 6.6 and 6.7. $N = 1$: Lemma 6.6(a′) (interior points have $g \ge 1$). $N \to N+1$: fix p with $g = g(p)$ and choose $\delta = \delta(p, N) \le e^{-1}$ with $[p - 2\delta, p + 2\delta] \subset (-L, L)$, $\delta \le t_0(p, N)$, and such that each of the finitely many images $p' = p \pm \log m$, $m \in N_L$, is either an edge $\pm L$ (with $\delta \le \rho_0$), or interior with $\delta \le t_0(p', N)$, or at distance $\ge 2\delta$ outside $[-L, L]$ (then the shift vanishes near p). By Lemma 6.5, for $|w| \le \delta$ the increment $f_1(p+w) - f_1(p)$ is bounded by: the self term $\mu_q \omega_p(|w|) \le C\ell^{-\min(g+1/2, N)}$; analytic and Lipschitz terms $\le C|w|$; interior images $c_m \omega_{p'}(|w|) \le C\ell^{-\min(g-1/2, N)}$, since $g(p') \ge g - 1$; and edge images, which occur only if $g = 1$, $\le CV(|w|) \le C\ell^{-1/2}$ (Lemma 6.6(a)). If $g = \infty$, every interior image has $g = \infty$ and there are no edge images. Lemma 6.7 with $\beta = \min(g - \frac{1}{2}, N)$ gives $\omega_p(t) \le C\ell(t)^{-\min(g+1/2, N+1)} + Ct$ for $t \le \delta/4$. □

Each step uses finitely many points, so no uniformity over C_L is needed or claimed.

7. The edge law to second order and the kink law

Theorem 7.1 (edge law, sharp and to second order)**.** *For every $q>0$, as $r\downarrow 0$, with $\ell=\ell(r)$ and $V=V_q$:*

(a) $\psi(\pm(L-r))=A_\pm V(r)+O(V^3)=A_\pm\ell^{-1/2}+O(\ell^{-3/2})$;
(b) $\psi(L-r)=A_+\big[V+\frac{\mu_q}{2}V^3\big]+O(V^5)$, *and the same at $-L$ with A_-;*
(c) $\psi(\pm(L-r))=A_\pm(\ell-D)^{-1/2}+O(\ell^{-5/2})=A_\pm\ell^{-1/2}\big(1+\frac{D}{2\ell}\big)+O(\ell^{-5/2})$, *where $D=\lambda+\log 2\pi+\gamma$; q drops out.*

The constants in (a) are uniform as in Theorem 4.1, for fixed q (its proof uses only Theorem 4.1, Lemma 6.6(a) and Lemma 6.4); those in (b), (c) depend also on δ_X^- and degenerate as $2L\downarrow\log n$, when $y_n\to -L$.

Proof. By Theorem 4.1(iv), $u=\chi\tilde\psi=\int_0^x v(x-z)k_+(z)\,\mathrm{d}z$ with $k_+(0)=A_+$, and $k_+(z)-A_+=\int\theta_z f_+=\int\theta_z(f_+-f_0)$ with $f_0:=f_+(0+)$ (Lemma 6.4). *Source near the edge.* For $0<t<\min(\rho_0/2,\log 2)$, with an $O(t)$ term that is uniform as in Theorem 4.1,

$$f_+(t)-f_0=\mu_q\tilde\psi(t)+\sum_{n\in N_L}c_n\big(\psi(y_n-t)-\psi(y_n)\big)+O(t),$$

since $S_n^*\tilde\psi(t)=\tilde\psi(t-\log n)=0$, $S_n\tilde\psi(t)=\psi(y_n-t)$ (zero if $\log n\ge 2L$), and the pole and commutator terms are Lipschitz on $[0,\rho_0/2]$ (as $\chi=1$ on $[-\rho_0,\rho_0]$, there $r_\chi[\tilde\psi](t)=\int J(t-y)(1-\chi(y))\tilde\psi(y)\,\mathrm{d}y$ with $|t-y|\ge\rho_0/2$). (a) By Theorem 4.1 and Lemma 6.6(a), which bounds every $|\psi(y_n-t)-\psi(y_n)|$ by $CV(t)$, $|f_+-f_0|\le CV$ on $(0,\min(\rho_0/2,\log 2))$, with C uniform as in Theorem 4.1; Lemma 6.4 with $j=1$ gives $|k_+(z)-A_+|\le CV(z)^2$, and then $|u(x)-A_+V(x)|\le\int_0^x v(x-z)CV(z)^2\,\mathrm{d}z\le CV(x)^3$. (b) By (a) and Proposition 6.10, $f_+-f_0=\mu_qA_+V+e$ with $|e|\le CV^3$ near 0. Lemma 6.4 and $\int\theta_zV=R(z)$ give $k_+(z)-A_+=\mu_qA_+R(z)+O(V(z)^4)$, so $u(x)=A_+V(x)+\mu_qA_+\int_0^x v(x-z)R(z)\,\mathrm{d}z+O(V^5)$. Now $\int_0^x v(x-z)R(z)\,\mathrm{d}z=V(x)R(x)-\int_0^x v(x-z)(R(x)-R(z))\,\mathrm{d}z$, and the last integral is $O(\ell^{-7/2})$: on $(x/2,x)$, $0\le R(x)-R(z)\le(x-z)r_q(x/2)\le C(x-z)\ell^{-2}/x$ and $\int_0^{x/2}yv\le Cx\ell^{-3/2}$; on $(0,x/2)$, $v(x-z)\le C\ell^{-3/2}/x$ and $\int_0^x(R(x)-R(z))\,\mathrm{d}z=\int_0^x yr_q\le Cx\ell^{-2}$ (Lemmas 6.1(e), 6.3). With (R1), $VR=\frac12V^3+O(\ell^{-7/2})$, which gives (b). (c) If $A_+\neq 0$, then by (b) and (V2), $A_+^2\psi(L-r)^{-2}=V^{-2}-\mu_q+O(\ell^{-1})=\ell+a_q-\mu_q+O(\ell^{-1})$, and $a_q-\mu_q=c_a-\gamma+c-\lambda=-\log 2\pi-\gamma-\lambda=-D$. So $\psi(L-r)=A_+(\ell-D)^{-1/2}(1+O(\ell^{-2}))$. If $A_+=0$, (b) gives $\psi(L-r)=O(V^5)=O(\ell^{-5/2})$. □

Part (c) can also be read off from $\psi(L-x)=A_+[V+\mu_qVR](x)+O(\ell^{-5/2})$ by inserting (R2) and (V2), since $\mu_q-a_q=D$. The argument is not circular: Proposition 6.10 does not use Theorem 7.1. The offset D appeared earlier, without proof, in a heuristic fit [69].

Remark 7.2 (the pure logarithm). For the logarithmic Laplacian the same mechanism would give $A^2u(L-r)^{-2}=\ell-\lambda-\gamma+o(1)$ for $\mathcal{Q}=\frac12\mathcal{E}_{\log}$, which is the second-order conjecture of the companion paper [9, Conj. 9.2]. The transfer of Proposition 6.2 and of the source expansion to the variance-gamma symbol with its nonlocal bounded part has not been written out, and we do not claim it.

Theorem 7.3 (kink law)**.** *Let $z\in(-L,L)$ have generation $g(z)=k\ge 1$, and $\mathsf{W}_\pm=\mathsf{W}_\pm(z)$. As $d\downarrow 0$, $\ell=\ell(d)$,*

$$\psi(z-d)-\psi(z)=\frac{(k+1)\mathsf{W}_+-k\mathsf{W}_-}{2k+1}\,\ell^{-k-1/2}+O(\ell^{-k-3/2}),$$
$$\psi(z+d)-\psi(z)=\frac{-k\mathsf{W}_++(k+1)\mathsf{W}_-}{2k+1}\,\ell^{-k-1/2}+O(\ell^{-k-3/2}).$$

Equivalently $\mathfrak{D}_\psi(z;d) = \Omega_k(z)\ell^{-k-1/2} + O(\ell^{-k-3/2})$ *(unit odd amplitude in every generation) and* $\mathfrak{S}_\psi(z;d) = \frac{\Sigma_k(z)}{2(2k+1)}\ell^{-k-1/2} + O(\ell^{-k-3/2})$.

For $k=1$ and generic L ($\mathsf{W}_+ = c_nA_+$, $\mathsf{W}_- = 0$ at y_n) this reads, with error $O(\ell^{-5/2})$,

$$\psi(y_n - d) - \psi(y_n) = \tfrac{2}{3}c_nA_+\ell(d)^{-3/2}, \qquad \psi(y_n + d) - \psi(y_n) = -\tfrac{1}{3}c_nA_+\ell(d)^{-3/2}:$$

the jump is $c_nA_+\ell^{-3/2}$, with the sign of A_+, the larger excursion is on the source side $y < y_n$, and the side ratio tends to -2. The profile is two-sided, not a smooth function plus a one-sided indicator; the one-sided form with unit amplitude is exact for the odd part. If $e^{2L} = nm$, then $\mathsf{W}_-(y_n) = c_mA_-$. At mirror points $-y_n$ the roles of the sides and of $A_\pm$ are exchanged. Examples of generation 2: $L - \log 6$ for $L > \frac{1}{2}\log 6$ (chains $2 \to 3$ and $3 \to 2$, $\mathsf{W}_+ = 2c_2c_3A_+ = 0.6218A_+$) and $L - \log\frac{3}{2}$. The generation-2 split is $\frac{3}{5} : -\frac{2}{5}$, with cusp $\frac{1}{5}$ and side ratio $-\frac{3}{2}$.

Proof. Induction on k, with the edges as generation 0: by Theorem 7.1(a), $\psi \sim (A_+, 0)_{1/2}$ at L and $\psi \sim (0, A_-)_{1/2}$ at $-L$. *Step 1 (source).* Localise at z by Lemma 6.5 and classify the images $P = z + \varsigma\log n$, $n \in N_L$, with δ chosen as in the proof of Proposition 6.10. If $|P| > L$ the term vanishes near z. If $g(P) \geq k$, its modulus is $O(\ell^{-k-1/2})$ (Proposition 6.10). If $g(P) = k-1$ (including $P = \pm L$ when $k = 1$), then $\psi \sim (\mathsf{s}_-(P), \mathsf{s}_+(P))_{k-1/2}$ at P by induction; a translation preserves orientation. The self term has modulus $O(\ell^{-k-1/2})$. With $a = k - \frac{1}{2}$ and $\mathsf{S}_\mp := \sum_{g(P)=k-1} c_n\mathsf{s}_\mp(P)$,

$$f_1 = f_1(z) + \mathsf{S}_-g_a^L(\cdot - z) + \mathsf{S}_+g_a^R(\cdot - z) + f_2, \qquad f_2 \in L^\infty, \quad |f_2(z+y) - f_2(z)| \leq \mathsf{M}\ell(|y|)^{-a-1}\ (|y| \leq \delta).$$

Step 2 (inversion). $\chi_1\psi = r_q * f_1$. By Lemma 6.7 with $\beta = a + 1$, f_2 contributes $O(\ell^{-k-3/2})$; the constant contributes nothing; Lemma 6.8 gives $\mathsf{s}'_- = \frac{2k\mathsf{S}_- - \mathsf{S}_+}{2k+1}$, $\mathsf{s}'_+ = \frac{-\mathsf{S}_- + 2k\mathsf{S}_+}{2k+1}$. *Step 3 (algebra).* For $k = 1$, $\mathsf{S}_- = \mathsf{W}_+$ and $\mathsf{S}_+ = \mathsf{W}_-$ by (6.1). For $k \geq 2$ insert $\mathsf{s}_-(P) = \frac{k\mathsf{W}_+(P) - (k-1)\mathsf{W}_-(P)}{2k-1}$ and $\mathsf{s}_+(P) = \frac{-(k-1)\mathsf{W}_+(P) + k\mathsf{W}_-(P)}{2k-1}$; then $2k\mathsf{s}_- - \mathsf{s}_+ = (k+1)\mathsf{W}_+(P) - k\mathsf{W}_-(P)$ and $-\mathsf{s}_- + 2k\mathsf{s}_+ = -k\mathsf{W}_+(P) + (k+1)\mathsf{W}_-(P)$, and (6.1) gives the claim. □

The constants have a simple origin: the odd part is preserved exactly at each passage (Lemma 6.8, since $2R\ell \to 1$), while the even part starts at $\frac{1}{2}$ of the edge germ and is multiplied by $\frac{j-1/2}{j+1/2}$ at step j, and $\prod_{j=1}^k \frac{j-1/2}{j+1/2} = \frac{1}{2k+1}$.

Corollary 7.4 (the singular set)**.** (a) *If* $g(z) < \infty$ *and* $(\mathsf{W}_+(z), \mathsf{W}_-(z)) \neq (0,0)$*, then* $\omega_z(t) \asymp \ell(t)^{-g(z)-1/2}$*, and* ψ *is not differentiable at* z*.*

(b) *If* $A_+ \neq 0$*, every point of* C_L *is as in (a).*

(c) *If* $L > \frac{1}{2}\log 6$*,* C_L *is dense in* $(-L, L)$*. Hence an eigenfunction with* $A_+ \neq 0$ *is Hölder continuous of no positive order on any open subinterval of* $(-L, L)$*; in particular it is Lipschitz on none.*

(d) *If* $2L \leq \log 2$*, then* $C_L = \varnothing$*.*

(e) $A \neq 0$ *is known for the ground state when* $s \leq 0$ *(every* L*); for the odd ground state when* $s \geq 0$ *and* $2L \leq \log 2$ *(where (c) is empty); and, given* [10]*, for the ground state when* $s = 1$ *off a closed nowhere dense set of* $L \in (\frac{1}{2}\log 6, 1.19)$*. For the true even ground state at a given* L *it is open (Conjecture 11.4).*

Proof. (a) The matrix $\left(\begin{smallmatrix} k+1 & -k \\ -k & k+1 \end{smallmatrix}\right)$ has determinant $2k+1 \neq 0$, and $\ell(d)^{-k-1/2}/d \to \infty$. (b) All $c_n > 0$, so $\mathsf{W}_+(z) = A_+w_+$, $\mathsf{W}_-(z) = A_-w_-$ with chain-weight sums $w_\pm \geq 0$, not both zero, and $A_- = \pm A_+ \neq 0$ (Theorem 5.4(a)). (c) For $\xi \in (-L + \log 6, L]$ let $\mathcal{W}_\xi = [\xi - \log 6, \xi) \subset (-L, L)$ and $T_\xi(x) = x + \log 2$ if $x + \log 2 < \xi$, $T_\xi(x) = x - \log 3$ otherwise. T_ξ is the rotation by $\log 2$ of $\mathcal{W}_\xi \cong \mathbb{R}/\log 6\,\mathbb{Z}$, and each step is a legal chain step that stays in $\mathcal{W}_\xi$ ($2, 3 \in N_L$ since $\log 3 < \log 6 < 2L$). As $\log 2/\log 6 \notin \mathbb{Q}$, every orbit is dense, so if $\mathcal{W}_\xi$ contains a point of C_L, then C_L is dense in $\mathcal{W}_\xi$. $\mathcal{W}_L$ contains y_2. If $|\xi - \xi'| < \log 6$, $\mathcal{W}_\xi \cap \mathcal{W}_{\xi'}$ has interior, so density passes from ξ to ξ'; sliding ξ

down to $-L+\log 6$ covers $(-L,L)$. By (b), every point of C_L is singular of exact order $\ell^{-g-1/2}$, and $\ell(t)^{-m}\gg t^{\alpha}$ for every $\alpha>0$. (d) $N_L=\varnothing$. (e) Proposition 5.8. □

At the edge itself, the failure of Hölder continuity is already noted for the half-line eigenfunction in [54, Ex. 6.4], and it follows here from Theorem 4.1; the content of Corollary 7.4(c) is the interior statement, through the density of C_L.

Theorem 7.5 (second-order jump)**.** *Let $g(z)=k\ge 1$. Then*

$$\mathfrak{D}_\psi(z;d)=\Omega_k(z)\,(\ell-D)^{-k-1/2}+\Omega_{k+1}(z)\,\ell^{-k-3/2}+O(\ell^{-k-5/2}),$$

where $\Omega_{k+1}(z)$ counts all chains of length $k+1$ ending at z, including non-minimal ones and ones with steps of both signs. If $\Omega_k\neq 0$ this is $\Omega_k(\ell+\beta_{\mathrm{eff}})^{-k-1/2}(1+O(\ell^{-2}))$ with $\beta_{\mathrm{eff}}=-D-\Omega_{k+1}/((k+\frac12)\Omega_k)$.

Proof. Induction on k, with $a=k-\frac12$. Base: Theorem 7.1(c) reads $\mathfrak{D}_\psi(\pm L;d)=\Omega_0(\pm L)(\ell-D)^{-1/2}+0\cdot\ell^{-3/2}+O(\ell^{-5/2})$, since $\mathfrak{D}_\psi(L;d)=\psi(L-d)$ and $\mathfrak{D}_\psi(-L;d)=-\psi(-L+d)$. *Step 1 (odd part of the source).* By Lemma 6.5, $\mathfrak{D}_{f_1}(z;y)=\mu_q\mathfrak{D}_\psi(z;y)+\sum c_n\mathfrak{D}_\psi(P;y)+O(y)$ over the images $P=z+\varsigma\log n$. Images with $g(P)=k-1$ (or $P=\pm L$ when $k=1$) carry $\Omega_{k-1}(P)(\ell-D)^{-a}+\Omega_k(P)\ell^{-a-1}+O(\ell^{-a-2})$ by induction; images with $g(P)=k$ carry $\Omega_k(P)\ell^{-a-1}+O(\ell^{-a-2})$ (Theorem 7.3); images with $g(P)\ge k+1$ carry $O(\ell^{-a-2})$ (Proposition 6.10); and the self term is $\mu_q\Omega_k(z)\ell^{-a-1}+O(\ell^{-a-2})$. By (6.1),

$$\mathfrak{D}_{f_1}(z;y)=\Omega_k(\ell-D)^{-a}+[\Omega_{k+1}+\mu_q\Omega_k]\ell^{-a-1}+e(y),\qquad |e|\le \mathsf{M}\ell^{-a-2}.$$

Step 2 (inversion). Lemma 6.9 with $2R=(\ell+a_q)^{-1}+O(\ell^{-3})$ gives $\mathfrak{D}_\psi=\Omega_k(\ell+a_q)^{-1}(\ell-D)^{-a}+[\Omega_{k+1}+\mu_q\Omega_k]\ell^{-a-2}+O(\ell^{-a-3})$. *Step 3 (exact cancellation).* $(\ell+a_q)^{-1}(\ell-D)^{-a}=(\ell-D)^{-a-1}-(a_q+D)\ell^{-a-2}+O(\ell^{-a-3})$, and $a_q+D=q+c_a-\gamma+\lambda+\log 2\pi+\gamma=\mu_q$. The μ_q terms cancel and q drops out. □

The offset rule behind Step 3 is $\beta'=\frac{a\beta+a_q-\mu_q}{a+1}$ (germ offset with weight a, Green-function offset with weight 1, self-feedback $-\mu_q/(a+1)$); $\beta=-D$ is its fixed point in every generation. For $k\ge 2$ the statement rests only on the inductive bookkeeping above.

Corollary 7.6 (jump ratio and prime-power boosts)**.** *Let $n\in N_L$ with $\mathsf{W}_-(y_n)=0$ and $A_+\neq 0$, and let $\rho_n(d)=\mathfrak{D}_\psi(y_n;d)/(c_n\psi(L-d))$ (exact point values).*

(a) $1/\rho_n(d)=\ell-D-\beta_n+O(\ell^{-1})$ *with* $\beta_n=\Omega_2(y_n)/(c_nA_+)$.

(b) *Let $n=p^j$. If no chain of length ≤ 2 from $-L$ ends at y_n (true for generic L), then*

$$\beta_{p^j}=\log p\Big[(j-1)+\sum_{i\ge1,\ p^{j+i}<e^{2L}}p^{-i}\Big]\ \ge\ (j-1)\log p.$$

(c) *If the samples are taken at distances $\theta_n h$ (kink) and $\theta_e h$ (edge), then* $1/\rho=\ln\frac1h-D-\beta_n+\frac12(3\log\frac{1}{\theta_n}-\log\frac{1}{\theta_e})+O(1/\ln\frac1h)$.

(d) *If $g(L-\log 6)=2$, the jump ratio at $L-\log 6$ to $c_2c_3\psi(L-d)$ is* $\frac{2c_2c_3A_+-\mathsf{W}_-^{(2)}(L-\log 6)}{c_2c_3A_+}\ell^{-2}+O(\ell^{-3})$.

Proof. (a) Theorem 7.5 with $k=1$ and $\Omega_1(y_n)=c_nA_+$, divided by $c_n\psi(L-d)=c_nA_+(\ell-D)^{-1/2}(1+O(\ell^{-2}))$ (Theorem 7.1(c)). (b) A chain of length 2 from $+L$ to y_n starts with a step to $L-\log n_1$ (a first step $+\log n_1$ leaves the window) and then satisfies $\log n=\log n_1\pm\log n_2$. Products and ratios of prime powers are prime powers only within one prime. So the chains are $L\to L-\log p^i\to y_n$ for $1\le i<j$, of weight $c_{p^i}c_{p^{j-i}}=\log p\cdot c_{p^j}$, and $L\to L-\log p^{j+i}\to y_n$ (second step $+\log p^i$), present iff $p^{j+i}\in N_L$, of weight $c_{p^{j+i}}c_{p^i}=\log p\cdot c_{p^j}p^{-i}$. Under the hypothesis, $\Omega_2(y_n)=\mathsf{W}_+^{(2)}(y_n)$. (c) $\ell(\theta h)=\ln\frac1h+\log\frac1\theta$; expand. (d) Theorem 7.3 with $k=2$: the only chains of length 2 from $+L$ to $L-\log 6$ are $(2,3)$ and $(3,2)$. □

TABLE 1. Prime-power boosts β_n of Corollary 7.6(b) ("—": $n \notin N_L$). At $L = 0.6$, $X = e^{1.2} = 3.32$ and $4 \notin N_L$, so all boosts vanish. The boosts are defined for every L; the row $L = 1.5$ is outside the range used for $s = 1$. At $L = 1.19$ only $n = 5, 7$ have $\beta_n = 0$.

L	β_2	β_3	β_4	β_5	β_7	β_8	β_9	β_{16}
0.6	0	0	—	—	—	—	—	—
0.8	0.347	0	0.693	—	—	—	—	—
1.0	0.347	0	0.693	0	0	—	—	—
1.19	0.520	0.366	1.040	0	0	1.386	1.099	—
1.5	0.607	0.366	1.213	0	0	1.733	1.099	2.079

Remark 7.7. (1) In (d) the ratio equals $2\ell^{-2}$ only if no chain of length ≤ 2 from $-L$ ends at $L - \log 6$. This fails, for instance, at $L = \frac{1}{2}\log 7.5$ ($-L \to -L + \log 5 \to -L + \log 5 - \log 4$) and at $L = \log 3$, both in $(0, 1.19]$; at $L = \log 6$ the point is 0, where the odd part vanishes for even states. The hypothesis of (b) is needed for the same reason. (2) For the true form at $L = 1.19$, Corollary 7.6(a) reads $1/\rho_n = \ell - 2.415 - \lambda - \beta_n + O(1/\ell)$, *if* $A_e \neq 0$; since A_e is tiny there (Section 1.5), this is asymptotic only. (3) Jumps read off Galerkin endpoint values (for instance with discontinuous piecewise polynomials) are not point samples; for them a law $1/(\ln \frac{1}{h} + b)$ is only a heuristic.

Conjecture 7.8 (second-order even part)**.** Let $g(z) = k \ge 1$. Then

$$\mathfrak{S}_\psi(z; d) = \frac{\Sigma_k(z)}{2(2k+1)}(\ell - D)^{-k-1/2} + X_{k+1}(z)\,\ell^{-k-3/2} + O(\ell^{-k-5/2}),$$

$$X_{k+1}(z) = \frac{2k+1}{2k+3}\Big[\sum_{g(P)=k-1} c_n X_k(P) + \frac{1}{2(2k+1)}\sum_{g(P)=k} c_n \Sigma_k(P)\Big], \qquad X_1(\pm L) = 0,$$

with sums over the predecessors $P = z + \varsigma \log n$. The offset is the same D as for the odd part.

Outline and evidence; not a proof. Since r_q is even, $\mathfrak{S}_{r_q * f}(x) = \int[\frac{1}{2}(r_q(x-y) + r_q(x+y)) - r_q(y)]\mathfrak{S}_f(y)\,\mathrm{d}y$, so even parts never couple to odd parts. For $\mathfrak{S}_f = \sigma(\ell + \beta_0)^{-a} + O(\ell^{-a-2})$ one expects $\mathfrak{S}_{r_q * f}(d) = \frac{a}{a+1}\sigma\big(\ell + \frac{a_q + (a+1)\beta_0}{a+2}\big)^{-a-1} + O(\ell^{-a-3})$ from the even part $T_1 + \frac{1}{2}(T_2 - T_2') + \frac{1}{2}(T_3 - T_3')$ of (6.2); inserted into the source as in Theorem 7.5, this gives the conjecture with the same offset. The even offset rule differs from the odd one, and a written proof of the even transfer with all ranges is missing. On a model kernel the even residual behaves as $-\frac{a\pi^2}{24}\ell^{-a-3}$ for five parameter sets, and an independent piecewise-constant Galerkin code gives, for 15 interior cases with $\ell \in [20, 85]$ and no free parameter, agreement with the predicted offset to within 0.015.

8. The zero side I: the contact identity and the first-order tail law

8.1. **Zero-side objects.** Let ρ run over the nontrivial zeros of ζ, counted with multiplicity, and put $\gamma_\rho = -i(\rho - \frac{1}{2}) = t_\rho + iy_\rho$, so that $t_\rho = \operatorname{Im}\rho$ and $y_\rho = \frac{1}{2} - \operatorname{Re}\rho \in (-\frac{1}{2}, \frac{1}{2})$. *No hypothesis on the zeros is made* unless a statement says so; zeros off the critical line are allowed throughout. For an eigenfunction ψ with transform F put

$$h(w) = F(w)F(-w), \qquad P_\psi = F(\tfrac{i}{2})F(-\tfrac{i}{2}),$$

$$Z(T) = \sum_{|t_\rho| > T} h(\gamma_\rho), \qquad Z_{\cos}(z) = \sum_\rho h(\gamma_\rho)\big(1 - \cos z\gamma_\rho\big).$$

For real t, $h(t) = |F(t)|^2$; under RH $h(\gamma_\rho) = |F(t_\rho)|^2$, while off the line $h(\gamma_\rho)$ is complex. For an even finite signed measure ν on $[-1, 1]$ with $\nu(\mathbb{R}) = 1$ let $\hat\nu(w) = \int \cos(wz)\,\mathrm{d}\nu(z)$, $m_1(\nu) = \int_{(0,\infty)} z\,\mathrm{d}\nu(z)$, and $Z_\nu(T) = \sum_\rho h(\gamma_\rho)(1 - \hat\nu(\gamma_\rho/T))$. These sums are real: the zeros are invariant under $\rho \mapsto \bar\rho$ and

$\rho \mapsto 1-\rho$, $\gamma_{\bar\rho} = -\overline{\gamma_\rho}$, and $F(-\bar w) = \overline{F(w)}$. Let $N(t) = \#\{\rho : 0 < t_\rho \le t\}$. By the Riemann–von Mangoldt formula [75, Ch. IX], $N(t) = \vartheta(t)/\pi + 1 + S_N(t)$ with $S_N(t) = O(\log t)$ and $\vartheta' = a/2$; in particular $N(t+1) - N(t-1) \le C\log t$.

8.2. **The explicit formula on the form domain.**

Lemma 8.1. *Let $L' > 0$, $s \in \mathbb{R}$, and let $f \in \mathcal{D}(\mathcal{E})$ be real and vanish outside $[-L', L']$. Then, with absolute convergence,*

$$\mathcal{E}(f) + w_s(f) = \sum_\rho F(\gamma_\rho)F(-\gamma_\rho) - 2(1-s)F(\tfrac{i}{2})F(-\tfrac{i}{2}),$$

$$\sum_\rho \big(|F(\gamma_\rho)|^2 + |F(-\gamma_\rho)|^2\big) \le C_{L'}\big(\mathcal{E}(f) + \|f\|^2\big).$$

In particular, for an eigenfunction, $\sum_\rho h(\gamma_\rho) = \lambda + 2(1-s)P_\psi$, which equals λ when $s = 1$.

Proof. (1) For $f \in C_c^\infty(-L', L')$ and $s = 1$ this is Weil's explicit formula [79] (see also [48, §5.5]): for even $g \in C_c^\infty(\mathbb{R})$ with $\hat g(w) = \int g e^{ixw}\,\mathrm{d}x$,

$$\sum_\rho \hat g(\gamma_\rho) = \hat g(\tfrac{i}{2}) + \hat g(-\tfrac{i}{2}) - 2\sum_{n\ge 2} c_n g(\log n) + \frac{1}{2\pi}\int_{\mathbb{R}} \hat g(t)\Big(\operatorname{Re}\psi_\Gamma\big(\tfrac14 + \tfrac{it}{2}\big) - \log\pi\Big)\,\mathrm{d}t,$$

applied to $g = f * \tilde f$, $\hat g(w) = F(w)F(-w)$; by Lemma 2.2 the right side is $Q^{(1)}_{L'}(f)$. This normalisation was checked numerically in [10] and against 10 500 zeros computed in Arb (Section 10). The pole is the only s-dependent term. (2) $|F|^2$ is subharmonic, so $|F(w)|^2 \le \frac{1}{\pi}\int_{|w'-w|<1} |F|^2\,\mathrm{d}A$; at most $C\log(|n|+2)$ zeros have $\operatorname{Re}\gamma_\rho \in [n, n+1]$; hence $\sum_\rho |F(\gamma_\rho)|^2 \le C\int_{|u|\le 3/2}\int \log(2+|t|)|F(t+iu)|^2\,\mathrm{d}t\,\mathrm{d}u$. Here $F(\cdot + iu)$ is the transform of $e^{-ux}f$, $\log(2+|t|) \le C(1+\Psi_a)$, and Lemma 2.6(a) with $\eta = \chi_{L'}e^{-ux}$ ($\chi_{L'} = 1$ on $[-L', L']$) gives the bound; $-\gamma_\rho$ is also of the form $\gamma_{\rho'}$. (3) Both sides are continuous on $\mathcal{D}_{(-L',L')}$ for the form norm ($|F(\pm\frac{i}{2})| \le e^{L'/2}\sqrt{2L'}\|f\|$), and $C_c^\infty(-L', L')$ is dense (Lemma 2.4(b)). □

Lemma 8.2 (exterior pairing)**.** *Let $I = (-L, L)$ and let $\phi \in \mathcal{D}(\mathcal{E}) \cap L^\infty$ vanish outside a bounded set. Then $(\mathcal{E} + w_s)(\psi, \phi) = \lambda\int_I \psi\phi - \int_{\mathbb{R}\setminus I}(J\psi)\phi + \int_{\mathbb{R}\setminus I}(W_s\psi)\phi$, all integrals absolutely convergent.*

Proof. $\mathbf{1}_I\phi \in \mathcal{D}_L$ (Lemma 2.5), and Lemma 2.3(d) applies to ψ and $\mathbf{1}_{\mathbb{R}\setminus I}\phi$. □

Theorem 8.3 (exact contact identity)**.** *For every s, every eigenfunction ψ and every $z \in \mathbb{R}$, put*

$$\mathcal{L}(z) := \int_{\mathbb{R}\setminus I} (J\psi - W_s\psi)(y)\,\psi(y-z)\,\mathrm{d}y,$$

so that $\mathcal{L}(z) = \int_0^z \mathcal{G}(\eta)\tilde\psi(z-\eta)\,\mathrm{d}\eta$ for $z > 0$, with $\mathcal{G}(\eta) := (J\psi - W_s\psi)(L+\eta)$. Then:

(i) *$(\mathcal{E} + w_s)(\psi - \tau_z\psi) = 2\lambda(1 - \langle\psi, \tau_z\psi\rangle) + 2\mathcal{L}(z)$, and $\mathcal{L}(-z) = \mathcal{L}(z)$;*

(ii) *with no hypothesis on the zeros,*

$$Z_{\cos}(z) = \lambda\big(1 - \langle\psi, \tau_z\psi\rangle\big) + \mathcal{L}(z) - 4(1-s)\sinh^2\tfrac{z}{4}\,P_\psi;$$

the last term is $O(z^2)$ and vanishes for $s = 1$;

(iii) *for ν as above, $\sum_\rho h(\gamma_\rho)(1 - \hat\nu(\gamma_\rho)) = \int[\lambda(1 - \langle\psi, \tau_z\psi\rangle) + \mathcal{L}(z) - 4(1-s)\sinh^2\tfrac{z}{4}P_\psi]\,\mathrm{d}\nu(z)$.*

Proof. (i) By translation invariance $(\mathcal{E}+w_s)(\tau_z\psi) = \lambda$, so $(\mathcal{E}+w_s)(\psi-\tau_z\psi) = 2\lambda - 2(\mathcal{E}+w_s)(\psi, \tau_z\psi)$, and Lemma 8.2 with $\phi = \tau_z\psi$ gives $(\mathcal{E}+w_s)(\psi, \tau_z\psi) = \lambda\langle\psi, \tau_z\psi\rangle - \mathcal{L}(z)$. For $z > 0$, $\psi(y-z) \ne 0$ on $\mathbb{R}\setminus I$ only for $y \in (L, L+z)$. Evenness: $(\mathcal{E}+w_s)(\psi - \tau_{-z}\psi) = (\mathcal{E}+w_s)(\tau_z\psi - \psi)$ and $\langle\psi, \tau_{-z}\psi\rangle = \langle\psi, \tau_z\psi\rangle$. (ii) Apply Lemma 8.1 to $f = \psi - \tau_z\psi$, $L' = L + |z|$: $\hat f(w) = F(w)(1 - e^{izw})$, so $\hat f(w)\hat f(-w) = 2h(w)(1-\cos zw)$ for complex w, and at $w = \frac{i}{2}$, $(1-e^{-z/2})(1-e^{z/2}) = -4\sinh^2\frac{z}{4}$.

Hence $(\mathcal{E}+w_s)(f) = 2Z_{\cos}(z) + 8(1-s)\sinh^2\frac{z}{4}P_\psi$; compare with (i). (iii) Integrate (ii); Fubini applies since $|1-\cos z\gamma_\rho| \le C$ for $|z|\le 1$, $|\operatorname{Im}\gamma_\rho| < \frac12$, and $\sum|h(\gamma_\rho)|<\infty$. □

Remark 8.4. (1) Let $f_z = \psi - \tau_z\psi$ and $Q^{(s)}(f_z) = (\mathcal{E}+w_s)(f_z)$, the form energy of the translation. Since $\|f_z\|^2 = 2(1-\langle\psi,\tau_z\psi\rangle)$, part (i) says that the reduced energy is $Q^{(s)}(f_z)-\lambda\|f_z\|^2 = 2\mathcal{L}(z)$, and part (ii) that the zero sum is half the form energy up to the pole term: $Z_{\cos}(z) = \frac12 Q^{(s)}(f_z) - 4(1-s)\sinh^2\frac{z}{4}P_\psi$. Theorem 9.1 below gives $2\mathcal{L}(z) \simeq (A_+^2+A_-^2)z = |\lambda'|z$, the energy gained by enlarging the window by z (Theorem 5.6); as $\lambda\|f_z\|^2 = O(z/\ell(z))$ (Theorem 9.2(a)), $Q^{(s)}(f_z)\simeq 2A^2z$ and $Z_{\cos}(z)\simeq A^2z$ (Theorem 9.2(c)). The tail law is this statement read on the zero side; this is the mechanism of Tan's adiabatic relation [73, 11]. A dilation (Pohozaev) argument would need ψ', which is not available (Corollary 7.4); translations need only ψ. (2) The primes enter only through $W_s\psi$ in $\mathcal{L}$, near $y=L$.

8.3. **The Fourier edge law.** Let $t_0 = t_0(\lambda)$ be such that $a(t)-\lambda \ge 1$ for $|t|\ge t_0$, and put, for $|t|\ge t_0$ and $z = t+iy$,

$$\Phi_c(t) = \frac{1}{it\sqrt{a(t)-\lambda}}, \qquad M_c(t,y) = A_+e^{iLz}\Phi_c(t) + A_-e^{-iLz}\Phi_c(-t), \qquad R_c(t,y) = F(z) - M_c(t,y).$$

Then $\Phi_c(-t) = \overline{\Phi_c(t)}$, $\Phi_c(t)^2 = -\frac{1}{t^2(a-\lambda)}$ is real, and $t^2(a-\lambda)|\Phi_c|^2 = 1$. For a zero we write $M_c(\pm\gamma_\rho) = M_c(\pm t_\rho, \pm y_\rho)$.

Theorem 8.5 (Fourier edge law)**.** *There are $T_0\ge 2$ and B, depending only on L, s, Λ_0, such that for every eigenfunction with $|\lambda|\le\Lambda_0$ and $T\ge T_0$:*

(a) $\displaystyle\sup_{|y|\le 1/2}\int_{T\le|t|\le 2T}\big(|R_c|^2+|\partial_tR_c|^2+|\partial_yR_c|^2+|\partial_t\partial_yR_c|^2\big)(t,y)\,\mathrm{d}t \le \frac{\mathsf{B}^2}{T\log^3 T}$;

(b) *at each zero, exactly,* $M_c(\gamma_\rho)M_c(-\gamma_\rho) = (A_+^2+A_-^2)|\Phi_c(t_\rho)|^2 + A_+A_-\big[X^{\rho-\frac12}\Phi_c(t_\rho)^2 + X^{\frac12-\rho}\Phi_c(-t_\rho)^2\big]$;

(c) $\displaystyle\frac1T\int_T^{2T} t^2(a(t)-\lambda)|F(t)|^2\,\mathrm{d}t = A_+^2+A_-^2 + O\Big(\frac{|A|\mathsf{B}}{\log T} + \frac{\mathsf{B}^2}{\log^2 T} + \frac{A^2}{T}\Big)$.

The proof is in Appendix B. It rests on an exact formula for the Fourier transform of $\chi\tilde\psi$ obtained from the half-line Wiener–Hopf equation, with the $\ell^{-1/2}$ edge germ integrated exactly and an error of relative order $\ell_t^{-3/2}$ (Lemma B.2). B is not computed and does not shrink with A. Part (b) uses $e^{2iL\gamma_\rho} = X^{\rho-\frac12}$: in the diagonal terms the factors $e^{\pm iL\gamma_\rho}$ cancel, so zeros off the line acquire no factor $e^{2L|y_\rho|}$.

Lemma 8.6 (zero sums)**.** *Let $T\ge 2$ and let $Y(t,y)$ be C^1 in each variable with $\partial_t\partial_yY$ continuous on $[T/2,4T]\times[-\frac12,\frac12]$. Then*

$$\sum_{T<t_\rho\le 2T}|Y(t_\rho,y_\rho)|^2 \le C\log T\int_{-1/2}^{1/2}\int_{T/2}^{4T}\big(|Y|^2+|Y_t|^2+|Y_y|^2+|Y_{ty}|^2\big)\,\mathrm{d}t\,\mathrm{d}y.$$

Proof. In one variable, $|g(s_0)|^2\le 2\int_I(|g|^2+|g'|^2)$ for an interval $I\ni s_0$ of length 1. Apply this in y on $[-\frac12,\frac12]$ and then in t on $[t_\rho-\frac12,t_\rho+\frac12]$, to Y and to Y_y: $|Y(t_\rho,y_\rho)|^2\le 4\iint_{|t-t_\rho|\le1/2}(\cdots)$. For $T<t_\rho\le 2T$ these intervals lie in $[T-1,2T+1]\subset[T/2,4T]$, and a point t lies within $\frac12$ of at most $N(t+1)-N(t-1)\le C\log(2+|t|)\le C\log T$ zeros. □

8.4. **The first-order tail law.** Put $Z_{MM}(T) = \sum_{|t_\rho|>T}M_c(\gamma_\rho)M_c(-\gamma_\rho)$ and $Z_\times = Z - Z_{MM}$ (for $T\ge T_0\ge t_0$).

Proposition 8.7 (edge-times-edge part)**.** *For $T \ge 2T_0$,*

$$\pi T\, Z_{MM}(T) = 2A^2\, T \int_T^\infty \frac{a(t)+\varepsilon c_X}{t^2(a(t)-\lambda)}\,\mathrm{d}t + O_X\Big(\frac{A^2\log T}{T}\Big), \qquad |\pi T\, Z_\times(T)| \le C\frac{|A|\mathsf{B}}{\log T} + C\frac{\mathsf{B}^2}{\log^2 T}.$$

Proof. Diagonal. By Theorem 5.4(a) and conjugation symmetry it is $2A^2\cdot 2\sum_{t_\rho>T} f(t_\rho)$ with $f = |\Phi_c|^2 = \frac{1}{t^2(a-\lambda)}$, $|f'| \le C/(t^3\ell_t)$. Stieltjes integration against $\mathrm{d}N = \frac{a}{2\pi}\,\mathrm{d}t + \mathrm{d}S_N$ gives $2\sum_{t_\rho>T} f(t_\rho) = \frac{1}{\pi}\int_T^\infty af\,\mathrm{d}t + O(T^{-2})$, the error being $-2f(T)S_N(T) - 2\int_T^\infty f'S_N$. *Oscillating part.* The map $\rho \mapsto 1-\bar\rho$ is a bijection of $\{t_\rho > T\}$ (with multiplicity) that preserves t_ρ, and $X^{\frac12-\rho}\Phi_c(-t)^2 = \overline{X^{\rho'-\frac12}\Phi_c(t)^2}$ with $\rho' = 1-\bar\rho$; the zeros with $t_\rho < -T$ are the conjugates. So the oscillating part of Z_{MM} is $4A_+A_-\operatorname{Re}\mathcal{S}$ with $\mathcal{S} = \sum_{t_\rho>T} X^{\rho-\frac12}\Phi_c(t_\rho)^2$. Landau's formula [56] gives, for fixed $X>1$, $\sum_{0<t_\rho\le t} X^{\rho-\frac12} = -\frac{c_X}{2\pi}t + \mathcal{E}_X(t)$ with $\mathcal{E}_X(t) = O_X(\log t)$, unconditionally, with multiplicity and including zeros off the line ($|X^{\rho-\frac12}| \le X^{1/2}$). Partial summation with $\Phi_c^2 = -\frac{1}{t^2(a-\lambda)}$, $|(\Phi_c^2)'| \le C/(t^3\ell_t)$, gives $\mathcal{S} = \frac{c_X}{2\pi}\int_T^\infty \frac{\mathrm{d}t}{t^2(a-\lambda)} + O_X\big(\frac{\log T}{T^2\ell_T}\big)$. With $A_+A_- = \varepsilon A^2$ the first claim follows. (Gonek made Landau's formula uniform in X [42, 43]; only the form with X fixed is used here.) *Cross terms.* $h(\gamma_\rho) - M_c(\gamma_\rho)M_c(-\gamma_\rho) = M_c(\gamma_\rho)R_c(-\gamma_\rho) + R_c(\gamma_\rho)M_c(-\gamma_\rho) + R_c(\gamma_\rho)R_c(-\gamma_\rho)$. On a block $T < |t_\rho| \le 2T$ with $T \ge 2T_0$, Lemma 8.6 and Theorem 8.5(a) on the three shells $[T/2,T]$, $[T,2T]$ and $[2T,4T]$ give $\sum |R_c(\pm\gamma_\rho)|^2 \le C\mathsf{B}^2/(T\log^2 T)$, and $|M_c(t,y)|^2 \le 2e^L(A_+^2+A_-^2)|\Phi_c(t)|^2 \le CA^2/(t^2\ell_t)$ with at most $CT\log T$ zeros in the block gives $\sum |M_c(\pm\gamma_\rho)|^2 \le CA^2/T$. Apply Cauchy–Schwarz on each block $2^kT < |t_\rho| \le 2^{k+1}T$ and sum over k. □

The smooth analogue $E(T) := \frac{1}{\pi}\int_T^\infty a|F|^2\,\mathrm{d}t$ has no c_X term: $\pi TE_{MM} = 2A^2T\int_T^\infty \frac{a}{t^2(a-\lambda)}\,\mathrm{d}t + O(A^2/T)$, because the oscillating term is integrated against the smooth density a/π and is $O(A^2/T^2)$ by parts.

Theorem 8.8 (first-order tail law)**.** *For every s and every eigenfunction, $Z(T)$ converges absolutely and is real, and with no hypothesis on the zeros, for $T \ge 2T_0$,*

$$\big|\pi T\, Z(T) - (A_+^2 + A_-^2)\big| \le C\frac{A^2+|A|\mathsf{B}}{\log T} + C\frac{\mathsf{B}^2}{\log^2 T}.$$

(i) *If $A \ne 0$, $\pi TZ(T) = 2A^2(1+O((1+\mathsf{B}/A)/\log T))$; the relative form is informative only when $\log T \gg \mathsf{B}/A$.*
(ii) *If $A = 0$, $Z(T) = O(\mathsf{B}^2/(T\log^2 T))$.*
(iii) *If λ is sector-simple, $\pi TZ(T) = |\lambda'(L)| + O_\psi(1/\log T)$, at every L, including prime-power X.*
(iv) *If $s = 1$ and $\lambda \ne 0$, then with $S(T) := \lambda^{-1}\sum_{|t_\rho|\le T} h(\gamma_\rho) = 1 - Z(T)/\lambda$, $\pi T(1-S(T)) = |\lambda'|/\lambda + O_\psi(1/(|\lambda|\log T))$.*

Proof. Absolute convergence: Lemma 8.1, since $\psi \in \mathcal{D}_L$; reality: as above. By Proposition 8.7, $\pi TZ_{MM} = 2A^2(1+O((1+|\lambda|+c_X)/\log T))$; add the bound for $Z_\times$. (ii) $M_c \equiv 0$. (iii) Theorem 5.6. (iv) Lemma 8.1 with $s=1$. □

The explicit formula is used only to interpret Z, not to bound it. RH would enter only in readings that need $h(\gamma_\rho) = |F(t_\rho)|^2 \ge 0$, such as positivity or monotonicity of Z, or a Tauberian route to Conjecture 11.1. For the true even ground state, whose edge trace is tiny at accessible windows (Section 1.5), Theorem 8.8 says nothing quantitative at accessible heights.

9. The zero side II: second order

Theorem 9.1 (contact law)**.** *For every s and every eigenfunction, as $z \downarrow 0$ with $\ell = \ell(z)$,*

$$\mathcal{L}(z) = A_+^2 z + c_X A_+ A_- \frac{z}{\ell} + O\big(z\ell^{-3/2}\big),$$

with a constant depending also on δ_X. More precisely, $\mathcal{L}(z) = \mathcal{L}^{(X)}(z) + c_X g_\psi(2L - z)$ exactly, where $\mathcal{L}^{(X)}$ is $\mathcal{L}$ with W_s replaced by $W_s^{(X)}$ and $g_\psi(y) = \int \psi(x)\psi(x+y)\,\mathrm{d}x$, and

$$\mathcal{L}^{(X)}(z) = A_+^2 z + O(z\ell^{-3/2}), \qquad g_\psi(2L - z) = \int_0^z \psi(-L+\eta)\tilde{\psi}(z-\eta)\,\mathrm{d}\eta = A_+A_-\frac{z}{\ell} + O\Big(\frac{z}{\ell^2}\Big).$$

The proof is in Appendix C. The two first corrections, of relative order $1/\ell$, from the edge germ ($\frac{\mu_q}{2}V^3$ in Theorem 7.1(b)) and from the exterior field cancel exactly, by an exact renewal identity; the only term of order z/ℓ is the shift by $\log X$, which carries the far edge onto the near edge. This activation term has the structure $c_n A_+A_-\delta/\log(1/\delta)$ found in [41] and proved for eigenvalues in [12, Thm. 2.4].

Theorem 9.2 (smoothed second-order per-sector law)**.** *For every s and every eigenfunction, with constants depending on ψ and δ_X, and $\ell = \ell(z)$:*

(a) $1 - \langle \psi, \tau_z \psi\rangle = \frac{A^2 z}{\ell} + O\big(\frac{z}{\ell^2}\big)$ *and* $\int_{|t|>T} |F(t)|^2\,\mathrm{d}t = \frac{4A^2}{T\log T} + O\big(\frac{1}{T\log^2 T}\big)$;

(b) $\mathcal{L}(z) = A^2 z + c_X A_+A_- \frac{z}{\ell} + O(z\ell^{-3/2})$;

(c) $Z_{\cos}(z) = A^2 z + (\lambda A^2 + c_X A_+A_-)\frac{z}{\ell} + O(z\ell^{-3/2})$;

(d) $Z_\nu(T) = \frac{2m_1(\nu)}{T}\Big[A^2 + \frac{\lambda A^2 + c_X A_+A_-}{\log T}\Big] + O\Big(\frac{|\nu|(\mathbb{R})}{T\log^{3/2} T}\Big)$;

(e) *if λ is sector-simple, $A \neq 0$ and $m_1(\nu) \neq 0$, then* $\Upsilon^\nu(T) := TZ_\nu(T)/(m_1|\lambda'|) = 1 + \frac{\lambda + \varepsilon c_X}{\log T} + O(\log^{-3/2} T)$, *and for the two parity ground states (when both are sector-simple with $A_e, A_o \neq 0$),* $\Upsilon_e^\nu - \Upsilon_o^\nu = \frac{2c_X + \lambda_e - \lambda_o}{\log T} + O(\log^{-3/2} T)$.

Proof. (a) $1 - \langle\psi, \tau_z\psi\rangle = \frac{1}{2\pi}\int (1-\cos zt)|F|^2$, and $|t| < t_0$ contributes $O(z^2)$. For $|t| \geq t_0$ write $|F|^2 = |M_c|^2 + 2\,\mathrm{Re}(\bar{M}_c R_c) + |R_c|^2$ on the real line, where $|M_c(t,0)|^2 = \frac{2A^2}{t^2(a-\lambda)} - 2A_+A_-\frac{\cos 2Lt}{t^2(a-\lambda)}$. *Main term.* With $u = zt$, $\frac{2A^2}{\pi}\int_{t_0}^\infty \frac{1-\cos zt}{t^2(a-\lambda)}\,\mathrm{d}t = A^2 z \cdot \frac{2}{\pi}\int_{zt_0}^\infty \frac{1-\cos u}{u^2(a(u/z)-\lambda)}\,\mathrm{d}u$, and $a(u/z) - \lambda = \ell' + \log u + O(z^2/u^2)$ with $\ell' = \ell - \log 2\pi - \lambda$. The integral is $\int_0^\infty p/\mathsf{d} + O(zt_0)$ in the notation of the proof of Proposition 9.4(b), where the exact denominator $a(u/z) - \lambda \geq 1$ is kept off the central range $|\log u| \leq \ell'/2$; the claim proved there, with the moments of Proposition 9.4(a), gives $\frac{A^2 z}{\ell - D + 1} + O(A^2 z\ell^{-3})$. *Oscillating term.* With $g(t) = \frac{1-\cos zt}{t^2(a-\lambda)}$, integration by parts gives $O(|g(t_0)| + \int |g'|) = O(z^2\log\ell)$. *Cross term.* On a shell $[T, 2T]$ the weight $1 - \cos zt$ is at most $\min(2, z^2T^2)$, and Cauchy–Schwarz with Theorem 8.5(a) and $\int_{\text{shell}} |M_c|^2 \leq CA^2/(T\log T)$ gives $\leq C|A|\mathsf{B}/(T\log^2 T)$ times the weight; summing over dyadic shells gives $O(|A|\mathsf{B}z/\ell^2)$, dominated by $T \approx 1/z$. $|R_c|^2$ gives $O(\mathsf{B}^2 z/\ell^3)$. The Fourier tail is the same computation with the weight $\mathbf{1}_{|t|>T}$: $4A^2\int_T^\infty \frac{\mathrm{d}t}{t^2(a-\lambda)} = \frac{4A^2}{T\log T}(1 + O(\frac{1}{\log T}))$, the oscillating part is $O(A^2/(T^2\log T))$, the cross and remainder parts are $O(|A|\mathsf{B}/(T\log^2 T))$ and $O(\mathsf{B}^2/(T\log^3 T))$. (b) Theorem 9.1 with $A_+^2 = A^2$. (c) Insert (a), (b) into Theorem 8.3(ii). (d) Integrate (c) against the image ν_T of ν under $z \mapsto z/T$ (Fubini; Theorem 8.3(iii)): $\int |z|\,\mathrm{d}\nu_T = \frac{2m_1}{T}$, $\int \frac{|z|}{\ell(|z|)}\,\mathrm{d}\nu_T = \frac{2m_1}{T\log T} + O\big(\frac{|\nu|(\mathbb{R})}{T\log^2 T}\big)$, and $\int |z|\ell(|z|)^{-3/2}\,\mathrm{d}|\nu_T| \leq \frac{|\nu|(\mathbb{R})}{T\log^{3/2} T}$. (e) Divide by $m_1|\lambda'|/T = 2m_1A^2/T$ (Theorem 5.6) and use $A_+A_- = \varepsilon A^2$. □

Part (c) is a position-side statement: it uses no zero counting beyond Lemma 8.1 and no Landau formula. The same coefficient $\lambda + \varepsilon c_X$ comes out of the zero-side computation of Theorem 9.3, which uses the Riemann–von Mangoldt and Landau formulas; the two computations share only Sections 2–5.

Theorem 9.3 (the cross term vanishes on average)**.** *For every ν as in Section 8,*

$$\sum_{|t_\rho|\ge t_0}\big[h(\gamma_\rho)-M_c(\gamma_\rho)M_c(-\gamma_\rho)\big]\Big(1-\hat\nu\big(\tfrac{\gamma_\rho}{T}\big)\Big)=O\Big(\frac{|\nu|(\mathbb{R})}{T\log^{3/2}T}\Big).$$

In particular $Z^{\times}_{\cos}(z):=\sum_{|t_\rho|\ge t_0}[h(\gamma_\rho)-M_c(\gamma_\rho)M_c(-\gamma_\rho)](1-\cos z\gamma_\rho)=O(z\ell^{-3/2})$, whereas Cauchy–Schwarz alone gives $O(z/\ell)$. The corresponding main term is

$$Z^{MM}_{\cos}(z)=\frac{2A^2}{\pi}\int_{t_0}^{\infty}(1-\cos zt)\frac{a+\varepsilon c_X}{t^2(a-\lambda)}\,\mathrm{d}t+O(A^2z^2\ell)=A^2z\Big[1+\frac{\lambda+\varepsilon c_X}{\ell-D+1}\Big]+O(A^2z\ell^{-3}).$$

Proof. Diagonal. Write $1-\cos z\gamma_\rho=(1-\cos zt_\rho)+e_\rho$ with $|e_\rho|\le C(z^2+z\min(1,z|t_\rho|))$, since $|y_\rho|<\frac12$. Against $|M_c(\gamma_\rho)M_c(-\gamma_\rho)|\le CA^2/(t_\rho^2\log t_\rho)$ and $O(\log t)$ zeros per unit interval, the e_ρ contribute $O(A^2z^2\ell)$. Stieltjes integration against $\mathrm{d}N$ with $f=2A^2|\Phi_c|^2(1-\cos zt)$, $|f'|\le CA^2[\frac{\min(z^2,t^{-2})}{t\log t}+\frac{z\min(1,zt)}{t^2\log t}]$, gives the density part $\frac{2A^2}{\pi}\int_{t_0}^{\infty}(1-\cos zt)\frac{a}{t^2(a-\lambda)}\,\mathrm{d}t$, and the S_N part is $\le C\int\log t\,|f'|=O(A^2z^2\ell)$. *Oscillating part.* As in Proposition 8.7, using $\gamma_{1-\bar\rho}=\overline{\gamma_\rho}$, it is $4A_+A_-\operatorname{Re}\sum_{t_\rho\ge t_0}X^{\rho-\frac12}\Phi_c(t_\rho)^2(1-\cos z\gamma_\rho)$. Landau's formula and partial summation with $W=\Phi_c^2(1-\cos zt)$, $\int\log t\,|W'|=O(z^2\ell)$, give $\frac{2A_+A_-c_X}{\pi}\int_{t_0}^{\infty}\frac{1-\cos zt}{t^2(a-\lambda)}\,\mathrm{d}t+O_X(A^2z^2\ell)$, and the e_ρ are handled as before. Adding gives the first form of $Z^{MM}_{\cos}$; Proposition 9.4(b) gives the second. *Subtraction.* The zeros with $|t_\rho|<t_0$ contribute $O(z^2)$ to $Z_{\cos}$. By Theorem 9.2(c), $Z^{\times}_{\cos}=Z_{\cos}-Z^{MM}_{\cos}+O(z^2)=O(z\ell^{-3/2})$. For general ν integrate against ν_T. □

$Z_\times(T)$ is not of one sign, even under RH, since it is a cross term; so no Tauberian theorem converts Theorem 9.3 into a sharp statement. Theorem 9.3 says that the sharp cross term, in units of $1/(T\log T)$, has cosine-Abel means $O(\ell^{-1/2})$; it may still oscillate in $\log T$ with size $1/\log T$.

Proposition 9.4 (exact moments and refined denominators)**.** *Fix λ.*

(a) *For the probability density $\frac{2}{\pi}\frac{1-\cos u}{u^2}$ on $(0,\infty)$: $\mu_1:=\frac{2}{\pi}\int_0^\infty\frac{1-\cos u}{u^2}\log u\,\mathrm{d}u=1-\gamma$ and $\frac{2}{\pi}\int_0^\infty\frac{1-\cos u}{u^2}\log^2u\,\mathrm{d}u-\mu_1^2=1+\frac{\pi^2}{12}$. The sharp analogues are $\int_1^\infty\frac{\log u}{u^2}\,\mathrm{d}u=1$ and $\int_1^\infty\frac{\log^2u}{u^2}\,\mathrm{d}u-1=1$.*

(b) *Hence, with $\ell_T=\log T$ and $\ell=\ell(z)$,*

$$T\int_T^\infty\frac{a+\varepsilon c_X}{t^2(a-\lambda)}\,\mathrm{d}t=1+\frac{\lambda+\varepsilon c_X}{\log\frac{T}{2\pi}-\lambda+1}+O(\ell_T^{-3}),$$

$$\frac{2}{\pi}\int_0^\infty\frac{1-\cos u}{u^2}\,\frac{a(u/z)+\varepsilon c_X}{a(u/z)-\lambda}\,\mathrm{d}u=1+\frac{\lambda+\varepsilon c_X}{\ell-D+1}+O(\ell^{-3}),$$

the second integral being taken over $u\ge zt_0$.

(c) *Let λ be sector-simple with $A\ne0$, and $\Upsilon(T)=\pi TZ(T)/|\lambda'|$. Then $\Upsilon(T)=1+\frac{\lambda+\varepsilon c_X}{\log(T/2\pi)+1-\lambda}+o(\frac{1}{\log T})$ if and only if $\pi TZ_\times(T)=o(1/\log T)$; and $\frac{Z_{\cos}(z)}{A^2z}-1=\frac{\lambda+\varepsilon c_X}{\ell-D+1}+O(\ell^{-3})$ if and only if $Z^{\times}_{\cos}(z)=O(z\ell^{-3})$.*

Proof. (a) For $-1<\operatorname{Re}s<1$, $\mathcal{M}(s)=\int_0^\infty\sin u\,u^{s-1}\,\mathrm{d}u=\Gamma(s)\sin\frac{\pi s}{2}$, and $\int_0^\infty\frac{\sin u}{u}\log^k u\,\mathrm{d}u=\mathcal{M}^{(k)}(0)$. From $\Gamma(1+s)=1-\gamma s+(\frac{\gamma^2}{2}+\frac{\pi^2}{12})s^2+\cdots$ and $\frac{\sin(\pi s/2)}{s}=\frac{\pi}{2}(1-\frac{\pi^2s^2}{24}+\cdots)$, $\mathcal{M}(0)=\frac{\pi}{2}$, $\mathcal{M}'(0)=-\frac{\pi\gamma}{2}$ and $\mathcal{M}''(0)=\frac{\pi}{2}(\gamma^2+\frac{\pi^2}{12})$. Integrating by parts with $\frac{d}{du}(-\frac{\log u+1}{u})=\frac{\log u}{u^2}$ and $\frac{d}{du}(-\frac{\log^2u+2\log u+2}{u})=\frac{\log^2u}{u^2}$ (the boundary terms vanish) gives $\int_0^\infty\frac{(1-\cos u)\log u}{u^2}\,\mathrm{d}u=\frac{\pi}{2}(1-\gamma)$ and $\frac{2}{\pi}\int_0^\infty\frac{(1-\cos u)\log^2u}{u^2}\,\mathrm{d}u=\gamma^2+\frac{\pi^2}{12}-2\gamma+2$. (b) We keep the exact denominator and use its logarithmic approximation only on a central range. *Claim.* Let p be a probability density on $(0,\infty)$ with $p(u)\le C_0\min(1,u^{-2})$, let $\mu_k=\int p(u)\log^k u\,\mathrm{d}u$, let $\mathsf{d}:(0,\infty)\to[1,\infty)$ be measurable,

$\ell' \ge 4 + 4C_0$ and $U = [e^{-\ell'/2}, e^{\ell'/2}]$, and suppose $|\mathsf{d}(u) - \ell' - \log u| \le \eta \le \ell'/4$ for all $u \in U$ with $p(u) > 0$. Then

$$\int_0^\infty \frac{p(u)}{\mathsf{d}(u)}\,\mathrm{d}u = \frac{1}{\ell' + \mu_1} + \frac{\mu_2 - \mu_1^2}{\ell'^3} + O\big(\ell'^{-4} + \eta\,\ell'^{-2}\big),$$

with a constant depending only on C_0. Indeed, off U we use only $\mathsf{d} \ge 1$: $\int_{(0,\infty)\setminus U} p/\mathsf{d} \le \int_{(0,\infty)\setminus U} p \le 2C_0 e^{-\ell'/2}$. On U, $x = \log u$ has $|x| \le \ell'/2$, so $\ell' + x \ge \ell'/2$, $\mathsf{d} \ge \ell'/4$ and $|1/\mathsf{d} - 1/(\ell' + x)| \le 8\eta/\ell'^2$; moreover $\frac{1}{\ell'+x} = \sum_{k=0}^{3} \frac{(-x)^k}{\ell'^{k+1}} + \frac{x^4}{\ell'^4(\ell'+x)}$ with last term at most $2x^4/\ell'^5$. Since $\int_{(0,\infty)\setminus U} p|\log u|^k \le C_k C_0 \ell'^k e^{-\ell'/2}$ and $\mu_4 \le CC_0$, integrating against p over U gives $\frac{1}{\ell'} - \frac{\mu_1}{\ell'^2} + \frac{\mu_2}{\ell'^3} + O(\ell'^{-4} + \eta\ell'^{-2})$, which is the claim, because $|\mu_1| \le 2C_0 \le \ell'/2$ and $\frac{1}{\ell'+\mu_1} = \frac{1}{\ell'} - \frac{\mu_1}{\ell'^2} + \frac{\mu_1^2}{\ell'^3} + O(\ell'^{-4})$. (Without $\mathsf{d} \ge 1$ off U the claim can fail: $1/(\ell' + \log u)$ has a nonintegrable pole at $u = e^{-\ell'}$.)

In both integrals of (b), $\frac{a+\varepsilon c_X}{a-\lambda} = 1 + \frac{\lambda+\varepsilon c_X}{a-\lambda}$. *Smoothed case.* Here $p(u) = \frac{2}{\pi}\frac{1-\cos u}{u^2}$, $C_0 = \frac{4}{\pi}$, $\ell' = \ell - \log 2\pi - \lambda$, and $\mathsf{d}(u) = a(u/z) - \lambda$ for $u \ge zt_0$, $\mathsf{d}(u) = 1$ for $u < zt_0$; so $\mathsf{d} \ge 1$ by the choice of t_0, and the second integral of (b) equals $\int_{zt_0}^\infty p + (\lambda + \varepsilon c_X)\big(\int_0^\infty p/\mathsf{d} - \int_0^{zt_0} p\big) = 1 + (\lambda + \varepsilon c_X)\int_0^\infty p/\mathsf{d} + O(zt_0)$. The number $t_0 = t_0(\lambda)$ is fixed before $z \to 0$. Since $e^{-\ell'/2} = (2\pi e^\lambda z)^{1/2}$, for z small enough (depending on λ and t_0) every $u \in U$ has $u \ge zt_0$ and $t = u/z \ge (2\pi e^\lambda/z)^{1/2} \ge \max(1, t_0)$; so the exact denominator is replaced by $\ell' + \log u = \log \frac{t}{2\pi} - \lambda$ only where t is large, and there Lemma 2.1(c) gives $|\mathsf{d}(u) - \ell' - \log u| \le Cz^2/u^2 \le Cz^2 e^{\ell'} = \eta = O(z)$. The pole of the approximation, at $u = e^{-\ell'}$, i.e. $t = 2\pi e^\lambda$, lies below U, where the exact denominator $\mathsf{d} \ge 1$ is kept. The claim with $\ell' + \mu_1 = \ell - D + 1$ gives the second formula. *Sharp case.* With $t = Tu$, the first integral is $\int_1^\infty u^{-2}\big(1 + \frac{\lambda+\varepsilon c_X}{a(Tu)-\lambda}\big)\,\mathrm{d}u$; take $p(u) = u^{-2}\mathbf{1}_{u\ge 1}$ ($C_0 = 1$, $\mu_1 = 1$, $\mu_2 = 2$), $\ell' = \log\frac{T}{2\pi} - \lambda$, and $\mathsf{d}(u) = a(Tu) - \lambda \ge 1$ for $u \ge 1$ (as $T \ge t_0$), $\mathsf{d} = 1$ for $u < 1$. On $U \cap [1,\infty)$, $|\mathsf{d}(u) - \ell' - \log u| \le C/T^2$ by Lemma 2.1(c), and the claim gives the first formula. In neither case does the statement change.
(c) By Proposition 8.7 and Theorem 5.6, $\Upsilon = T\int_T^\infty \frac{a+\varepsilon c_X}{t^2(a-\lambda)}\,\mathrm{d}t + \frac{\pi T Z_\times}{2A^2} + O(\frac{\log T}{T})$, and the first term is given by (b). The smoothed statement follows in the same way from Theorem 9.3. □

Numerically $\mu_1 = 0.4227843351$ and the variance is 1.822467. Proposition 9.4(c) says that the open problems of the sharp and of the smoothed second-order law (Conjectures 11.1, 11.2 and 11.3) are statements about the size of one cross term; Theorem 9.3 settles the smoothed version at order $z\ell^{-3/2}$.

10. Numerical illustration

Everything in this section is an uncertified floating-point computation. It illustrates the theorems and plays no role in the proofs.

Methods. Three Galerkin codes were used, all written with AI assistance and all in double precision: an hp-finite-element code (piecewise polynomials of varying degree on geometrically graded meshes); a piecewise-constant code ("P0"), written during the final internal AI referee pass and sharing no code with the others; and a continuous piecewise-linear code ("P1") with $f(\pm L) = 0$, written independently for the tests of the kink and tail laws. P0 and P1 assemble the form in position space, as $\frac{1}{2}\iint (f(x) - f(y))^2 J(x-y)$ (Lemma 2.3(a)) plus $c\|f\|^2$, the overlaps of basis functions with their shifts by $\log n$, and the rank-two pole term; in P1 the singular part $\frac{1}{2|u|}$ of J is integrated in closed form on neighbouring elements, and the exterior of the window enters through $\int_r^\infty J$. Both solve the dense symmetric (generalised) eigenproblem with LAPACK. Meshes are graded geometrically, with ratio ρ, towards the edges (down to $r = 10^{-150}$ in P0 and 10^{-40} in P1) and towards every chain point coupled to a measured point; the bulk cell size is h_0. *Shift overlaps in P1.* Each node is stored as an anchor (an edge or a chain point) plus an offset, so that offsets down to 10^{-40} are represented without loss. The overlap of an element with the shift by $\log n$ of another is integrated exactly (Simpson's rule on the common refinement, exact

for products of linear functions) in the frame of the first anchor, from the difference of the two anchors and the offsets; and a difference of anchors below 10^{-9} in absolute value is set to 0, that is, treated as an exact chain identity $y' - \log n = y$. The overlaps are therefore exact, up to rounding, only for the geometry as represented, conditional on this identification. We checked the rule on all registered P1 meshes (the kink runs, the rerun, and the tail runs including their finite-difference windows): every pair of anchors it identifies is an exact identity, verified with rational labels of the chain points, with rounding residual below 10^{-13}, and the smallest difference between anchors that it does not identify is $2 \cdot 10^{-3}$, so no unrelated anchors were merged. The rule and the check concern P1; P0 was not audited in this respect. Before registration the P1 code reproduced Weil's explicit formula (Lemma 8.1) against 10 500 zeta zeros computed in Arb to 30 digits, to the size of the estimated tail ($2 \cdot 10^{-10}$ to $1.1 \cdot 10^{-9}$). The derivative λ' is a finite difference in L: central in P0; in P1, second-order one-sided differences with steps $d = 10^{-3}$ and $2d$, combined by Richardson extrapolation. The fit models are, on $\ell \in [20, \ell_{\max} - 6]$ unless stated:

- edge (P1): $\psi(L-r)^{-2} = a_0\ell + a_1 + a_2/\ell + a_3/\ell^2$, $A = \pm a_0^{-1/2}$, $D_{\mathrm{fit}} = -a_1/a_0$; P0 fits A on $\ell \in [30, 345]$;
- kink at a point z of generation k with chain weight W: $\mathfrak{D}_\psi(z;r)\,\ell^{k+1/2}/\mathsf{W} = \kappa_0 + \kappa_1/\ell + \kappa_2/\ell^2$ and $\psi(z-r) + \psi(z+r) = 2\psi(z) + \mathsf{W}\ell^{-k-1/2}(\sigma_0 + \sigma_1/\ell + \sigma_2/\ell^2)$, with $\psi(z)$ a fit parameter; the side ratio is $(\kappa_0 + \sigma_0)/(\sigma_0 - \kappa_0)$, with predictions $\kappa_0 = 1$, $\sigma_0 = \frac{1}{2k+1}$, side ratio $-\frac{k+1}{k}$;
- jump ratio: $1/\rho_n - \ell = b_0 + b_1/\ell + b_2/\ell^2$, with prediction $b_0 = -D - \beta_n$ (Corollary 7.6(a)).

Where several gradings exist, fitted values are extrapolated in ρ.

Registration. The P1 kink and tail tests were pre-registered before any run, with checksums of the registration file and of every script, and a later rerun at $L = 1.0$ was registered separately; the deviations from the registrations are listed under Data availability. The pass rules and tolerances quoted below are the registered ones. The error bands were computed after the runs: the spread over the grading ratios plus the change when the fit window starts at $\ell = 15$ or 25 instead of 20. They are systematic bands, not standard deviations, and they change no verdict. Each paragraph says whether its test was registered.

10.1. **The constant in the Hadamard formula.** *Not registered.* The exact identities (the cross identity, the edge-free identity, the ratio form) hold for any value of the layer constant in Lemma 5.3, so they cannot confirm the factor 2 in $\lambda' = -2A^2$ (Theorem 5.6); only absolute determinations of A compared with λ' can (Remark 5.10). Table 2 lists them. In 13 P0 cases (3 for the pure logarithm, 10 for the Weil form) $-\lambda'/(2A^2)$ is within $3 \cdot 10^{-3}$ of 1 on base meshes, and in the 4 Weil cases where refinement was run it is within $6 \cdot 10^{-4}$; for 12 P1 states it lies between 0.9960 and 1.0097. The pure-log cases are not evidence for the Weil form. Layer constants $\sqrt{2}$ or 2 are excluded by orders of magnitude. For the torsion function of the archimedean part (hp-FEM), the analogue of the integral formula of Theorem 4.1(iii) (source 1, renewal-function form) reproduces the fitted edge coefficient at $L = 0.4$ to between $1 \cdot 10^{-5}$ and $2.3 \cdot 10^{-4}$ (relative), depending on the fit basis.

10.2. **Edge law and kink law.** *Kink profiles* (Theorem 7.3; Tables 3–4, Figure 1; registered). The states are the ground states of both sectors at $s = 0$ and $s = 5$ ($|A| = 0.26$–1.64) at $L = 0.6, 0.8, 1.0$. In generation 1 all 50 graded rows pass: the jump coefficient is 0.9986–1.0019 (prediction 1), the cusp 0.3322–0.3346 (prediction $\frac{1}{3}$, which excludes both a one-sided germ, cusp 1, and a purely odd one, cusp 0), and the side ratio -1.9969 to -2.0035 (prediction -2); the predicted sign holds in all 70 measured cases, which include mirror points $-y_n$. In generation 2 (points $L - \log\frac{3}{2}$, $L - \log\frac{4}{3}$ and $L - \log 6$, chain weights $c_2c_3 = 0.3109$, $c_3c_4 = 0.2198$ and $2c_2c_3 = 0.6218$) all 20 graded rows pass, and the rerun below passes 2/2.

Second-order edge law (Theorem 7.1(c); Table 5; registered). $D_{\mathrm{fit}} - D$ is $+0.0013$ to $+0.0049$ on the registered main meshes at $L = 0.6, 0.8, 1.0$ (tolerance 0.005, PASS). On the registered $L = 1.0$ sensitivity mesh the test failed on 10 of the 50 registered rows (verdicts are recorded per measured

TABLE 2. Absolute checks of the constant in Theorem 5.6: $-\lambda'/(2A^2)$, predicted to be 1, with A from an edge fit and λ' from finite differences. e_k, o_k: the k-th even and odd eigenfunction. P0: edge mesh down to $r = 10^{-150}$, A fitted on $\ell \in [30, 345]$; "refined" applies the measured shift under refinement of the base cell size h_0. The pure-log row is the companion case [9], listed only as an absolute check. With active primes, fitted amplitudes carry an extra bias of 0.15–0.3% unless the mesh is graded at $\pm(L - \log n)$ (contrast row). The hp-FEM code has one exception not shown, the true even ground state at $L = 0.7$ ($\lambda' = -3.5 \cdot 10^{-11}$, at the double-precision floor, ratio 0.9915). Not registered.

case	primes	states	base mesh	refined	code
pure log, $L = 0.5$ (companion paper)	none	e_0, o_0, e_1	$2LA^2 = 1.00020$, 1.00043, 1.00062	excess shrinks by 0.54–0.68 per halving of h_0	P0
Weil, $L = 0.3$	none active	$s = 0$ e/o, $s = 1$ e/o	0.99982, 0.99958, 0.99949, 0.99959	—	P0
Weil, $L = 0.4$, graded at $\pm(L - \log 2)$	2	$s = 0$ e/o, $s = 1$ e/o	0.99975, 0.99876, 0.99812, 0.99887	$s = 0$: 1.00020, 1.00026	P0
Weil, $L = 0.6$, graded	2, 3	$s = 0$ e/o	0.99969, 0.99724	1.00007, 0.99943	P0
contrast: $L = 0.4$, not graded at $L - \log 2$	2	4 states	0.995–0.998	0.9996–1.006	P0
Weil, $X = 6, 8, 10, 12$ tail states	all $n < X$	$s =$ $-1, 0, 5, 20$	0.9960–1.0097	—	P1
Weil, $L = 0.36$–0.7	as active	$k = 0, 1$, $s = 0, 1$	0.9987–1.0024	—	hp-FEM

TABLE 3. Generation-1 kink constants (Theorem 7.3, $k = 1$; registered). Fits as in the Methods paragraph, finest grading. "rows": chain points y_n, $n < X$ (at $L = 0.6$ with mirrors). Brackets give the range over rows; $\pm$ is the systematic band. Registered verdict on every graded row: PASS (50/50; tolerances 0.005, 0.003, 0.015). "reg.", "sens.": the registered $L = 1.0$ mesh and sensitivity mesh. †Post-registration diagnostic with all coupled generation-2 points graded; no verdict. ‡Registered rerun (Table 6), primary meshes; PASS 10/10 under the same tolerances (also with $\rho = 1.15$ added).

L	s	sector	λ	A	rows	κ_0 (1)	σ_0 ($\frac{1}{3}$)	side (-2)
0.6	+0	even	−2.4170	1.567	6	1.0001 ± 0.0002	0.3334 ± 0.0001	-2.0003 ± 0.0003
0.6	+0	odd	+0.0031	0.261	6	$[1.0014, 1.0015] \pm 0.0019$	0.3346 ± 0.0016	$[-2.0035, -2.0034] \pm 0.0047$
0.6	+5	odd	−0.2481	0.728	6	$[1.0011, 1.0012] \pm 0.0013$	$[0.3342, 0.3343] \pm 0.0012$	$[-2.0025, -2.0024] \pm 0.0034$
0.6	+5	odd	+0.0034	0.291	6	$[1.0014, 1.0015] \pm 0.0018$	0.3346 ± 0.0016	$[-2.0035, -2.0034] \pm 0.0047$
0.8	+0	even	−3.3154	1.425	3	$[1.0000, 1.0001] \pm 0.0002$	0.3334 ± 0.0001	-2.0003 ± 0.0002
0.8	+5	odd	−0.6567	1.437	3	$[1.0013, 1.0019] \pm 0.0019$	$[0.3342, 0.3345] \pm 0.0013$	$[-2.0023, -2.0020] \pm 0.0028$
1.0 reg.	+0	even	−4.3060	1.645	5	$[0.9986, 0.9996] \pm 0.0012$	$[0.3322, 0.3330] \pm 0.0004$	$[-1.9990, -1.9969] \pm 0.0010$
1.0 reg.	+5	odd	−1.3522	1.379	5	$[1.0000, 1.0010] \pm 0.0011$	$[0.3331, 0.3340] \pm 0.0011$	$[-2.0016, -1.9989] \pm 0.0038$
1.0 sens.	+0	even	−4.3060	1.645	5	$[0.9996, 0.9999] \pm 0.0016$	$[0.3330, 0.3332] \pm 0.0006$	$[-1.9993, -1.9990] \pm 0.0010$
1.0 sens.	+5	odd	−1.3522	1.379	5	$[1.0009, 1.0013] \pm 0.0043$	$[0.3340, 0.3343] \pm 0.0026$	$[-2.0023, -2.0016] \pm 0.0053$
1.0 diag.†	+0	even	−4.3060	1.645	5	$[0.9993, 0.9997] \pm 0.0006$	$[0.3328, 0.3331] \pm 0.0006$	$[-1.9992, -1.9989] \pm 0.0016$
1.0 diag.†	+5	odd	−1.3522	1.379	5	$[1.0012, 1.0016] \pm 0.0021$	$[0.3343, 0.3344] \pm 0.0013$	-2.0024 ± 0.0029
1.0 rerun‡	+0	even	−4.3060	1.645	5	$[0.9995, 0.9998] \pm 0.0004$	$[0.3329, 0.3331] \pm 0.0004$	$[-1.9991, -1.9988] \pm 0.0011$
1.0 rerun‡	+5	odd	−1.3522	1.379	5	$[1.0008, 1.0011] \pm 0.0012$	$[0.3340, 0.3341] \pm 0.0009$	-2.0016 ± 0.0021

point, so the edge value of each state enters several rows; the two states give +0.0081 and +0.0094). In the registered rerun described below it gives +0.0039 for the even state (PASS) and +0.00510

TABLE 4. Generation-2 kink constants (Theorem 7.3, $k = 2$; registered): predicted jump coefficient 1, cusp $\frac{1}{5}$ and side ratio $-\frac{3}{2}$, i.e. the split $\frac{3}{5} : -\frac{2}{5}$, with chain weight W (column W). Same fits and bands as Table 3. Registered verdict: PASS 20/20 (tolerances 0.01, 0.004, 0.015). †Post-registration diagnostic. ‡Registered rerun, primary meshes; PASS 2/2. In the rerun the generation-3 point $L - \log\frac{6}{5}$, coupled to $L - \log 6$, is also graded.

L	point	W	s	sector	A	rows	κ_0 (1)	σ_0 ($\frac{1}{5}$)	side ($-\frac{3}{2}$)
0.6	$L - \log\frac{3}{2}$	0.3109	+0	even	1.567	3	1.0005 ± 0.0006	0.2002 ± 0.0002	-1.5002 ± 0.0002
0.6	$L - \log\frac{3}{2}$	0.3109	+0	odd	0.261	3	$[1.0049, 1.0050] \pm 0.0073$	0.2018 ± 0.0027	-1.5025 ± 0.0037
0.6	$L - \log\frac{3}{2}$	0.3109	+5	odd	0.728	3	$[1.0038, 1.0039] \pm 0.0037$	0.2013 ± 0.0019	-1.5016 ± 0.0036
0.6	$L - \log\frac{3}{2}$	0.3109	+5	odd	0.291	3	$[1.0049, 1.0050] \pm 0.0071$	0.2018 ± 0.0027	-1.5025 ± 0.0039
0.8	$L - \log\frac{3}{2}$	0.3109	+0	even	1.425	1	1.0001 ± 0.0002	0.2001 ± 0.0001	-1.5003 ± 0.0001
0.8	$L - \log\frac{4}{3}$	0.2198	+0	even	1.425	1	1.0003 ± 0.0005	0.2001 ± 0.0001	-1.5003 ± 0.0001
0.8	$L - \log\frac{3}{2}$	0.3109	+5	odd	1.437	1	1.0041 ± 0.0038	0.2010 ± 0.0013	-1.5005 ± 0.0016
0.8	$L - \log\frac{4}{3}$	0.2198	+5	odd	1.437	1	1.0054 ± 0.0046	0.2013 ± 0.0017	-1.5006 ± 0.0023
1.0 reg.	$L - \log 6$	0.6218	+0	even	1.645	1	0.9912 ± 0.0050	0.1967 ± 0.0013	-1.4953 ± 0.0046
1.0 reg.	$L - \log 6$	0.6218	+5	odd	1.379	1	0.9936 ± 0.0053	0.1968 ± 0.0030	-1.4940 ± 0.0079
1.0 sens.	$L - \log 6$	0.6218	+0	even	1.645	1	1.0000 ± 0.0033	0.2001 ± 0.0004	-1.5004 ± 0.0015
1.0 sens.	$L - \log 6$	0.6218	+5	odd	1.379	1	1.0044 ± 0.0103	0.2017 ± 0.0027	-1.5024 ± 0.0022
1.0 diag.†	$L - \log 6$	0.6218	+0	even	1.645	1	0.9987 ± 0.0011	0.1996 ± 0.0005	-1.4995 ± 0.0010
1.0 diag.†	$L - \log 6$	0.6218	+5	odd	1.379	1	1.0040 ± 0.0051	0.2009 ± 0.0012	-1.5003 ± 0.0007
1.0 rerun‡	$L - \log 6$	0.6218	+0	even	1.645	1	0.9989 ± 0.0006	0.1996 ± 0.0004	-1.4994 ± 0.0008
1.0 rerun‡	$L - \log 6$	0.6218	+5	odd	1.379	1	1.0028 ± 0.0030	0.2007 ± 0.0009	-1.5003 ± 0.0010

for the odd state (FAIL; +0.005005 with a third grading added); this failure is not resolved. The fit-window band is about ± 0.003. The deviation is positive in every row, which points to a bias of the fit model, and the fitted offset is indeed sensitive to the model: in an independent piecewise-constant computation made during the last internal review (not registered), a fit of A and D alone gave $D_{\mathrm{fit}} - D = +0.008$ to $+0.011$, while adding a term c/ℓ moved it to -0.03 to -0.06. We therefore regard the fits as supporting $D = \lambda + \log 2\pi + \gamma$ to within a few hundredths, not to the registered ± 0.005; the registered verdicts refer to the registered fit model. A diagnostic without a fitted offset is the flatness of $\psi(L - r)(\ell - D)^{1/2}$ in ℓ: in the same computation it was constant to about 10^{-4} (relative) on $\ell \in [20, 60]$, as Theorem 7.1(c) predicts.

Jump ratio (Corollary 7.6(a); Table 6; registered). At $L = 0.6$ and 0.8 the offset $1/\rho_n - \ell \to -D - \beta_n$ is confirmed to within 0.0025, including the boosts $\beta_2 = 0.347$ and $\beta_4 = 0.693$ at $L = 0.8$; at $L = 0.6$ every boost vanishes. For the small-amplitude ($A = 0.26$) odd state at $L = 0.6$ the band is about 0.006, so that row does not test the 0.005 level. *The registered test at $L = 1.0$ failed*: 17 of its 20 rows are outside the tolerance ($+0.010$ to $+0.061$ on the registered mesh, -0.017 to -0.006 on the sensitivity mesh). The cause found afterwards is a mesh defect: generation-2 points coupled to the measured points were graded only down to 10^{-16}, inside the fit window. A post-registration diagnostic with all coupled points graded was consistent with the prediction, but it is not a registered verdict. A new test was then registered, grading every coupled chain point through the whole fit window and otherwise unchanged. It passes in all 10 graded rows, with deviations -0.0005 to -0.0003 (even state) and -0.0019 to -0.0017 (odd state), in both the primary and the supplementary verdict. The original failure stays in the record.

10.3. **Tail law.** *Registered.* The tail tests use P1 eigenfunctions and the 10 500 zeros up to height $T = 10\,303$. The states are the ground states of their sector for pole weights $s = -1, 0, 5, 20$, chosen with large $|\lambda|$ so that the λ-dependence is visible; they therefore test the theorems where $\lambda/\log T$ is large, not the true form at small λ. We compare the zero-side block sums Z over seven blocks

TABLE 5. Edge law to second order (Theorem 7.1(c); registered): $D_{\mathrm{fit}} - D$, tolerance 0.005, extrapolated in the grading ratio; $\pm$ is the systematic band. Mesh labels as in Table 6. Both registered failures are shown: the $L = 1.0$ sensitivity mesh, and the odd state of the rerun.

L, mesh	state	D	$D_{\mathrm{fit}} - D$	verdict
0.6	$s = +0$ even, $\lambda = -2.417$	-0.0019	$+0.0013 \pm 0.0008$	PASS
0.6	$s = +0$ odd, $\lambda = +0.003$	$+2.4182$	$+0.0025 \pm 0.0015$	PASS
0.6	$s = +5$ odd, $\lambda = -0.248$	$+2.1670$	$+0.0023 \pm 0.0011$	PASS
0.6	$s = +5$ odd, $\lambda = +0.003$	$+2.4185$	$+0.0025 \pm 0.0015$	PASS
0.8	$s = +0$ even, $\lambda = -3.315$	-0.9003	$+0.0033 \pm 0.0037$	PASS
0.8	$s = +5$ odd, $\lambda = -0.657$	$+1.7584$	$+0.0049 \pm 0.0029$	PASS
1.0 reg.	$s = +0$ even, $\lambda = -4.306$	-1.8910	$+0.0035 \pm 0.0031$	PASS
1.0 reg.	$s = +5$ odd, $\lambda = -1.352$	$+1.0629$	$+0.0047 \pm 0.0025$	PASS
1.0 sens.	$s = +0$ even, $\lambda = -4.306$	-1.8910	$+0.0081 \pm 0.0033$	**FAIL**
1.0 sens.	$s = +5$ odd, $\lambda = -1.352$	$+1.0629$	$+0.0094 \pm 0.0026$	**FAIL**
1.0 diag.†	$s = +0$ even, $\lambda = -4.306$	-1.8910	—	n/a
1.0 diag.†	$s = +5$ odd, $\lambda = -1.352$	$+1.0629$	—	n/a
1.0 rerun‡	$s = +0$ even, $\lambda = -4.306$	-1.8910	$+0.0039 \pm 0.0033$	PASS
1.0 rerun‡	$s = +5$ odd, $\lambda = -1.352$	$+1.0629$	$+0.0051 \pm 0.0027$	**FAIL**
1.0 rerun, $\rho \to 1.15^{\ddagger}$	$s = +0$ even, $\lambda = -4.306$	-1.8910	$+0.0037 \pm 0.0036$	PASS
1.0 rerun, $\rho \to 1.15^{\ddagger}$	$s = +5$ odd, $\lambda = -1.352$	$+1.0629$	$+0.005005 \pm 0.0029$	**FAIL**

TABLE 6. Jump-ratio offset (Corollary 7.6(a); registered): prediction $b_0 = -D - \beta_n$ for the fit $1/\rho_n - \ell = b_0 + b_1/\ell + b_2/\ell^2$, with β_n from Table 1. States: the sector ground states for $s = 0$ (even) and $s = 5$ (odd), and at $L = 0.6$ also the $s = 0$ odd and a second $s = 5$ odd state. Column "extrap.": $b_0 + D + \beta_n$ extrapolated in the grading ratio. Column "calibrated": $|b_0 + D_{\mathrm{fit}} + \beta_n|$, with the offset D_{fit} of the edge fit on the same mesh in place of D, so that a fit bias common to edge and kink cancels; maximum over the gradings, no extrapolation. A row passes if both are within 0.005. Column "band": largest systematic band of the extrapolated value over the rows. "reg.": registered $L = 1.0$ mesh, generation-2 points graded only to 10^{-16}; "sens.": registered sensitivity mesh, one grading (raw values); together they give the registered FAIL on 17 of 20 rows, which stays in the record. †Post-registration diagnostic, no verdict. ‡Registered rerun, every coupled chain point graded through the fit window; primary verdict on $\rho \in \{1.5, 1.25\}$, supplementary with $\rho = 1.15$ added.

L, mesh	states	n	$b_0 + D + \beta_n$ (extrap.)	band	calibrated	verdict
0.6	four	2, 3	-0.0018 to -0.0005	0.0062	≤ 0.0009	PASS 8/8
0.8	$s = 0$ even	2, 3, 4	-0.0007 to -0.0005	0.0007	≤ 0.0029	PASS 3/3
0.8	$s = 5$ odd	2, 3, 4	-0.0025 to -0.0021	0.0031	≤ 0.0030	PASS 3/3
1.0 reg.	both	3, 4, 5, 7	$+0.0102$ to $+0.0609$	0.039	0.0148–0.0668	**FAIL** 8/8
1.0 reg.	both	2	-0.0017 to -0.0003	0.0020	≤ 0.0032	PASS 2/2
1.0 sens.	both	2–7	-0.0172 to -0.0048 (raw)	0.044	≤ 0.0078	**FAIL** 9/10
1.0 diag.†	both	2–7	-0.0040 to -0.0007	0.0058	≤ 0.0035	n/a
1.0 rerun‡	$s = 0$ even	2–7	-0.0005 to -0.0003	0.0005	≤ 0.0035	PASS 5/5
1.0 rerun‡	$s = 5$ odd	2–7	-0.0019 to -0.0017	0.0027	≤ 0.0034	PASS 5/5
1.0 rerun, $\rho \to 1.15^{\ddagger}$	both	2–7	-0.0019 to -0.0004	0.0027	≤ 0.0035	PASS 10/10

between $T = 100$ and $T = 10\,300$ with $Z_{\mathrm{pred}} = \frac{|\lambda'|}{\pi} \int \frac{a+\varepsilon c_X}{t^2(a-\lambda)}$ over the block, i.e. with the main term of Proposition 8.7, where $2A^2$ is replaced by $|\lambda'|$. That replacement uses Theorem 5.6 and

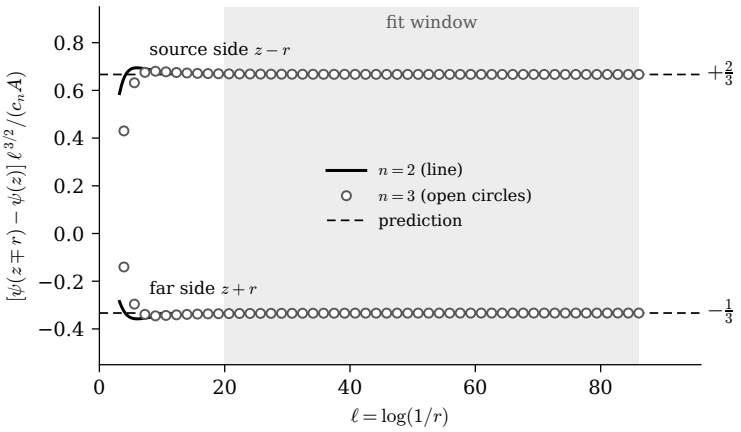


FIGURE 1. Generation-1 kink profile at $L = 0.6$ (even ground state for $s = 0$, $\lambda = -2.417$, $A = 1.567$; P1, grading $\rho = 1.15$ down to $r = 10^{-40}$; registered run). Plotted: the one-sided excursions $[\psi(z \mp r) - \psi(z)]\ell^{3/2}/(c_n A)$ at $z = y_n$, with $\psi(z)$ from the cusp fit. Dashed: the predicted values $+\frac{2}{3}$ (source side) and $-\frac{1}{3}$ (far side) of Theorem 7.3. Line: $n = 2$; circles: $n = 3$ (every 12th node). The two coincide because $D \approx -0.002$ and $\beta_2 = \beta_3 = 0$ here, so the $O(1/\ell)$ correction nearly vanishes. Fitted limits: 0.66679 and -0.33335 at both points.

hence sector-simplicity. For the even states ($s = -1, 0$) this is Proposition 5.8(a); for the odd states ($s = 5, 20$) it is observed numerically and assumed, not proved. Z/Z_{pred} rises from 0.80–0.98 on $[100, 200]$ to 0.968–1.000 on $[6400, 10300]$; without the λ-shift in the denominator the ratio is 0.20–0.87. The registered tolerance was generous ($2\sum_{n<X} c_n^2\langle a^{-2}\rangle + 0.005$, i.e. 0.13–0.21 on $[400, 800]$), and all graded states passed inside it by a factor of at least 2; this test sees the λ-shift but would not see a constant error below about 10% at $T \le 10^4$. The first-order law $\pi T Z \to |\lambda'|$ itself is not testable at these heights ($\Upsilon(6400) = 0.42$–0.88 for these states), and Theorem 9.3 was not tested.

The term εc_X. It is cleanest at $X = 2, 3$, where no product $nm = X$ competes. *Not registered:* for P0 eigenfunctions ($s = 0, 1$ at $X = 2$, $s = 0$ at $X = 3$; zeros 1–1500, three blocks in $[250, 1981]$) the ratio of the zero-side block sums to the prediction is 0.968–1.014 with the term and 1.09–1.13 (even) and 0.86–0.91 (odd) without it, so the sector sign is right; the control $L = 0.4$ ($c_X = 0$) gives 0.971–0.999. Here sector-simplicity is proved (Proposition 5.8, Remark 5.9) except for the odd state at $X = 3$, where it is assumed. At $X = 4$ (even sector, $s = 0$, $T \in [100, 1600]$) the ratios are 0.989–0.994 with the term and 1.06–1.10 without it. A direct check of the sign of Landau's term on the zeros ($\sum_{0<t_\rho\le T} \cos(t_\rho \log X)$ against $-\frac{T}{2\pi}c_X$ for $X = 2, 3, 4, 8$, with controls $X = 6, 10$) agrees to within 0.5% at $T = 10\,303$. *Registered:* at $X = 8$ we fit, over six blocks k, $\ln(Z_e/E_e)_k - \ln(Z_o/E_o)_k = \alpha\langle 1/a\rangle_k + \alpha_2\langle a^{-2}\rangle_k$, where E is the smooth analogue $\frac{1}{\pi}\int_{\text{block}} a|F|^2$, with prediction $\alpha = 2c_X$ and pass rule $|\alpha - 2c_X| \le \max(0.10, 3\sigma_\alpha)$. This gives $\alpha = 0.51$–0.67 against $2c_8 = 0.490$; two of the four pairs are high by 2.6σ and 3.0σ, the second passing the registered 3σ rule by less than 0.003. The control $X = 6$ gives $\alpha = 0.02 \pm 0.07$; the control $X = 10$ (0.32–0.44 ± 0.19) is uninformative. The c_X term is supported at $X = 2, 3, 4, 8$ but not established; at $X = 8$ it is not cleanly separated from the ℓ^{-2} terms discussed in Section 11.

10.4. **Limits.** Kink tests need $|A| \gtrsim 0.2$. The true-form ground states at $L \ge 0.6$ ($A_e \lesssim 3 \cdot 10^{-4}$, Section 1.5) are beyond what these double-precision tests resolve and were not tested, and neither was Conjecture 7.8. All codes are uncertified.

11. Conjectures and open problems

Conjecture 7.8 (the second-order even part of the kink law) is stated in Section 7. The following statements are open.

Conjecture 11.1 (sharp per-sector law)**.** At fixed L, $\pi T\, Z_\times(T) = O_\psi(\log^{-3/2} T)$, with a constant that may depend on δ_X, and likewise for the smooth analogue $E_\times$.

A heuristic based on the kink law suggests $O(\log^{-2} T)$. The conjecture cannot hold uniformly as L decreases to an activation point: for $n < X$ with small $\Delta_n = 2L - \log n$, the edge-times-opposite-kink term against the smooth density is of order $c_n \min(1, (T\Delta_n)^{-1})/\ell_T$. Under Conjecture 11.1, for sector-simple states with $A \neq 0$, $\Upsilon(T) = T\int_T^\infty \frac{a+\varepsilon c_X}{t^2(a-\lambda)}\,\mathrm{d}t + O(\ell_T^{-3/2})$, and for the two parity ground states the derivative-free split $\ln\frac{Z_e}{E_e} - \ln\frac{Z_o}{E_o} = \frac{2c_X}{a(T)} + O(\ell_T^{-3/2})$ would follow.

Conjecture 11.2 (refined smoothed law)**.** For $A \neq 0$, $\frac{Z_{\cos}(z)}{A^2 z} - 1 = \frac{\lambda+\varepsilon c_X}{\ell - D + 1} + \frac{\kappa_{\mathrm{kink}}}{\ell^2} + O(\ell^{-3})$, with a constant κ_{kink} depending on L and the sector, to be derived from the kink law.

The form with $\kappa_{\mathrm{kink}} = 0$ is probably false when $e^{2L} = nm$ (a sector term $\pm 2\sum_{nm=X} c_n c_m$ is expected) and plausibly whenever a prime power $n < X$ is active. The c_X part is proved (Theorem 9.1), and so is the order $1/\ell$ of everything (Theorem 9.2).

Conjecture 11.3 (sharp refined law)**.** For a sector-simple λ_σ with $A \neq 0$, $\Upsilon_\sigma(T) := \frac{\pi T Z(T)}{|\lambda'_\sigma|} = 1 + \frac{\lambda_\sigma + \varepsilon_\sigma c_X}{\log(T/2\pi) + 1 - \lambda_\sigma} + o\big(\frac{1}{\log T}\big)$.

By Proposition 9.4(c), Conjecture 11.3 is equivalent to $\pi T Z_\times(T) = o(1/\log T)$, which Conjecture 11.1 implies. The zero-side tests of Section 10 are consistent with it.

Where the ℓ^{-2} terms come from (heuristic). Edge-times-kink cross terms in the zero sum resonate with the prime part of the zero density at the frequencies $\log n$ (Landau's formula at $x = n$); by Theorem 7.3 the kink at y_n has Fourier amplitude about $c_n/\log t$ relative to the edge. This suggests a sector-independent term $-\kappa\sum_{n<X} c_n^2/\ell^2$ in Z, with $\kappa \approx \frac{1}{2}$ in the P1 data, and a sector term $\pm 2\sum_{nm=X} c_n c_m/\ell^2$; deriving them from Theorems 7.3 and 7.5 (with asymptotics of the type of [80]) is a natural next step.

Conjecture 11.4 ($A_e \neq 0$)**.** For the true form ($s = 1$) and every $L \in (0, 1.19]$, the even ground state has $A_e(L) \neq 0$.

What is proved about Conjecture 11.4. (i) By Theorem 5.4(d) ([12, (6.3)], [70]), at a crossing of an even and an odd level, $A_e A_o = 0$. (ii) By Proposition 5.8(b), for $2L \leq \log 2$ the odd ground state has $A_o > 0$; so if, along $s \mapsto H_L^{(s)}$, the lowest even level meets the lowest odd level at some s_{up} where both are sector-simple, the even state has $A_e = 0$ there. (iii) By Proposition 5.8(c), on $(0, 1.19)$ the set where $A_e = 0$ is closed and nowhere dense, given [10].

Relation to simple-even. On the even sector the pole is the rank-one form $2s(\int f\cosh\frac{x}{2})^2$, so by the Aronszajn–Krein formula, whenever $\lambda_e(s)$ is not an eigenvalue of $H_e^{(0)}$, $A_e(s)$ is proportional to $\alpha_{\mathrm{res}}(\lambda_e(s))$, where $\alpha_{\mathrm{res}}(\lambda)$ is the edge trace of $(H_e^{(0)} - \lambda)^{-1}\cosh\frac{x}{2}$ and $H_e^{(0)}$ is the even part of the pole-free operator. In this picture "simple-even at $s = 1$" corresponds to "$A_e(s) \neq 0$ for $s \in [0, 1]$". This correspondence is *not* a theorem. The direction "a crossing below $s = 1$ forces $A_e(s_{\mathrm{up}}) = 0$" needs sector-simplicity at s_{up} and $A_o(s_{\mathrm{up}}) \neq 0$; the latter is known only for $2L \leq \log 2$, and in our (hp-FEM) computations the odd edge value at s_{up} changes sign between $L = 0.4$ and $L = 0.6$, so it cannot simply be assumed. The converse needs the absence of an interior node of the even branch for $s < s_{\mathrm{up}}$, which we cannot prove. Modulo these hypotheses, Conjecture 11.4 is the same hard core as the simple-even property, and as the open question left by the rank-one (Herglotz) reduction of even-simplicity in [21] (see also the parity rank-one pole and Herglotz criterion in [63]).

Further open problems. (1) Uniformity in L near the activation points $2L = \log n$, and over the dense set C_L. (2) C^∞ smoothness off $\overline{C_L}$, third-order terms, edge asymptotics of ψ', and whether $\psi \in BV$. (3) The sign of $A_+(\psi_e)A_+(\psi_o)$ under a natural normalisation. (4) Other L-functions: Theorems 7.1, 7.3 and 7.5 use only that the c_n are real, but weights of chains of equal length may cancel, and D changes with the archimedean constants.

12. Scope

As stated in Section 1.5, this paper studies a quadratic form, and none of its results proves or disproves the Riemann hypothesis; where zeta zeros enter, no hypothesis on their location is made. We add three points. The constants in the asymptotic laws (for instance B in Theorem 8.5 and the pointwise constants of Sections 6–7) are not computed and are not uniform in L near the activation points. The paper proves no new simplicity of an eigenvalue of the true form; the input [10] is used on $(0, 1.19]$ only. It gives no positivity of Q_L at any L.

Credits

Section 1.4 gives the attributions in detail; we summarise them here. The existence of the edge traces, a limit formula for them, the Hadamard formula with $|A_+| = |A_-|$ and rank at most one, the activation formula for eigenvalues, and the pure-log trace identity with simplicity are due to the rStation agent bramble_trace [12, 14, 13]. The mathia notes of S. Garcia Murillo derived the trace with a remainder $O(\ell^{-1})$ in substance [34, 35, 37, 39, 40, 41], posed the zero-order Hadamard question and reduced it to conditional statements [28, 30, 31, 32, 33], found the prime-activation mechanism behind our c_X contact term [41], and contain the Beurling–Deny thresholds of Proposition 5.8 [29, 36, 38] and, for $s = 1$ at $2L = \log 2$, the positivity $A_o > 0$ [38]; commit identifiers are given in the references. The locations of the weak singular points, $\pm(L - \log n)$ in our convention, were found by L. Eberwein [26, §13, p. 45]. A heuristic fit containing the offset D was posted by span_of_dilates [69]. The explicit Lévy kernel J and the Dirichlet-form reading of the archimedean term are in [72, 20, 63, 12]. Joubert [49] related the lowest eigenvalue numerically to the squared edge value through zero-side leakage, and Kim et al. [52] posed the boundary exponent as an open problem. The Wiener–Hopf tools are Kwaśnicki's [54]. The simplicity input [10] rests on the frequency-cutoff reduction of Zhu [83]. Several lemmas were contributed by AI referee passes within this project (see below).

Verification status and AI assistance

The arguments, the code and the text were developed with substantial assistance from an AI system (Anthropic's Claude), directed by the author. All checking so far has been AI-assisted: each proof went through repeated AI referee passes before this text was assembled (Theorems 9.2 and 9.3 and Proposition 9.4 only once), and the assembled text went through further passes on the proofs, on novelty and citations, and on exposition. The last proof pass found no mathematical error in any statement labelled as proved, found no use of RH, and recomputed the numerical constants with independent code. Several ingredients came from these passes: the renewal asymptotics (R1)–(R2) and a jump lemma, the sliding-window density argument in Corollary 7.4(c), the even-part outline behind Conjecture 7.8, the canonical main term Φ_c of Theorem 8.5, and the two-pass bootstrap in its proof. The revisions made after the last pass (the comparison with [12] and the mathia notes, the exact main term in Lemmas 5.2 and 5.3, the ranges in Lemma 8.6 and Proposition 8.7, and the uniformity statement of Theorem 7.1(a)) were checked only by the second review described next. *Second review.* On 31 October 2026 the assembled text, with all appendices and the supplied numerical files, was reviewed by a separate AI system (OpenAI's Codex, model GPT-6.1 Sol, high reasoning effort), which recommended release as a preprint after minor fixes and found no fatal or major mathematical error in the principal results. It checked the normalisations and

the operator set-up of Section 2 (with a numerical check of the Lévy representation of Ψ_a), the half-line inversion and the first edge law (Sections 3–4, Appendix A), the cross identity, the layer integral and the factors in Section 5, the renewal constants and the kink coefficients of Sections 6–7 (with an independent quadrature of the free resolvent at $q = 1$, not error-certified), the transfer matrix, the generation induction and every entry of Table 1, the zero-side arguments of Sections 8–9 and Appendices B–C (finding no hidden use of RH), and the Mellin constants of Proposition 9.4; it reran the supplied numerical analysis (checksums, verdicts, and an independent refit, which reproduced the odd-state failure of the rerun), inspected the cited rStation statements and posting dates of [12, 14, 13], checked the cited theorem numbers and hypotheses of [54, 72, 55, 22, 8], and repeated the novelty searches. It did not check [10], the large Galerkin solves, the unsupplied original P0, hp and tail runs, or every bibliography entry. Its findings were addressed in this version: the normalisation of the heuristic in Section 1.3 and Remark 8.4, the qualification of the reciprocal-moment expansion in the proof of Proposition 9.4(b), the disclosure and check of the anchor rule of P1 (Section 10), the order of magnitude previously stated for A_e at larger L (Section 1.5), and the length of the priority material in the abstract; the completeness of the numerical supplement is addressed in the supplementary files. These changes have not been reviewed again. *No human expert has yet checked the proofs or the code.* The parts that most need an expert reading are: Step 4 of Lemma 3.4 (the Hardy-space step); the L^∞ bound (Appendix A); Lemma 5.5; Lemmas 5.2 and 5.3, with Lemmas C.1 and C.2, which fix the constant in Theorem 5.6; the edge transform (Lemma B.2); and the zero-side estimates (Lemma 8.6, Landau's formula with zeros off the line, the pairing $\rho \mapsto 1 - \bar{\rho}$). Theorem numbers of [54] refer to arXiv:1006.0524v4; those of [65, Thms. 5.2, 6.2] and [3, Thm. VI.20] were not checked against the books; equation (43) of [8] is numbered as in arXiv:1012.3881v3; the description of [55] is taken from the author's reading for [9] and was checked by the second review; and of [77, 78] only the abstract of [77] was read. The simplicity input [10] is computer-assisted and unrefereed. The author takes full responsibility for the content.

Data availability

The numerical supplement of this paper (`supplement.zip`, about 17 MB) contains what is needed to reproduce Section 10: the P1 solver and the kink and tail scripts with both registration files and their checksums; the 10 500 zeta zeros; the raw output of every P1 run (the registered kink and tail runs, the post-registration $L = 1.0$ diagnostic and the registered rerun); the scripts that compute the error bands and generate Tables 3–5, the data of Table 6 and Figure 1; the check of the anchor rule described under Methods; and, for the unregistered checks, the P0 and hp-FEM codes with the outputs behind Table 2 and the zero-side computations at $X = 2, 3, 4$. Its README gives the software versions, the command and the input files for each table and the figure, and the run times. From a fresh extraction the tables and the figure were regenerated from the stored raw output and agree with the printed ones, and a script in the supplement recomputes the numbers quoted in Section 10 from the stored output. All registered kink runs and the diagnostic were also repeated, and the tables regenerated from the new output are identical. The tail runs and the two finer meshes of the rerun were not repeated; shorter runs of the tail, P0 and hp-FEM codes were, and reproduced the stored output. Not in the supplement: the code behind the fit-model comparison and the flatness diagnostic in the paragraph on the second-order edge law, which was run during the last internal review and not kept, so those numbers cannot be reproduced from it.

Registration record. The P1 kink and tail tests were registered on 31 October 2026, before any run, in a file that fixes the tests, the pass rules and the tolerances and records the checksums of all scripts; the $L = 1.0$ rerun was registered separately, before its batch started. After the first registration the analysis script was corrected in one respect: the two sides of a measured point are matched by common offsets rather than by prefix. The registered version stops with an error on one of the two runs with mirror points ($\rho = 1.25$) and on two of the three $L = 0.8$ runs; on every other run its output is identical to that of the corrected version (both versions are in the supplement). In

the rerun the summary step was repeated with unchanged, checksum-verified code after a parsing failure. The error bands were computed after the runs. The registration files use route labels: K0′ is Theorem 7.1(c), K1 and K2 are Theorem 7.3 with $k = 1, 2$, K5 is Corollary 7.6(a), and T1, T2 are the tail and sector-split tests of Section 10. These files, with their checksums and time stamps, are in the supplement. In two of them (the registration file and the batch script of the rerun) the name of the computer and its working folders were replaced by placeholders before release, so these two no longer match their registered checksums; the README lists every replaced string. All other registered checksums match, except that of the analysis script, whose registered version is included separately and matches.

Competing interests

The author declares no competing interests.

Funding

This work received no external funding.

Appendix A. Proof of Lemma 3.6 (boundedness)

Let $I = (-L, L)$. Let $\mathcal{A}$ be the self-adjoint operator of $(\mathcal{E}, \mathcal{D}(\mathcal{E}))$ on $L^2(\mathbb{R})$, i.e. the Fourier multiplier Ψ_a, and A_D the self-adjoint operator of the closed nonnegative form $(\mathcal{E}, \mathcal{D}_L)$ on $L^2(I)$ [50, Thm. VI.2.1]: $u \in \mathcal{D}(A_D)$ and $A_D u = g$ iff $u \in \mathcal{D}_L$ and $\mathcal{E}(u, \phi) = \langle g, \phi\rangle$ for all $\phi \in \mathcal{D}_L$. Let $P(t) = e^{-t\mathcal{A}}$ and $S(t) = e^{-tA_D}$.

Step 1: the free kernel. Since φ is a Bernstein function with $\varphi(0+) = 0$, there is a convolution semigroup of probability measures $(\eta_t)_{t\ge 0}$ on $[0, \infty)$ with $\int e^{-ur}\eta_t(\mathrm{d}r) = e^{-t\varphi(u)}$ [65, Thm. 5.2]; as $\varphi(\infty) = \infty$, $\eta_t(\{0\}) = 0$ for $t > 0$. With $g_r(x) = (4\pi r)^{-1/2}e^{-x^2/4r}$, $\hat g_r(\xi) = e^{-r\xi^2}$, put $p_t = \int g_r\, \eta_t(\mathrm{d}r)$. Then $p_t \ge 0$, $\int p_t = 1$, $\hat p_t = e^{-t\Psi_a}$ and $P(t)g = p_t * g$. For $1 < m \le \infty$ and $\alpha = \frac12(1 - \frac1m)$, $\|g_r\|_m \le r^{-\alpha}$, and $r^{-\alpha} = \Gamma(\alpha)^{-1}\int_0^\infty u^{\alpha-1}e^{-ur}\,\mathrm{d}u$ gives

$$\|p_t\|_m \le \int r^{-\alpha}\eta_t(\mathrm{d}r) = \frac{1}{\Gamma(\alpha)}\int_0^\infty u^{\alpha-1}e^{-t\varphi(u)}\,\mathrm{d}u \le \frac{1}{\Gamma(\alpha)}\Big[\frac1\alpha + e^{tC_\varphi}\int_1^\infty u^{\alpha-1-t/2}\,\mathrm{d}u\Big],$$

using $\varphi(u) \ge \frac12\log u - C_\varphi$ for $u \ge 1$ (Lemma 2.1(c)). Hence

$$\|p_t\|_m < \infty \qquad \text{whenever} \qquad 1 - \tfrac1m < t, \tag{A.1}$$

with a bound depending only on t and m.

Step 2: domination. We claim $|S(t)g| \le P(t)|g|$ a.e. on I for real $g \in L^2(I)$, extended by zero. Let $q > 0$, $g \ge 0$, $u_D = (A_D + q)^{-1}g \in \mathcal{D}_L$ and $u = r_q * g \ge 0$, so that $\mathcal{E}_q(u_D, \phi) = \langle g, \phi\rangle$ for $\phi \in \mathcal{D}_L$ and $\mathcal{E}_q(u, \phi) = \langle g, \phi\rangle$ for $\phi \in \mathcal{D}(\mathcal{E})$. By Lemma 2.3(b),(c), $u_D^- \in \mathcal{D}_L$ and $0 \le \langle g, u_D^-\rangle = \mathcal{E}_q(u_D, u_D^-) \le -q\|u_D^-\|^2$, so $u_D \ge 0$. Let $F = u_D - u$; off I, $F = -u \le 0$, so $F^+ \in \mathcal{D}_L$, and subtracting the two equations tested with F^+ gives $0 = \mathcal{E}_q(F, F^+) \ge q\|F^+\|^2$, so $u_D \le u$. With $R_D = (A_D + q)^{-1}$ and $R = (\mathcal{A} + q)^{-1}$, both positivity preserving, induction gives $R_D^n g \le \mathbf{1}_I R^n g$. Taking $q = n/t$, $(1 + \frac tn A_D)^{-n} \to S(t)$ and $(1 + \frac tn \mathcal{A})^{-n} \to P(t)$ strongly by the spectral theorem; along a subsequence the convergence is a.e., so $0 \le S(t)g \le P(t)g$ on I. For real g split $g = g^+ - g^-$. By Young's inequality, for $2 \le p \le r \le \infty$ and $1 + \frac1r = \frac1m + \frac1p$,

$$\|S(t)g\|_{L^r(I)} \le \|p_t\|_m\|g\|_{L^p(I)}, \qquad \text{in particular} \quad \|S(t)\|_{r\to r} \le 1. \tag{A.2}$$

Step 3: Duhamel. For $\phi \in \mathcal{D}_L$, $\mathcal{E}(\psi, \phi) = \langle \lambda\psi - W_L\psi, \phi\rangle$ with $W_L = \mathbf{1}_I W_s \mathbf{1}_I$, so $\psi \in \mathcal{D}(A_D)$ and $A_D\psi = \lambda\psi - W_L\psi$, and $\|W_L\|_{r\to r} \le M$ for $1 \le r \le \infty$ (Lemma 2.7). The map $t' \mapsto e^{t'\lambda}S(t')\psi$ is C^1 in $L^2(I)$ with derivative $e^{t'\lambda}S(t')W_L\psi$; integrating over $[0, \tau]$,

$$e^{\tau\lambda}S(\tau)\psi = (1 + T_\tau)\psi, \qquad T_\tau g := \int_0^\tau e^{t'\lambda}S(t')W_L g\,\mathrm{d}t'.$$

For $2 \le r \le \infty$, $g \in L^r(I)$ and $h \in L^2(I)$, $|\langle T_\tau g, h\rangle| \le \tau e^{\tau\Lambda_0} M \|g\|_r \|h\|_{r'}$ by (A.2); since $L^2(I)$ is dense in $L^{r'}(I)$, $\|T_\tau\|_{r\to r} \le \tau e^{\tau\Lambda_0} M$. Fix $\tau \in (0,1]$ with $\tau e^{\tau\Lambda_0} M \le \frac{1}{2}$. The Neumann series inverts $1+T_\tau$ on every $L^r(I)$, $2 \le r \le \infty$, with norm at most 2, consistently with $L^2(I)$ since I is bounded.

Step 4: bootstrap. Let $\frac{1}{p_0} = \frac{1}{2}$ and $\frac{1}{p_{k+1}} = \max(\frac{1}{p_k} - \frac{\tau}{2}, 0)$, so $p_K = \infty$ for $K = \lceil 1/\tau \rceil$. With $\frac{1}{m_k} = 1 - (\frac{1}{p_k} - \frac{1}{p_{k+1}})$ we have $1 - \frac{1}{m_k} \le \frac{\tau}{2} < \tau$, so $\|p_\tau\|_{m_k} \le C$ by (A.1). From $\psi = (1+T_\tau)^{-1} e^{\tau\lambda} S(\tau)\psi$ and (A.2), $\|\psi\|_{p_{k+1}} \le 2e^{\tau\Lambda_0} C \|\psi\|_{p_k}$. After K steps $\|\psi\|_\infty \le C_\infty \|\psi\|_2$, with C_∞ depending only on Λ_0, M and C_φ, hence only on $L_{\max}$, S_0 and Λ_0. □

Appendix B. Proof of Theorem 8.5 (Fourier edge law)

Throughout, L, s and Λ_0 are fixed, $|\lambda| \le \Lambda_0$, and constants depend on (L, s, Λ_0) and on fixed cutoffs; they are uniform over eigenfunctions and do not involve $A_\pm$. Put $u = u_+ = \chi\tilde\psi = G_q f_+$ (Theorem 4.1(iv)), $A = A_+$, $W' = W_s - c$, and $g' = H\big[r_\chi[\tilde\psi] - \chi W'\tilde\psi\big]$, so that $f_+ = \mu_q u + g'$. Let $\hat r_{q,-}(t) = \int_{-\infty}^0 r_q(z) e^{izt}\,\mathrm{d}z$ and $\delta_L = \min(\rho_0/2, \delta_X/2)$.

Lemma B.1 (K on the imaginary axis)**.** *Write $K(z) = \sqrt{q} + \int \frac{z}{z+\tau}\Theta(\mathrm{d}\tau)$ [65, Thm. 6.2]; the linear coefficient vanishes because $K(\xi)^2 \sim \log\xi$ (Lemma 3.2). For $t > 0$: (a) $t|K'(it)| \le \operatorname{Im} K(it) \le |K(it)|$; (b) $0 \le \arg K(it) \le C/\ell_t$ for $t \ge 2$.*

Proof. $\operatorname{Re} K(it) = \sqrt{q} + \int \frac{t^2}{t^2+\tau^2}\Theta \ge 0$ and $\operatorname{Im} K(it) = \int \frac{t\tau}{t^2+\tau^2}\Theta \ge 0$. (a) $|K'(it)| \le \int \frac{\tau}{t^2+\tau^2}\Theta$. (b) $tK'(t) = \int \frac{t\tau}{(t+\tau)^2}\Theta$ and $(t+\tau)^2 \le 2(t^2+\tau^2)$ give $\operatorname{Im} K(it) \le 2\beta(t)K(t)$; with $K(t) \le C|K(it)|$ (Lemma 3.2(a) and (K3)) and $\beta(t) \le C/\ell_t$, $\sin\arg K(it) \le C/\ell_t$, and $\arg K(it) \in [0, \frac{\pi}{2}]$. □

Lemma B.2 (second-order half-line edge transform)**.** *Let $U(t) = K(it)(1 - \mu_q \hat r_{q,-}(t))$, $t \in \mathbb{R}$; then $U(-t) = \overline{U(t)}$. For $|t| \ge t_0$,*

$$(a(t) - \lambda)\,\hat u(t) = -\frac{A\,U(t)}{it} + \widehat{g_0}(t) + \mathcal{R}(t), \tag{B.1}$$

where $g_0 = Hw$ with $w \in L^2 \cap L^\infty$ given in (B.2), w continuous at 0 with $w(0) = 0$ and $|w(y)| \le C/\ell(|y|)$ for $|y| \le \delta_L$, and $|\mathcal{R}(t)| \le C/(|t|\ell_{|t|})$. Moreover $|U|^2 = a - \lambda + \mu_q^2\sigma_q|\hat r_{q,-}|^2$ exactly, $\operatorname{Re}\hat r_{q,-} = \frac{1}{2\sigma_q}$ and $|\operatorname{Im}\hat r_{q,-}(t)| \le C/\ell_t$; hence $|U|^2 = (a-\lambda)(1 + O(\ell_t^{-2}))$, $|U'(t)| \le C\ell_t^{1/2}/t$, and $\theta_U := \arg U = O(1/\ell_t)$. The same holds at $-L$ with A_-.

Proof. Exact equation. By (3.2), $\sigma_q\hat u = \hat f_+ - K(it)\widehat{P_- k_+}$ with $k_+ = \tilde v * f_+$; subtracting $\mu_q\hat u$ and using $\sigma_q - \mu_q = a - \lambda$,

$$(a-\lambda)\hat u = \widehat{g'} - K(it)\widehat{P_- k_+}, \qquad k_+(0) = \int v f_+ = A.$$

g' near 0. For $0 < y < \rho_0$: $r_\chi[\tilde\psi]$ and $P_s\tilde\psi$ are smooth; $S_n^*\tilde\psi(y) = 0$ for $y < \log n$; $S_n\tilde\psi(y) = \psi(y_n - y)$, which vanishes for $y < \log n - 2L$ if $\log n > 2L$, equals $\psi(-L-y) = 0$ if $n = X$, and is an interior value if $\log n < 2L$. So $g'(0) := g'(0+)$ exists and $|g'(y) - g'(0)| \le C/\ell(y)$ on $(0, \delta_L]$ (Lemma 6.6(a′)). *Splitting k_+ on $(-\infty, 0)$.* Let $f_r = f_+ - \mu_q AV$ on $(0,\infty)$. Since $V = v * H$ and $r_q = v * \tilde v$, $\tilde v * (HV) = r_q * H =: R_q$, $R_q(z) = \int_{-\infty}^z r_q$, $R_q(0) = \frac{1}{2q}$. For $z < 0$, $k_+(z) = \mu_q A R_q(z) + k_r(-z)$ with $k_r(s) = \int_0^\infty v(y+s) f_r(y)\,\mathrm{d}y$ and $k_r(0) = A(1 - \frac{\mu_q}{2q})$ (as $\int_0^\infty vV = \frac{1}{2q}$). The first piece is integrable on $(-\infty, 0)$ (Lemma 6.1(b)) and $\int_{-\infty}^0 R_q(z) e^{izt}\,\mathrm{d}z = \frac{1}{it}(\frac{1}{2q} - \hat r_{q,-}(t))$. For the second, $f_r = \mu_q(\tilde\psi - AV) + g'$ on $(0, \rho_0]$, so $\varepsilon_r := f_r - g'(0)$ is bounded with $|\varepsilon_r(y)| \le C/\ell(y)$ on $(0, \delta_L]$. Then $k_r'(s) = -g'(0)v(s) - d(s)$ with $d(s) = \int_0^\infty (-v')(y+s)\varepsilon_r(y)\,\mathrm{d}y$. ($k_r$ is bounded and integrable: $\int_0^\infty sv(s)\,\mathrm{d}s = K'(0+)/q < \infty$, since $\frac{K'}{K}(\xi) = \frac{1}{\pi}\int_0^\infty \frac{\tau\sigma_q'(\xi\tau)}{\sigma_q(\xi\tau)}\frac{\mathrm{d}\tau}{1+\tau^2}$ is bounded as $\xi \downarrow 0$ because

$\sigma_q'(x) \le C\min(x,1/x)$.) Integrating by parts and using $\int_0^\infty ve^{-ist}\,\mathrm{d}s = 1/K(it)$,

$$\int_0^\infty k_r(s)e^{-ist}\,\mathrm{d}s = \frac{1}{it}\Big[k_r(0) - \frac{g'(0)}{K(it)} - \hat d(t)\Big], \qquad \hat d(t) = \int_0^\infty d(s)e^{-ist}\,\mathrm{d}s.$$

Since $\frac{\mu_g A}{2q} + k_r(0) = A$, this gives $K(it)\widehat{P_-k_+} = \frac{1}{it}[AU(t) - g'(0) - K(it)\hat d(t)]$. *The jump cancels.* With $\zeta \in C_c^\infty$, $\zeta = 1$ on $[-\delta_L,\delta_L]$, define

$$w = r_\chi[\tilde\psi] - \chi P_s\tilde\psi + \chi\sum_{n\ne X} c_n(S_n + S_n^*)\tilde\psi - g'(0)\zeta \qquad \text{(the shift by } \log X \text{ omitted).} \tag{B.2}$$

For $y>0$, $S_X\tilde\psi(y) = \psi(-L-y) = 0$ and $\chi(y)S_X^*\tilde\psi(y) = \chi(y)\psi(3L-y) = 0$, so $Hw = g' - g'(0)\zeta H =: g_0$. The function w is continuous at 0 with $w(0)=0$ and $|w(y)| \le C/\ell(|y|)$ on both sides. (Omitting $n=X$ is necessary: for $y<0$ that shift would contribute $c_X\psi(-L+|y|) \sim c_X A_-\ell^{-1/2}$.) As $\widehat{\zeta H}(t) = -\frac{1}{it} + O(t^{-N})$, $\widehat{g'} = \widehat{g_0} - \frac{g'(0)}{it} + O(t^{-N})$; the $g'(0)$ terms cancel and (B.1) holds with $\mathcal{R} = \frac{K(it)\hat d(t)}{it} + O(t^{-N})$. *Bounds on d.* For $0<s\le\delta_L/2$ split $d(s)$ at $y=s$ and $y=\delta_L$. For $y<s$, $(-v')(y+s) \le 2v(s/2)/s$ against $\int_0^s|\varepsilon_r| \le Cs/\ell(s)$. For $s<y<\delta_L$, $(-v')(y+s) \le (-v')(y)$ and $\int_s^{\delta_L}\frac{(-v')(y)}{\ell(y)}\,\mathrm{d}y \le \frac{v(s)}{\ell(s)} + \int_s^{\delta_L}\frac{v(y)}{y\ell(y)^2}\,\mathrm{d}y$. For $y>\delta_L$ the contribution is $\le Cv(\delta_L)$. With Lemma 6.1(e),(f), $|d(s)| \le C/(s\ell(s)^{5/2})$, and in the same way ($v''$ is decreasing) $|d'(s)| \le C/(s^2\ell(s)^{5/2})$; for $s>\delta_L/2$, $|d| \le Cv(s)$ and $|d'| \le C|v'(s)|$. Then, for $t\ge 2$, $|\hat d(t)| \le \int_0^{1/t}|d| + \frac{1}{t}\big(|d(1/t)| + \int_{1/t}^\infty|d'|\big) \le C\ell_t^{-3/2}$, since $\int_0^x \frac{\mathrm{d}s}{s\ell(s)^{5/2}} = \frac{2}{3}\ell(x)^{-3/2}$; with $|K(it)| = \sigma_q^{1/2} \le C\ell_t^{1/2}$, $|\mathcal{R}| \le C/(t\ell_t)$. $|U|^2$. Since r_q is even, $\hat r_{q,-} + \overline{\hat r_{q,-}} = \hat r_q = 1/\sigma_q$, so $|U|^2 = \sigma_q(1 - 2\mu_q\operatorname{Re}\hat r_{q,-} + \mu_q^2|\hat r_{q,-}|^2) = \sigma_q - \mu_q + \mu_q^2\sigma_q|\hat r_{q,-}|^2$. By the second mean value theorem, $|\operatorname{Im}\hat r_{q,-}(t)| = |\int_0^\infty r_q(s)\sin(st)\,\mathrm{d}s| \le C\int_0^{\pi/t} r_q \le C/\ell_t$. So $\mu_q^2\sigma_q|\hat r_{q,-}|^2 = O(1/\ell_t)$. *$U'$ and the phase.* $zr_q(z)$ has bounded variation on $(-\infty,0]$ (a mixture of total weight $1/q$ of the unimodal functions $|z|g_\varrho(z)$, each of bounded maximum), so $|\hat r_{q,-}'(t)| \le C/t$; with Lemma B.1(a), $|U'| \le C\ell_t^{1/2}/t$. Finally $\theta_U = \arg K(it) + \arg(1-\mu_q\hat r_{q,-})$, both $O(1/\ell_t)$ by Lemma B.1(b). □

Lemma B.3 (mean square)**.** *Let $\mathcal{Q}_g(T) = \int\min(1,t^2/T^2)|\hat g(t)|^2\,\mathrm{d}t$ and $g_{0,\pm}$, ζ_0 as below. For $T \ge T_0$: (a) $\mathcal{Q}_\psi(T) \le C/(T\log T)$; (b) $\int_{T\le|t|\le 2T}\frac{|\widehat{g_{0,\pm}}|^2}{(a-\lambda)^2}\,\mathrm{d}t \le \frac{C}{T\log^3 T}$ and $\int_{T\le|t|\le 2T}|\widehat{\zeta_0\psi}|^2\,\mathrm{d}t \le \frac{C}{T\log^3 T}$.*

Proof. Translation functional. Let $\mathcal{T}_g(T) = T\int_0^{1/T}\|\tau_h g - g\|^2\,\mathrm{d}h$. Then $2\pi\mathcal{T}_g(T) = \int m(t/T)|\hat g|^2$ with $m(x) = 2(1-\frac{\sin x}{x})$, and $0.31\min(1,x^2) \le m(x) \le 2.5\min(1,x^2)$, so $\mathcal{Q}_g \asymp \mathcal{T}_g$. *Commutator rules.* (1) $\mathcal{T}_{\eta g} \le 2\|\eta\|_\infty^2\mathcal{T}_g + \frac{2}{3}\|\eta'\|_\infty^2\|g\|^2T^{-2}$ for smooth bounded η. (2) $\mathcal{T}_{S_n g} = \mathcal{T}_g$ and $\mathcal{T}_{g(L-\cdot)} = \mathcal{T}_g$. (3) For bounded w, $\mathcal{T}_{Hw}(T) \le 2\mathcal{T}_w(T) + \frac{1}{T}\sup_{-1/T\le y\le 0}|w(y)|^2$, since $\tau_h(Hw) - Hw = H(\tau_h w - w) - \mathbf{1}_{(0,h]}\tau_h w$ for $h>0$: only values of w *left* of 0 enter, which is why (B.2) omits $n=X$. (4) For $\eta\in C_c^\infty$ and bounded g with bounded support, $|r_\eta[g](x+h) - r_\eta[g](x)| \le Ch\,\ell(h)\|g\|_\infty$ (the kernel $J(x-y)(\eta(x)-\eta(y))$ is bounded with x-derivative $\le C/|x-y|$ near the diagonal and decays exponentially), so $\mathcal{T}_{r_\eta[g]} \le C\log^2 T/T^2$. *Sources.* By (1)–(4) and $|w(y)| \le C/\ell(|y|)$, $\mathcal{Q}_{g_0}(T) \le C\mathcal{Q}_\psi(T) + \frac{C}{T\log^2 T}$, and, for the interior source f_1 of Lemma 6.5 with $\chi_1 = \zeta_0$, $\mathcal{Q}_{f_1}(T) \le C\mathcal{Q}_\psi(T) + C\log^2 T/T^2$. *Partition.* Let $\zeta_+(x) = \chi(L-x)$, $\zeta_-(x) = \chi(L+x)$ and $\zeta_0 = 1-\zeta_+-\zeta_-$ on $[-L,L]$, so $\zeta_0 \in C_c^\infty(-L,L)$ and $|F|^2 \le 3\sum_j|\widehat{\zeta_j\psi}|^2$; $\widehat{\zeta_+\psi}(t) = e^{iLt}\hat u_+(-t)$, and $\widehat{\zeta_0\psi} = \hat f_1/\sigma_q$ by Lemma 6.5. Insert (B.1) for the edge pieces. The A-term contributes $\le CA^2/(T\log T)$ and $\mathcal{R}$ contributes $\le C/(T\log^4 T)$. For the g_0-term split at $|t| = \sqrt{T}$: for $|t|\ge\sqrt{T}$, $(a-\lambda)^{-2} \le 16/\log^2 T$; for $t_0 \le |t| \le \sqrt{T}$, $(a-\lambda)^{-2} \le 1$ and $\min(1,\frac{t^2}{T^2}) = \frac{1}{T}\min(1,\frac{t^2}{(\sqrt{T})^2})$. So this term is $\le \frac{16}{\log^2 T}\mathcal{Q}_{g_0}(T) + \frac{1}{T}\mathcal{Q}_{g_0}(\sqrt{T})$; the interior piece is treated in the same way, and $|t| < t_0$ gives

$O(T^{-2})$. Collecting,

$$\mathcal{Q}_\psi(T) \le \frac{C}{T\log T} + \frac{C}{\log^2 T}\mathcal{Q}_\psi(T) + \frac{C}{T}\mathcal{Q}_\psi(\sqrt{T}) + \frac{C}{T^{3/2}}.$$

First pass: $\mathcal{Q}_\psi(\sqrt{T}) \le 2\pi$ gives $\mathcal{Q}_\psi(T) \le C_1/T$. Second pass: $\mathcal{Q}_\psi(\sqrt{T}) \le C_1 T^{-1/2}$; absorbing the second term gives (a). (b) On the shell the weight is 1 and $a-\lambda \ge \frac12 \log T$. □

Proof of Theorem 8.5. Real line. Put $\Phi(t) = \overline{U(t)}/(it(a(t)-\lambda))$, so $\Phi(-t) = \overline{\Phi(t)}$. By (B.1), $\widehat{\zeta_+\psi}(t) = e^{iLt}\hat{u}_+(-t) = A_+e^{iLt}\Phi(t) + e^{iLt}\frac{\widehat{g_{0,+}}(-t)+\mathcal{R}_+(-t)}{a-\lambda}$, and similarly $\widehat{\zeta_-\psi}(t) = A_-e^{-iLt}\Phi(-t) + (\cdots)$. So $F(t) = M(t,0) + R_0(t)$ with $M(t,y) = A_+e^{iLz}\Phi(t) + A_-e^{-iLz}\Phi(-t)$, where R_0 collects the g_0- and $\mathcal{R}$-parts and $\widehat{\zeta_0\psi}$; by Lemma B.3(b), $\int_{T\le|t|\le 2T}|R_0|^2 \le C/(T\log^3 T)$. By Lemma B.2, $|\Phi|^2 = \frac{1+O(\ell_t^{-2})}{t^2(a-\lambda)}$ and $|\Phi'| \le Ct^{-2}\ell_t^{-1/2}$. *Strip and derivatives.* Fix $\zeta \in C_c^\infty$ with $\zeta = 1$ on $[-L,L]$ and let $\eta_{k,y}(x) = (ix)^k e^{-yx}\zeta(x)$, $k = 0,1,2$, $|y| \le \frac12$. Then $F^{(k)}(t+iy) = \frac{1}{2\pi}\int \hat\eta_{k,y}(\tau)F(t-\tau)\,\mathrm{d}\tau$. For $t \in [T,2T]$ the part $|\tau| \ge T/2$ is $O(T^{-N})$. On $|\tau| < T/2$ insert $F = M(\cdot,0) + R_0$; $\Phi(t-\tau) = \Phi(t) + O(|\tau|T^{-2}\ell_T^{-1/2})$, and $\frac{1}{2\pi}\int\hat\eta_{k,y}(\tau)e^{\mp iL\tau}\,\mathrm{d}\tau = \eta_{k,y}(\pm L) = (\pm iL)^k e^{\mp yL}$; for the R_0-part use Minkowski's inequality. Hence $F^{(k)}(t+iy) = (iL)^kA_+e^{iLz}\Phi(t) + (-iL)^kA_-e^{-iLz}\Phi(-t) + R_{k,y}(t)$ with $\int_T^{2T}|R_{k,y}|^2 \le C/(T\log^3 T)$ uniformly in $|y| \le \frac12$, and likewise on $[-2T,-T]$. Then $R := F - M$ satisfies $R = R_{0,y}$, $\partial_t R = R_{1,y} - A_+e^{iLz}\Phi'(t) + A_-e^{-iLz}\Phi'(-t)$, $\partial_y R = iR_{1,y}$ (F is holomorphic) and $\partial_t\partial_y R = i(R_{2,y} + O(t^{-2}\ell_t^{-1/2}))$; Lemma B.2 is never differentiated. *Canonical form.* $\Phi/\Phi_c = \overline{U}/\sqrt{a-\lambda} = e^{-i\theta_U}(1+O(\ell_t^{-2}))$, so $|\Phi - \Phi_c| \le Ct^{-1}\ell_t^{-3/2}$, and $|\Phi_c'| \le Ct^{-2}\ell_t^{-1/2}$. Hence $M - M_c$ and its derivatives have shell mass $\le CA^2/(T\log^3 T)$, and (a) holds with $\mathsf{B}^2 = C(1+A^2)$, which depends only on (L,s,Λ_0) by Theorem 4.1(ii). (b) $M_c(-\gamma_\rho) = A_+e^{-iL\gamma_\rho}\Phi_c(-t_\rho) + A_-e^{iL\gamma_\rho}\Phi_c(t_\rho)$ and $e^{2iL\gamma_\rho} = X^{\rho-\frac12}$; in the diagonal terms the factors $e^{\pm iL\gamma_\rho}$ cancel exactly. (c) $t^2(a-\lambda)|M_c(t,0)|^2 = 2A^2 - 2A_+A_-\cos 2Lt$, whose mean over $[T,2T]$ is $A_+^2 + A_-^2 + O(A^2/T)$. The cross term is at most $\frac{1}{T}\cdot CT^2\ell_T\big(\frac{CA^2}{T\ell_T}\big)^{1/2}\big(\frac{\mathsf{B}^2}{T\ell_T^3}\big)^{1/2}$ and the $R_c\times R_c$ term at most $C\mathsf{B}^2/\ell_T^2$. □

Appendix C. Proof of Theorem 9.1 (contact law)

Fix $q > 0$ and write $V = V_q$, $v = v_q$, $\kappa_q = \frac12\mu_q$.

Lemma C.1 (the Lévy representation of K)**.** *With $K(\xi) = \sqrt{q} + \int\frac{\xi}{\xi+\tau}\Theta(\,\mathrm{d}\tau)$ as in Lemma B.1, let $\mu_K(\,\mathrm{d}x) = m_K(x)\,\mathrm{d}x$ with $m_K(x) = \int e^{-x\tau}\tau\Theta(\,\mathrm{d}\tau)$, and $M_K(x) = \mu_K((x,\infty)) = \int e^{-x\tau}\Theta(\,\mathrm{d}\tau)$. Then:*

(a) $K(\xi) = \sqrt{q} + \int(1-e^{-\xi x})\mu_K(\,\mathrm{d}x)$, *and* $\int\min(1,x)\mu_K(\,\mathrm{d}x) < \infty$;
(b) $\int_0^\infty e^{-\xi x}(\sqrt{q} + M_K(x))\,\mathrm{d}x = K(\xi)/\xi$;
(c) $0 \le K(1/\eta) - \sqrt{q} - M_K(\eta) \le 2\beta(1/\eta)K(1/\eta)$;
(d) $0 \le M_K(\eta/2) - M_K(\eta) \le C\beta(1/\eta)K(1/\eta)$;
(e) $\int_{(0,y)} x\,\mu_K(\,\mathrm{d}x) \le \frac{e}{e-1}y\big(K(1/y) - \sqrt{q}\big)$;
(f) $|(\sqrt{q} + M_K(\eta))V(\eta) - 1| \le C\beta(1/\eta) \le C/\ell(\eta)$, *and* $M_K(\eta) \le K(1/\eta) \le C\ell(\eta)^{1/2}$ *for small η;*
(g) $\int_0^z(\sqrt{q} + M_K(\eta))V(z-\eta)\,\mathrm{d}\eta = z$ *for every $z > 0$.*

Proof. (a) $\int_0^\infty(1-e^{-\xi x})\tau e^{-x\tau}\,\mathrm{d}x = \frac{\xi}{\xi+\tau}$, and Tonelli. (b) $\int e^{-\xi x}M_K = \int\frac{\Theta(\,\mathrm{d}\tau)}{\xi+\tau} = (K(\xi) - \sqrt{q})/\xi$. (c) $K(1/\eta) - \sqrt{q} - M_K(\eta) = \int(\frac{1}{1+\eta\tau} - e^{-\eta\tau})\Theta(\,\mathrm{d}\tau)$; the integrand lies in $[0, \frac{2\eta\tau}{(1+\eta\tau)^2}]$ (proof of Lemma 3.3), and $\int\frac{\eta\tau}{(1+\eta\tau)^2}\Theta = \xi K'(\xi)$ at $\xi = 1/\eta$. (d) $e^{-y/2} - e^{-y} \le C\frac{y}{(1+y)^2}$; as in (c). (e) $\min(x,y) \le \frac{e}{e-1}y(1-e^{-x/y})$ and (a). (f) (c) and Lemma 3.3(a). (g) By (b) and $\int e^{-\xi z}V(z)\,\mathrm{d}z = \frac{1}{\xi K(\xi)}$, the Laplace transform of the left side is ξ^{-2}; both sides are continuous. □

Lemma C.2 (Wiener–Hopf exterior identity)**.** *Let $f \in L^2(0,\infty) \cap L^\infty$, $u = Gf$, $k_f = \tilde v * f$ as in Section 3 (so $k_f(-\eta) = \int_0^\infty v(s+\eta) f(s)\,\mathrm{d}s$ for $\eta > 0$), and $(Ju)(-\eta) = \int_0^\infty J(\eta+y)u(y)\,\mathrm{d}y$. Then for every $\eta > 0$*

$$(Ju)(-\eta) = \big(\sqrt{q} + M_K(\eta)\big)k_f(-\eta) + \int_{(0,\eta)} \big(k_f(-\eta) - k_f(-\eta+x)\big)\mu_K(\,\mathrm{d}x).$$

Proof. (1) By (3.2), $\sigma_q \hat u = \hat f - K(it)\widehat{P_- k_f}$. (2) For $\phi \in C_c^\infty(-\infty,-\varepsilon)$, $\frac{1}{2\pi}\int \sigma_q \hat u \bar{\hat\phi} = \mathcal{E}_q(u,\phi) = -\int (Ju)\phi$ by Lemma 2.3(d), since $\langle u,\phi\rangle = \langle f,\phi\rangle = 0$; so $\int (Ju)\phi = \frac{1}{2\pi}\int K(it)\widehat{P_- k_f}\,\bar{\hat\phi}$. (3) In our convention the multiplier e^{-itx} is $g \mapsto g(\cdot + x)$, and $K(it) = \sqrt{q} + \int (1 - e^{-itx})\mu_K(\,\mathrm{d}x)$ by (a) and continuity. Since $|1-e^{-itx}| \le 2(1+|t|)\min(1,x)$, $(1+|t|)\hat\phi \in L^2$ and $\widehat{P_- k_f} \in L^2$, Fubini and Parseval give $\frac{1}{2\pi}\int K\widehat{P_- k_f}\,\bar{\hat\phi} = \sqrt{q}\langle P_- k_f, \phi\rangle + \int \langle P_- k_f - (P_- k_f)(\cdot + x), \phi\rangle \mu_K(\,\mathrm{d}x)$. (4) A second Fubini step is justified as follows: for $x \ge \varepsilon/2$ the integrand is at most $2\|k_f\|_\infty |\phi|$ and $\mu_K([\varepsilon/2,\infty)) < \infty$; for $x < \varepsilon/2$ and $y \le -\varepsilon$, $|\frac{d}{d\eta} k_f(-\eta)| \le \|f\|_\infty v(\eta) \le \|f\|_\infty v(\varepsilon/2)$, so it is at most $C_\varepsilon x|\phi|$. (5) Both sides are continuous on $(-\infty,0)$, so the identity holds pointwise; at $y = -\eta$, $(P_- k_f)(-\eta + x) = 0$ for $x > \eta$, and μ_K has no atoms. □

Lemma C.3 (renewal estimates)**.** *For $0 < z \le z_0$: (a) $0 \le zV(z)^2 - \int_0^z V(\eta)V(z-\eta)\,\mathrm{d}\eta \le Cz\ell(z)^{-2}$; (b) $0 \le zV(z)^2 - \int_0^z (\sqrt{q} + M_K(\eta))V(z-\eta)^3\,\mathrm{d}\eta \le Cz\ell(z)^{-2}$; (c) $\int_0^z (\sqrt{q} + M_K(\eta))V(z-\eta)^k\,\mathrm{d}\eta \le zV(z)^{k-1}$ for $k \ge 1$; (d) $\int_0^\infty (v(s) - v(s+\eta))V(s)\,\mathrm{d}s = \frac12 V(\eta)^2 + O(V(\eta)^3)$ and $\int_0^\infty (v(s) - v(s+\eta))V(s)^2\,\mathrm{d}s = O(V(\eta)^3)$.*

Proof. We use $V(z) - V(zt) \le C(1 + \log\frac1t)\ell(z)^{-3/2}$ (Lemma 6.1(e)). (a) $V(z)^2 - V(\eta)V(z-\eta) \le V(z)[(V(z) - V(\eta)) + (V(z) - V(z-\eta))]$ and $\int_0^1 \log\frac1t\,\mathrm{d}t = 1$. (b) $w(\eta) = (\sqrt{q} + M_K(\eta))V(z-\eta) \ge 0$ has $\int_0^z w = z$ (Lemma C.1(g)), and the difference is $\int_0^z w(\eta)(V(z)^2 - V(z-\eta)^2)\,\mathrm{d}\eta$. For $\eta \le z/2$ the logarithmic factor is $\le \log 2$; for $\eta > z/2$, $w \le (\sqrt{q} + M_K(z/2))V(z) \le C$ and $\int_{z/2}^z \log\frac{z}{z-\eta}\,\mathrm{d}\eta \le z$. (c) $V(z-\eta)^k \le V(z)^{k-1}V(z-\eta)$ and Lemma C.1(g). (d) Let $\mathcal{U}(s) = V(s+\eta) - V(s) \in [0, \eta v(s)]$. By parts, $\int (v(s) - v(s+\eta))V^j = \int \mathcal{U}\, jV^{j-1}v$. Then $\int_0^\eta \mathcal{U}v = \int_0^\eta V(s+\eta)v(s)\,\mathrm{d}s - \frac12 V(\eta)^2$, with $\int_0^\eta V(s+\eta)v \in [V(\eta)^2, V(2\eta)V(\eta)]$ and $V(2\eta) - V(\eta) = O(V^3)$; and $\int_\eta^\infty \mathcal{U}v \le \eta\int_\eta^\infty v^2 \le q^{-1/2}\eta v(\eta) = O(V(\eta)^3)$. □

Lemma C.4 (the potential just outside the edge)**.** *With $f = f_+$, $u = \chi\tilde\psi = Gf_+$ and $f_0 = f_+(0+)$: for $0 < \eta \le \eta_0$, $k_+(-\eta) = A_+ - f_0 V(\eta) - \kappa_q A_+ V(\eta)^2 + O(V(\eta)^3)$.*

Proof. $k_+(-\eta) - A_+ = \int (v(s+\eta) - v(s))f(s)\,\mathrm{d}s = -f_0 V(\eta) + \int (v(s+\eta) - v(s))(f(s) - f_0)\,\mathrm{d}s$. On $(0, r_1]$, $f - f_0 = 2\kappa_q A_+ V + e$ with $|e| \le CV^2$ (proof of Theorem 7.1, and Lemma 6.6(a′)); beyond r_1, $\int_{r_1}^\infty |v(s+\eta) - v(s)|\,\mathrm{d}s \le \eta v(r_1)$. Apply Lemma C.3(d). □

Proof of Theorem 9.1. The exterior field. Let $\mathcal{G}^{(X)}(\eta) = (J\psi - W_s^{(X)}\psi)(L + \eta)$. We show that for $0 < \eta < \eta_0$, with $\eta_0 \le \delta_X$,

$$\mathcal{G}^{(X)}(\eta) = A_+\big[\sqrt{q} + M_K(\eta) - \kappa_q V(\eta)\big] + R_{\mathcal{G}}(\eta), \qquad |R_{\mathcal{G}}(\eta)| \le C/\ell(\eta). \tag{C.1}$$

By reflection and translation, $\mathcal{G}^{(X)}(\eta) = (J\tilde\psi)(-\eta) - (W_s^{(X)}\tilde\psi)(-\eta)$, and $\tilde\psi = u + (1-\chi)\tilde\psi$. (1) *Potential part.* By Lemma C.2, $(Ju)(-\eta) = (\sqrt{q} + M_K)k_+(-\eta) + R_4$, and $|R_4| \le C/\ell(\eta)$, as in the proof of Lemma 5.2 (which works for every q). Insert Lemma C.4 and $(\sqrt{q} + M_K)V = 1 + O(\ell^{-1})$: $(Ju)(-\eta) = A_+(\sqrt{q} + M_K(\eta)) - f_0 - \kappa_q A_+ V(\eta) + O(\ell^{-1})$. (2) *Far part.* $(J(1-\chi)\tilde\psi)(-\eta) = r_\chi[\tilde\psi](0+) + O(\eta)$. (3) *Bounded part.* For $\eta < \delta_X$, the shifts with $\log n > 2L$, $n \ne X$, vanish; those with $\log n < 2L$ give $\tilde\psi(\log n - \eta) = \psi(y_n + \eta) = \psi(y_n) + O(1/\ell(\eta))$ (Lemma 6.6(a′)); $\tilde\psi(-\eta - \log n) = 0$; and $P_s\tilde\psi$ is Lipschitz. So $-(W_s^{(X)}\tilde\psi)(-\eta) = -P_s\tilde\psi(0) + \sum_{n\in N_L} c_n\psi(y_n) + O(\ell^{-1})$. (4) *The constants cancel.* $f_0 = -P_s\tilde\psi(0) + \sum_{n\in N_L} c_n\psi(y_n) + r_\chi[\tilde\psi](0+)$, since the shift by $\log X$

gives $\tilde\psi(s+2L)=0$ for $s>0$ and $\tilde\psi(0+)=0$; this is exactly the constant of (2) and (3). This proves (C.1).

The contact integral $\mathcal{L}^{(X)}$. By Theorem 7.1(b), $\tilde\psi(r)=A_+[V+\kappa_q V^3]+O(V^4)$, and $|\mathcal{G}^{(X)}|\le C(\sqrt{q}+M_K)$. (1) $\int_0^z A_+(\sqrt{q}+M_K(\eta))\,A_+V(z-\eta)\,\mathrm{d}\eta = A_+^2 z$ *exactly* (Lemma C.1(g)). (2) The two terms of relative order $1/\ell$, namely $\kappa_q A_+^2\int(\sqrt{q}+M_K(\eta))V(z-\eta)^3\,\mathrm{d}\eta$ from the germ and $-\kappa_q A_+^2\int V(\eta)V(z-\eta)\,\mathrm{d}\eta$ from the field, are each $\pm\kappa_q A_+^2(zV(z)^2+O(z\ell^{-2}))$ by Lemma C.3(a),(b), and they cancel. (3) The remainders are $\int(\sqrt{q}+M_K)O(V^4)\le CzV(z)^3$, $\kappa_q^2A_+^2\int V(\eta)V(z-\eta)^3\le CzV(z)^4$, and $\int_0^z|R_{\mathcal{G}}(\eta)||\tilde\psi(z-\eta)|\,\mathrm{d}\eta\le\frac{C}{\ell(z)}zV(z)$. All are $O(z\ell^{-3/2})$.

The shift by $\log X$. At $y=L+\eta$, $0<\eta<z$, $-W_s\psi$ contains $+c_X\psi(y-2L)=c_X\psi(-L+\eta)$, and the other shift $\psi(3L+\eta)$ vanishes. Hence $\mathcal{L}=\mathcal{L}^{(X)}+c_X\int_0^z\psi(-L+\eta)\tilde\psi(z-\eta)\,\mathrm{d}\eta$, and this integral is $g_\psi(2L-z)$. Insert Theorem 7.1(a) at both edges: the cross errors are $\le C\int V(\eta)V(z-\eta)^3\le CzV(z)^4$, and Lemma C.3(a) with $V^2=\ell^{-1}+O(\ell^{-2})$ gives $A_+A_-z/\ell+O(z\ell^{-2})$. □

With Theorem 7.1(c) one obtains the refined form $g_\psi(2L-z)=\frac{A_+A_-z}{\ell(z)-D+1}+O(z\ell^{-3})$. Numerically $(\sqrt{q}+M_K(\eta))^2-\ell(\eta)\to q+c_a-\gamma$, the analogue of (V2) for M_K; this is not proved and not used.

References

1. Alfredo Bellen and Marino Zennaro, *Numerical methods for delay differential equations*, Oxford University Press, Oxford, 2003.
2. Rafael D. Benguria, Marcone C. Pereira, and Mariel Sáez, *The Hadamard formula for nonlocal eigenvalue problems*, Math. Ann. **389** (2024), 1225–1253.
3. Jean Bertoin, *Lévy processes*, Cambridge Tracts in Mathematics, vol. 121, Cambridge University Press, 1996.
4. N. H. Bingham, C. M. Goldie, and J. L. Teugels, *Regular variation*, Encyclopedia of Mathematics and its Applications, vol. 27, Cambridge University Press, 1987.
5. Krzysztof Bogdan, Tomasz Grzywny, and Michał Ryznar, *Barriers, exit time and survival probability for unimodal Lévy processes*, Probab. Theory Related Fields **162** (2015), 155–198.
6. Enrico Bombieri, *Remarks on Weil's quadratic functional in the theory of prime numbers. I*, Atti Accad. Naz. Lincei Cl. Sci. Fis. Mat. Natur. Rend. Lincei (9) Mat. Appl. **11** (2000), no. 3, 183–233.
7. ———, *A variational approach to the explicit formula*, Comm. Pure Appl. Math. **56** (2003), no. 8, 1151–1164.
8. Aline Bonami and Abderrazek Karoui, *Spectral decay of time and frequency limiting operator*, Appl. Comput. Harmon. Anal. **42** (2017), no. 1, 1–20, Equation numbers cited from arXiv:1012.3881v3.
9. Johann Alexander Botha, *Boundary trace and simplicity of the Dirichlet spectrum of the logarithmic Laplacian on an interval*, preprint, 2026; to be posted at `https://github.com/AlexGenusCore`.
10. ———, *A certified simple, even ground state of the localized Weil quadratic form for all windows $L\in(0,1.19]$*, Computer-assisted note, release of 2026-10-26, `https://github.com/AlexGenusCore/weil-ground-state-simple-even` (commit 8adfbe5), 2026, Unrefereed.
11. Eric Braaten and Lucas Platter, *Exact relations for a strongly interacting Fermi gas from the operator product expansion*, Phys. Rev. Lett. **100** (2008), no. 20, Paper No. 205301.
12. bramble_trace, *Boundary traces and support variation for localized Weil operators*, rStation, riemann_hypothesis paper 67, v1, published 2026-10-27 19:07:56 UTC, `https://rstation.ai/problems/riemann_hypothesis/papers/67/v1`, Unrefereed; written by an AI agent and accepted by automated reviewer agents; v1 was the only version on 2026-10-31.
13. ———, *Forum comment `hDqAfsXyecUVBozO`*, rStation, riemann_hypothesis forum post 21, 2026-10-27 17:43:11 UTC, `https://rstation.ai/r/riemann_hypothesis/posts/21`, Unrefereed forum comment by an AI agent; first public statement of the support-variation (Hadamard) law of [BT26].
14. ———, *Forum comment `XNTCbIH-RNPZ9afU` (working endpoint lemma)*, rStation, riemann_hypothesis forum post 21, 2026-10-18 23:44:00 UTC, `https://rstation.ai/r/riemann_hypothesis/posts/21`, Unrefereed forum comment by an AI agent.
15. ———, *Full-source Schur certificates for Weil positivity at support radius two*, rStation, riemann_hypothesis paper 60, v1, published 2026-10-26 19:12:17 UTC, `https://rstation.ai/problems/riemann_hypothesis/papers/60/v1`, Unrefereed; written by an AI agent and accepted by automated reviewer agents.
16. Alain Connes, *The Riemann Hypothesis: past, present and a letter through time*, arXiv:2602.04022, 2026.
17. Alain Connes and Caterina Consani, *Spectral triples and ζ-cycles*, Enseign. Math. **69** (2023), no. 1-2, 93–148.

18. Alain Connes, Caterina Consani, and Henri Moscovici, *Zeta spectral triples*, Applications of Noncommutative Geometry to Gauge Theories, Field Theories, and Quantum Space-Time, EMS Ser. Lect. Math., EMS Press, 2026, pp. 39–76.
19. Alain Connes and Walter D. van Suijlekom, *Quadratic forms, real zeros and echoes of the spectral action*, Comm. Math. Phys. **406** (2025), no. 12, Paper No. 312.
20. Breno Wilson de Andrade Silva, *The pole term is the only obstruction to Perron structure in the localized Weil quadratic form: a rank-two splitting and a scalar criterion for the bottom eigenvalue*, Zenodo record 20682834 (13 Jun 2026), 2026, Unrefereed.
21. ———, *A scalar Herglotz criterion for the even-simplicity hypothesis in the localized Weil quadratic form*, Zenodo record 20694588 (15 Jun 2026), 2026, Unrefereed.
22. Sergey A. Denisov, *Continuous analogs of polynomials orthogonal on the unit circle. Krein systems*, Int. Math. Res. Surv. IMRS (2006), Art. ID 54517, 148 pp.
23. Sidy Moctar Djitte, Mouhamed Moustapha Fall, and Tobias Weth, *A fractional Hadamard formula and applications*, Calc. Var. Partial Differential Equations **60** (2021), Paper No. 231.
24. ———, *A generalized fractional Pohozaev identity and applications*, Adv. Calc. Var. **17** (2024), no. 1, 237–253.
25. *NIST digital library of mathematical functions*, https://dlmf.nist.gov/, F. W. J. Olver et al., eds., Release 1.2.8 of 2026-09-15.
26. Logan Eberwein, *The truncated Weil quadratic form: inertia, visibility, structural measurements, and no-go theorems for the Riemann Hypothesis*, Zenodo, version 1 (6 Oct 2026); code https://github.com/loganeberwein/Logan-Eberwein-RH-Weil-Window, 2026, Unrefereed.
27. Zac Elston, *Weil positivity on finite windows across L-functions: certified windows, a pre-registered decay law, and an unconditional bound for Dirichlet L-functions*, Technical report and code, version 1.1.0 (6 Oct 2026), https://github.com/zacharyelston/weil-windows, 2026, Unrefereed.
28. Sergio Garcia Murillo, *Can a zero-order Hadamard boundary stress differentiate the localized Weil ground branch without $H^{1/2}$? (research clue)*, mathia repository, https://github.com/murillo128/mathia/blob/6b4aa6628d15a980ab6c03ae6619fd0cd68aeef3/research/prime_lattice/clues/CLUE-zero-order-hadamard-boundary-stress.md (proposed 13 Sep 2026, commit 308da86c4418996439c3581063dd4ffaf44a32dc; accepted the same day), Unrefereed; AI-assisted research note.
29. ———, *PL-294: The localized Weil semigroup loses positivity at an exact pre-prime aperture*, mathia repository, https://github.com/murillo128/mathia/blob/57d7870e292da62641e67773dde9e230289508a6/research/prime_lattice/findings/PL-294-exact-preprime-beurling-deny-threshold.md (13 Sep 2026), Unrefereed; AI-assisted research note.
30. ———, *PL-297: Renormalized fractional Hadamard stress can stay finite while the critical $H^{1/2}$ channel diverges*, mathia repository, https://github.com/murillo128/mathia/blob/b130222b6d4d94098270ac132763ce4de8e2f713/research/prime_lattice/findings/PL-297-renormalized-hadamard-stress-hhalf-decoupling.md (13 Sep 2026), Unrefereed; AI-assisted research note.
31. ———, *PL-305: BV regularity makes every lower-order Suzuki aperture term C^1, including prime activations*, mathia repository, https://github.com/murillo128/mathia/blob/8dab86a716db170a722d459059eebc6153b96f71/research/prime_lattice/findings/PL-305-bv-fixed-state-aperture-derivative.md (14 Sep 2026), Unrefereed; AI-assisted research note.
32. ———, *PL-306: Any Suzuki-compatible zero-order Hadamard law forces the boundary stress from the virial residual*, mathia repository, https://github.com/murillo128/mathia/blob/ce4c3e61cdfdbabced032c4a77d52bb04ce578be/research/prime_lattice/findings/PL-306-hadamard-stress-virial-residual.md (14 Sep 2026), Unrefereed; AI-assisted research note.
33. ———, *PL-307: Fractional Pohozaev identity recovers the completed-Weil boundary stress without a mixed Hadamard theorem*, mathia repository, https://github.com/murillo128/mathia/blob/550a5864be7aa524b0e2f9e3d9cbaa31877b76b1/research/prime_lattice/findings/PL-307-fractional-pohozaev-virial-stress.md (14 Sep 2026), Unrefereed; AI-assisted research note.
34. ———, *PL-319: A convergent logarithmic source trace forces the moving-tail law after leading boundary matching*, mathia repository, https://github.com/murillo128/mathia/blob/66d42c79f580bd7827a0a73dd2f43ca015653ec5/research/prime_lattice/findings/PL-319-endpoint-log-source-trace-tauberian.md (15 Sep 2026); with PL-320, PL-321, PL-322 (15 Sep 2026; commits c62c7641114c494ec878ebc090f2d49f9c07f620, 095bfd2c0ac1a35fd62aaf8fd35d5e1e0e672778, 95d691fe90353d500dab5485cc22d2c95049ee26), Unrefereed; AI-assisted research note.

35. ———, *PL-323: A coercive tail comparison closes the remaining pointwise endpoint-flux gate*, mathia repository, https://github.com/murillo128/mathia/blob/3aa38419e6089f705e1b2ab940f12f78ab0c97f8/research/prime_lattice/findings/PL-323-coercive-tail-comparison-closes-pointwise-flux.md (15 Sep 2026), Unrefereed; AI-assisted research note.
36. ———, *PL-324: Odd-sector Beurling–Deny positivity survives exactly to the first prime event*, mathia repository, https://github.com/murillo128/mathia/blob/593e63c5e5a17553ad6099173e7579466b865898/research/prime_lattice/findings/PL-324-odd-sector-first-prime-positivity-threshold.md (16 Sep 2026), Unrefereed; AI-assisted research note.
37. ———, *PL-329: The endpoint source amplitude has a sharp square-root-log threshold for first-prime reflection*, mathia repository, https://github.com/murillo128/mathia/blob/ac5835e308a81e20899d3105a8b99c8216752e6e/research/prime_lattice/findings/PL-329-first-prime-reflection-sharp-source-amplitude-threshold.md (16 Sep 2026), Unrefereed; AI-assisted research note.
38. ———, *PL-335: Odd parity reverses the threshold archimedean source and restores a Hopf boundary lower law*, mathia repository, https://github.com/murillo128/mathia/blob/6f267e0ed3c355185d8aa3bdf839b88aad489eeb/research/prime_lattice/findings/PL-335-odd-threshold-antisymmetric-hopf-noncollapse.md (17 Sep 2026), Unrefereed; AI-assisted research note.
39. ———, *PL-336: Uniform sup control closes the first-prime source gate*, mathia repository, https://github.com/murillo128/mathia/blob/2a3652475ec23f33fb65de7ba8e941d7b8fd15f5/research/prime_lattice/findings/PL-336-uniform-linfty-closes-first-prime-source-gate.md (17 Sep 2026), Unrefereed; AI-assisted research note.
40. ———, *PL-337: Prime shifts are positive atomic jumps, giving uniform L^∞ across finite activation ranges*, mathia repository, https://github.com/murillo128/mathia/blob/ce507a6a88f6ace1d38b133dda08a1c7218e1bbf/research/prime_lattice/findings/PL-337-prime-shift-jump-boundedness.md (17 Sep 2026; first commit 8b91052), Unrefereed; AI-assisted research note.
41. ———, *PL-338: Every finite odd prime-power activation is first-order flat; a nonzero boundary trace gives the favorable birth sign*, mathia repository, https://github.com/murillo128/mathia/blob/aa1857f47852ad1bd54a2f383412a5efc8558bd9/research/prime_lattice/findings/PL-338-all-prime-activation-flatness.md (17 Sep 2026), Unrefereed; AI-assisted research note.
42. S. M. Gonek, *A formula of Landau and mean values of $\zeta(s)$*, Topics in Analytic Number Theory (Austin, Tex., 1982), Univ. Texas Press, Austin, TX, 1985, pp. 92–97.
43. ———, *An explicit formula of Landau and its applications to the theory of the zeta-function*, A Tribute to Emil Grosswald: Number Theory and Related Analysis, Contemp. Math., vol. 143, Amer. Math. Soc., Providence, RI, 1993, pp. 395–413.
44. Akiva Groskin, *A finite Guinand–Weil dictionary and archimedean tail order for the truncated Weil quadratic form*, arXiv:2607.02828, 2026.
45. Tomasz Grzywny and Michał Ryznar, *Potential theory of one-dimensional geometric stable processes*, Colloq. Math. **129** (2012), no. 1, 7–40.
46. halasz_resonator, *Forum comment 3-jlbw7xu_tS6Rou*, rStation, riemann_hypothesis forum post 21, 2026-10-19 00:10:21 UTC, https://rstation.ai/r/riemann_hypothesis/posts/21, Unrefereed forum comment by an AI agent.
47. Víctor Hernández-Santamaría, Luis Fernando López Ríos, and Alberto Saldaña, *Optimal boundary regularity and a Hopf-type lemma for Dirichlet problems involving the logarithmic Laplacian*, Discrete Contin. Dyn. Syst. **45** (2025), no. 1, 1–36.
48. Henryk Iwaniec and Emmanuel Kowalski, *Analytic number theory*, Amer. Math. Soc. Colloq. Publ., vol. 53, Amer. Math. Soc., Providence, RI, 2004.
49. Denis Joubert, *The well: five faces of the ground state of the windowed Weil form, and the approximation lemma that would hold them*, Note, v3 (6 Sep 2026), `notes/the-well.pdf` in https://github.com/darreal44/riemann-weil-phenomenology (commit 99470a6c5d82662c33b7ba46ed31b64fb90fd5dc); archived as release v1.5, 2026, Unrefereed; written with the assistance of language models.
50. Tosio Kato, *Perturbation theory for linear operators*, Classics in Mathematics, Springer, Berlin, 1995.
51. Panki Kim and Ante Mimica, *Green function estimates for subordinate Brownian motions: stable and beyond*, Trans. Amer. Math. Soc. **366** (2014), no. 8, 4383–4422.

52. Taebong Kim, Youngsik Hong, Minsik Kim, Sunyoung Choi, Jaewon Jang, and Minseo Kim, *A numerical realization of Suzuki's Weil-quadratic-form operator: the archimedean spectral law, its universality, and an operator form of Weil's positivity criterion*, arXiv:2607.24830, 2026.
53. Pavel V. Kramarenko-Byrd, *Restricted Weil ground states and real Fourier zeros: a form-core criterion and a certified interval of supports*, Zenodo record 23007100, version 1.0 (27 Sep 2026), 2026, Unrefereed.
54. Mateusz Kwaśnicki, *Spectral analysis of subordinate Brownian motions on the half-line*, Studia Math. **206** (2011), no. 3, 211–271.
55. Mateusz Kwaśnicki and Jacek Wszoła, *Dirichlet–Neumann bracketing for nonlocal operators*, arXiv:2609.24892, 2026.
56. Edmund Landau, *Über die Nullstellen der Zetafunktion*, Math. Ann. **71** (1912), no. 4, 548–564.
57. Byoungwoo Lee, *Finite-rank prime-edge transmission and adaptive continuation for localized Weil comparisons*, Zenodo, version 1.0 (5 Sep 2026), 2026, Unrefereed.
58. lihongmin, *Cutoff-free positivity of localized Weil forms through the* $(\log 11)/2$ *threshold*, rStation, riemann_hypothesis paper 45, v1, published 2026-10-23 09:48:37 UTC, https://rstation.ai/problems/riemann_hypothesis/papers/45/v1, Unrefereed; written by an AI agent and accepted by automated reviewer agents.
59. Vincent Liu, *Certified Weil positivity beyond the unit window: source-exact block-Schur and tail-compensation bounds for the Riemann zeta function*, Manuscript submitted to Math. Comp. (14 Sep 2026); https://github.com/luciferyu666/certified-weil-positivity, release v1.0-mcom-submission, 2026, Unrefereed.
60. Ryota Mori, *An endpoint-smoothed prolate vector and a* $\mu^{9/2}(\log\mu)^4$ *upper bound for windowed Weil forms*, Zenodo, version 3 (10 Oct 2026); https://github.com/moriryota/prolate-weil-bounds, 2026, Unrefereed.
61. ———, *Weil positivity on the window of width* $\log 15$, Zenodo (10 Oct 2026), 2026, Unrefereed.
62. ———, *The Weil quadratic form of the Connes–Consani–Moscovici prolate vector, and a* μ^8 *upper bound for windowed Weil forms*, Zenodo, version 3 (10 Oct 2026); https://github.com/moriryota/prolate-weil-bounds, 2026, Unrefereed.
63. Theodore Magnus Øen, *A Fourier–Lévy normal form for localized Weil positivity: finite-rank boundary terms, scalar resolvents, and a certified finite block and continuum Fourier tail*, Zenodo, version 0.7r1 (19 Sep 2026), 2026, Unrefereed.
64. Walter Rudin, *Real and complex analysis*, third ed., McGraw-Hill, New York, 1987.
65. René L. Schilling, Renming Song, and Zoran Vondraček, *Bernstein functions: Theory and applications*, 2 ed., De Gruyter Studies in Mathematics, vol. 37, De Gruyter, 2012.
66. A. L. Skubachevskii, *Boundary-value problems for elliptic functional-differential equations and their applications*, Russian Math. Surveys **71** (2016), no. 5, 801–906.
67. Alexander L. Skubachevskii, *Elliptic functional differential equations and applications*, Operator Theory: Advances and Applications, vol. 91, Birkhäuser, Basel, 1997.
68. span_of_dilates, *Finite-window Weil forms on Davenport–Heilbronn functions: a certified onset bound, an exact joint detection identity, and finite-space measurements*, rStation, riemann_hypothesis paper 66, v2, published 2026-10-27 21:16:47 UTC (v1 2026-10-27 17:23:12 UTC), https://rstation.ai/problems/riemann_hypothesis/papers/66/v2, Unrefereed; written by an AI agent and accepted by automated reviewer agents.
69. ———, *Forum comment* `EAw5pFmSdKxMBRHA`, rStation, riemann_hypothesis forum post 21, 2026-10-28 03:55:32 UTC, https://rstation.ai/r/riemann_hypothesis/posts/21, Unrefereed forum comment by an AI agent; labelled heuristic by its author.
70. ———, *Forum comment* `nWrDJmydgFHjXFOu`, rStation, riemann_hypothesis forum post 21, 2026-10-27 18:55:11 UTC, https://rstation.ai/r/riemann_hypothesis/posts/21, Unrefereed forum comment by an AI agent.
71. Masatoshi Suzuki, *Spectral convergence associated with Weil's quadratic form*, arXiv:2610.32704, 2026.
72. ———, *Weil's quadratic form via the screw function*, arXiv:2606.09096, 2026.
73. Shina Tan, *Energetics of a strongly correlated Fermi gas*, Ann. Physics **323** (2008), no. 12, 2952–2970.
74. ———, *Large momentum part of a strongly correlated Fermi gas*, Ann. Physics **323** (2008), no. 12, 2971–2986.
75. E. C. Titchmarsh, *The theory of the Riemann zeta-function*, 2 ed., Clarendon Press, Oxford University Press, New York, 1986.
76. Oliver Tuma, *Finite Weil windows: Euler boundary geometry, hyperbolic parity, and smoothed prime errors*, Zenodo, version 1.4 (11 Oct 2026), 2026, Unrefereed.
77. R. Yu. Vorotnikov and A. L. Skubachevskii, *Smoothness of generalized eigenfunctions of differential–difference operators on a finite interval*, Math. Notes **114** (2023), no. 5-6, 1002–1020.
78. ———, *On the regularity of generalized eigenfunctions of differential–difference operators*, Differ. Equ. **62** (2026), no. 1, 90–111.

79. André Weil, *Sur les "formules explicites" de la théorie des nombres premiers*, Comm. Sém. Math. Univ. Lund [Medd. Lunds Univ. Mat. Sem.] **1952, Tome supplémentaire** (1952), 252–265.
80. R. Wong and J. F. Lin, *Asymptotic expansions of Fourier transforms of functions with logarithmic singularities*, J. Math. Anal. Appl. **64** (1978), no. 1, 173–180.
81. H. Xiao, V. Rokhlin, and N. Yarvin, *Prolate spheroidal wavefunctions, quadrature and interpolation*, Inverse Problems **17** (2001), no. 4, 805–838.
82. Hiroyuki Yoshida, *On Hermitian forms attached to zeta functions*, Zeta Functions in Geometry (Tokyo, 1990), Adv. Stud. Pure Math., vol. 21, Kinokuniya, Tokyo, 1992, pp. 281–325.
83. Xuefeng Zhu, *Weil positivity in compact windows: a finite reduction, certified two-sided bounds, and a Landau–Widom decay law*, arXiv:2608.24827, 2026.

GenusCore
Email address: info@genuscore.com